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Chapter 9

GELFAND-P000603

15.9.75. Gelfand Sem.

There will be a series of lectures by Polyakov on physics.

Kirillov. Thm. J. Vey on deformation of the Poisson bracket.



Uncertain reading. The initial reads “J.”.

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(M, ω) — a sympl. manifold, $[u, v]$ — the Poisson bracket. There are essentially no other Lie alg. structures (Kirillov et al.). But if diff. oper. of order ∞ are allowed, the structure can be deformed.

Thm.: $\exists$ a formal deformation of the Poisson bracket, inequivalent to it.

$$[u, v]_t = \sum_{k=0}^{\infty} t^k [u, v]^{(k)}$$

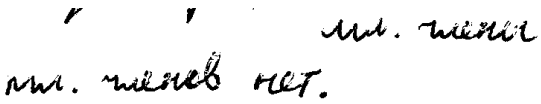
[stroke from $[u, v]^{(k)}$:] oper. of order $(2k + 1)$ in u and v.

nontriviality: a change of coordinates does not reduce it to the original Poisson bracket.

$$[u, v]^{(k)} = \frac{1}{(2k+1)!} \left(\frac{\partial^{2k+1} u}{\partial p^{2k+1}} \frac{\partial^{2k+1} v}{\partial q^{2k+1}} - \frac{\partial^{2k+1} u}{\partial q^{2k+1}} \cdot \frac{\partial^{2k+1} v}{\partial p^{2k+1}} \right) + \dots$$

lower terms

On the plane (or torus) there are no lower terms.



Uncertain reading. The abbreviation in both places tentatively reads “мл.” (lower).

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Defined globally on M provided that $H^3(M, R) = 0$.

$$? \qquad pq - qp = t$$

GELFAND-P000604

Polyakov. (Alexander Markovich).

Elementary particle physics.

γ — photon. — a separate group

Leptons

$$\left.\begin{array}{ll} e^- & m = 0{,}5 \\ \mu^- & m = 105 \\ \nu_e & m = 0 \\ \nu_\mu & m = 0 \end{array}\right\} \text{ neutrinos}$$

Hadrons.

$$\begin{array}{ll} p & m = 940 \\ n & \text{— } \| \text{ —} \\ \pi & m = 140 \end{array}$$

etc.



Uncertain reading. The top stroke of π is separate from the lower strokes.

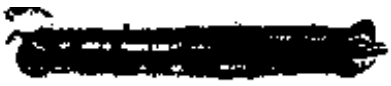
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Interactions: Intensity in units $x = \dfrac{E_{\text{int}}}{m}$

Gr	(gravit.)	$x \sim 10^{-38}$
W	(weak)	$x \sim 10^{-5}$
EM	(electromagn.)	$x \sim 10^{-2}$
S	(strong)	$x \sim 1$

The weaker the interact., the more accommodating it is (it violates fewer conservation laws).

They are distinguished only by symmetry (the strong from the weak, since both are [crossed out] short-range).



Undeciphered passage. The word after “both” is heavily crossed out.

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Leptons — all interact. except the strong.

Hadrons — all 4 interactions.

Quantum electrodynamics describes the electromagn. interact. of leptons. (the others are weaker and are ignored for now).

A particle is described by its own field. One must specify the action:

$$S = S[e(x), \mu(x), \ldots]$$

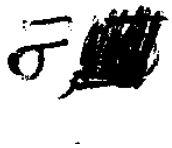
Invariance: rot., transl., Lorentz transf..

Exper. fact: under rot. the electron transf. according to the spinor representation

$$\begin{pmatrix} e_1' \\ e_2' \end{pmatrix} = \exp\left(\frac{i}{2}(\bar{\sigma}\,\bar{\Theta})\right)\begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$$

($e_j \in \mathbb{C}$ — components of the spinor)

$\bar{\sigma}$ [crossed out] — Pauli matrices (a set of 3 such matrices); $\bar{\Theta} \in \mathbb{R}^3$;



Uncertain reading. The entry after $\bar{\sigma}$ is heavily crossed out.

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$|\bar{\Theta}|$ — angle of rot., dir. of Θ — dir. of the axis of rotation.

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$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$



Uncertain reading. There is a short stroke before the upper right 1 in σ_x. It may be a minus.

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Calc. also the transf. under the action of the Lorentz group. 2 possibilities:



Uncertain reading. The initial abbreviation tentatively reads “Вычис.” (calc.).

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$$e'_L = \exp\left(\frac{\bar\sigma\bar\psi}{2}\right) e_L,$$

where $\bar\psi$ — a vector collinear with the velocity vect.:

$$v = \operatorname{th}|\bar\psi|, \qquad \frac{\vec v}{|\bar v|} = \frac{\bar\psi}{|\bar\psi|} \qquad (\bar\psi \text{ — also a 3-vector})$$

[below the line] “rapidity”

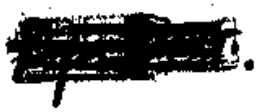
$$e'_R = \exp\left(-\frac{\bar\sigma\bar\psi}{2}\right) e_R$$

Explan. (G.) [crossed out]. $SO(3) \longrightarrow$ [crossed out]. $SU(2)$.



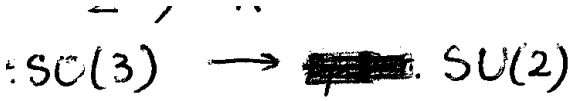
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Undeciphered passage. An entry before $SO(3)$ is crossed out.



Original source details: GELFAND-P000605

Undeciphered passage. An entry before $SU(2)$ is crossed out.



Original source details: GELFAND-P000605

Uncertain reading. The handwritten arrow points from $SO(3)$ to $SU(2)$.

$$\bar\Theta \text{ — rot. in } \mathbb{R}^3 \Longrightarrow \text{matr.} \in SU(2)$$

(representation of $SU(2)$)

This rep. extends to a rep. of the Lorentz group.

Lorentz transf. — the velocity vector $\vec v$ (rotation is omitted) (proper Lorentz transf.)

Rapidity $\bar\psi$ — introduced in place of $\bar v$

P.

$$\begin{cases} t' = t\,\mathrm{ch}\,\psi + z\,\mathrm{sh}\,\psi \\ z' = z\,\mathrm{ch}\,\psi + t\,\mathrm{sh}\,\psi \end{cases}$$



Original source details: GELFAND-P000605

Uncertain reading. In the first line z is written over an earlier mark.

[at right] — Lorentz transf. written in terms of ψ. (proper Lorentz transf. — with no added rot. or translation).

The electron is either e_L or e_R.

But there is a symmetry $\vec{x} \longrightarrow -\vec{x}$, $\vec{v} \longrightarrow -\vec{v}$, they turn into each other.

If reflection is adjoined, the rep. is only obtained in combined form:

$$\hat{p} \begin{pmatrix} e_L \\ e_R \end{pmatrix} = \begin{pmatrix} e_R \\ e_L \end{pmatrix}.$$

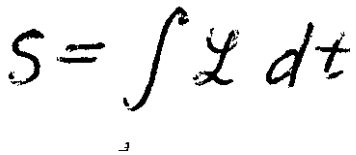
We need to introd. a bispinor: $\begin{pmatrix} e_L \\ e_R \end{pmatrix}$.

Then the Lagrangian density:

$$\mathcal{L}' = e_L^+(x) \left\{ \frac{\partial}{\partial t} - \left(\bar{\sigma} \frac{\partial}{\partial \bar{x}} \right) \right\} e_L(x) + e_R^+(x) \left\{ \frac{\partial}{\partial t} + \left(\bar{\sigma} \frac{\partial}{\partial \bar{x}} \right) \right\} e_R(x)$$

$(z^+ = \bar{z})$.

Action: $S = \int \mathcal{L}\, dt\, d^3\bar{x}$.



Original source details: GELFAND-P000605

Uncertain reading. No prime is visible on $\mathcal{L}$ in the action formula.

Invariance is easy to check.

Masslessness Lagr. eq.

$$\left(\frac{\partial}{\partial t} - \bar{\sigma} \frac{\partial}{\partial \bar{x}} \right) e_L = 0$$

Take the Fourier transf.: $(E - \bar{\sigma}\bar{p})e_L = 0$ and hence $E^2 = p^2$.

GELFAND-P000606

Hence E is arbitrarily small (for small p) $\Rightarrow m = 0$.

We must add a term with the mass. The inv. way is unique

e_L [crossed out] ~~me~~



Original source details: GELFAND-P000606

Uncertain reading. Under the crossing-out, the mark between e_L and me resembles a right-pointing arrow.

$$m(e_L^+ e_R + e_R^+ e_L).$$

Thus, the final form of the Lagrangian:

$$\mathcal{L} = \mathcal{L}' + m(e_L^+ e_R + e_R^+ e_L).$$

Equiv. form: let $e = \begin{pmatrix} e_L \\ e_R \end{pmatrix}$. (bispinor).

$$\mathcal{L} = \bar{e}\left(i\gamma_\mu \frac{\partial}{\partial x_\mu} - m\right) e$$

$$\gamma_\mu \frac{\partial}{\partial x_\mu} = \gamma_0 \frac{\partial}{\partial t} - \bar{\gamma}\frac{\partial}{\partial \bar{x}}, \qquad \bar{e} = e^+\gamma_0$$

γ_μ — Dirac matr., $\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2g_{\mu\nu}$,

$$g_{\mu\nu} = \begin{pmatrix} 1 & & & 0 \\ & -1 & & \\ & & -1 & \\ 0 & & & -1 \end{pmatrix}$$ — Minkowski metr..

γ_μ are written in terms of the matrices σ

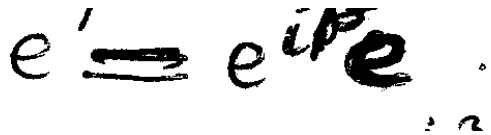
(References: Landau, Berestetskii, Lifshitz, Pitaevskii — Relativistic Quant. Theory, pt. 1).

Spectrum: $E = \sqrt{p^2 + m^2}$.

(In the Dirac eq. $\left(i\gamma_\mu \frac{\partial}{\partial x_\mu} - m\right) e = 0$ we pass to the Fourier transf. and obtain this res.)

— · —

Gauge group $e' \Rightarrow e^{i\beta}e$. $(\beta \in \mathbb{R})$.



Uncertain reading. There are two horizontal strokes between $e\prime$ and $e^{i\beta}e$. The upper stroke has a point at the right.

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$\mathcal{L}$ is also invariant with resp. to this group $e' = (2,7\ldots)^{i\beta} e$

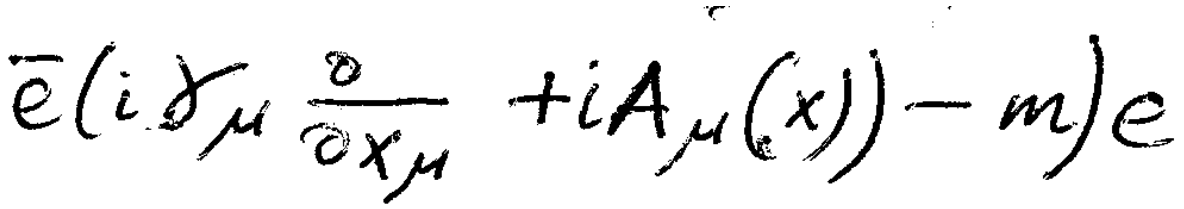
Hypothesis: vary the phase differently at different points and require covar. with resp. to these transformations (extreme locality)

$$e' = (2,7)^{i\beta(x)} e$$

Invar. is now lost:

$$\bar{e}\left(i\gamma_\mu\left(\frac{\partial}{\partial x_\mu} + i\partial_\mu\beta\right) - m\right)e.$$

$$\mathcal{L}' = \bar{e}\left(i\gamma_\mu\frac{\partial}{\partial x_\mu} + iA_\mu(x)\right) - m\Big)e$$



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Uncertain reading. A closing parenthesis follows $A_\mu(x)$, and another follows $-m$. No factor γ_μ is visible before $iA_\mu(x)$.

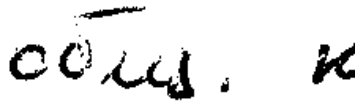
GELFAND-P000607

($A_\mu(x)$ — a connection).

$$A'_\mu(x) = A_\mu - \partial_\mu\beta.$$

$$\nabla_\mu = \partial_\mu + iA_\mu.$$

If we do this, general covariance will be restored.



Uncertain reading. The abbreviation before “covariance” reads “общ.” (general).

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From $\mathcal{L}'$ no eqs. arise for A_μ. We need the curvature

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

This is a tensor, and in [crossed out] $\mathcal{L}$ we introd. its square.



Uncertain reading. An entry before $\mathcal{L}$ is crossed out.

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$$\mathcal{L} = \text{[crossed out]}\ \mathcal{L}' + \left(\frac{1}{4e^2}\right) F_{\mu\nu}^2$$

Uncertain reading. A sign before $\mathcal{L}'$ on the right-hand side is crossed out.



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(e — the electron charge)

— · —

Scattering. $S = S[\varphi(x)]$.

B $\varphi_{\text{in}}(x) \longrightarrow \varphi_{\text{out}}(x)$

Uncertain reading. A separate mark to the left of the arrow resembles a B with a vertical stroke.



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$$\varphi_{\text{in}}(x) = \lim_{t\to-\infty} \varphi(\bar{x}, t), \qquad \varphi_{\text{out}}(x) = \lim_{t\to+\infty} \varphi(\bar{x}, t).$$

$$\int_{\Phi_{\text{in}}}^{\Phi_{\text{out}}} D\varphi(x)\, e^{iS[\varphi]} = M \text{ — transition amplitude}$$

(functional integration, $D\varphi$ — the natural measure, [below the line] (Lebesgue), integral over all $\Phi(x,t)$, $\Phi(x,+\infty) = \Phi_{\text{out}}$, $\varphi(x,-\infty) = \varphi_{\text{in}}$)

There are rules for calculation.

[crossed out] [inserted above the line and outlined] Without A_μ — a Gaussian integral (Fresnel)



Undeciphered passage. An entry before the insertion and the words “Gaussian integral” is heavily crossed out.

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Then — perturbation theory (take A_μ small); $eA_\mu e$ — a cubic term

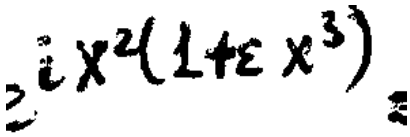
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G. $\int e^{i(x^2+\varepsilon x^3)}\,dx$ — your theory

Important:

$$\int e^{-\sum a_{ik}x_ix_k+\varepsilon\sum x_i^3}\,dx$$

$$\int e^{ix^2(1+\varepsilon x^3)} = \sum c_k \int x^k e^{ix^2}\,dx$$

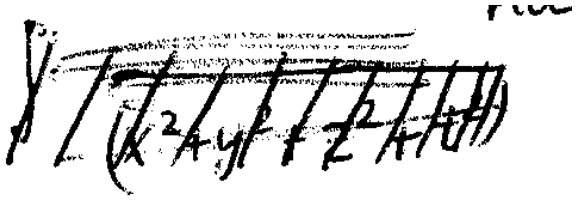


Uncertain reading. The exponent reads $ix^2(1+\varepsilon x^3)$.

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For this we use Feynman diagrs.

[crossed out at left] $\int \frac{}{(x^2+y^2+z^2+t\ldots)}$

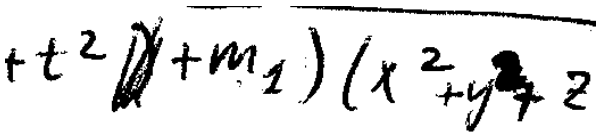


Undeciphered passage. In the crossed-out entry the integral sign, fraction bar, and part of the denominator are legible.

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E.g.

$$\int \frac{dx\,dy\,dz\,dt}{(x^2+y^2+z^2+t^2+m_1)(x^2+y^2+z^2+t^2+m_2)}$$



Uncertain reading. A parenthesis is crossed out after the first t^2. The exponent on y in the second factor is overwritten.

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(stolen by F. from Schwinger)

$$\frac{1}{ab} = \int_0^1 \frac{ds}{(sa+(1-s)b)^2}. \qquad \frac{1}{abc} \text{ analogously.}$$

GELFAND-P000608

P. one can also do $\frac{1}{a^\lambda b^\mu c^\gamma}$



Uncertain reading. The exponent on a is overwritten and tentatively reads λ.

Original source details: GELFAND-P000608

$$\frac{1}{a^\lambda b^\mu} = \int_0^1 \frac{s^{\lambda-1}(1-s)^{\mu-1}}{(sa+(1-s)b)^{\lambda+\mu}}\, ds$$

$$\frac{1}{\prod a_i^{\lambda_i}} = \int_{s_i \geq 0.} \frac{\prod ds_i\, \delta(\sum s_i - 1)\, s_i^{\lambda_i - 1}}{(\sum s_i a_i)^{\sum \lambda_j}}$$

$$\frac{1}{a^\lambda b^\mu} = \int_0^1 \frac{s^{\lambda-1}(1-s)^{\mu-1}}{(sa+(1-s)b)^{\lambda+\mu}}\, ds$$

$$\frac{1}{\prod a_i^{\lambda_i}} = \int_{s_i \geq 0.} \frac{\prod ds_i\, \delta(\sum s_i - 1)\, s_i^{\lambda_i - 1}}{(\sum s_i a_i)^{\sum \lambda_j}}$$

Original source details: GELFAND-P000608

Uncertain reading. Both formulas are written without a separate normalization factor.

—— · ——

We calculate by perturbation theory:

$x,\ x'$

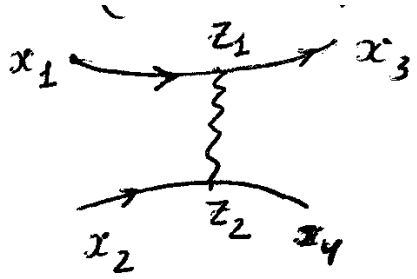
Amplitude of trans. from x to x': $K(x,x')$ according to Maxwell

$$\left(i\gamma_\mu \frac{\partial}{\partial x_\mu} - m\right) K(x,x') = \delta(x-x').$$



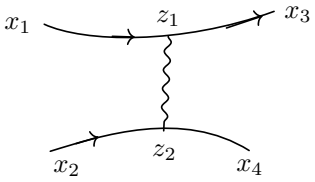
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Uncertain reading. The γ is heavily inked and may be overwritten.



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Diagram and typed description.

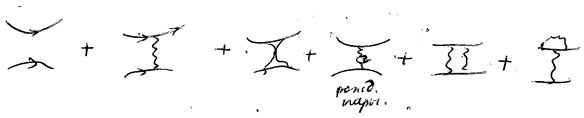


Emission and absorption of a photon are accomp. by a factor e.

$$\int K(x_3 - z_1)K(z_1 - x_1)K(x_4 - z_2)K(z_2 - x_2)$$
$$D(z_2 - z_1)\, d^4z_1\, d^4z_2$$

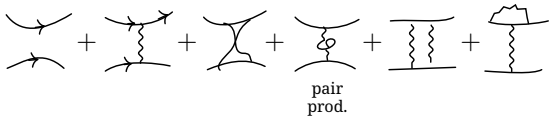
D — fund. soln. of Maxwell's eqs.

Other pictures:



Original source details: GELFAND-P000608

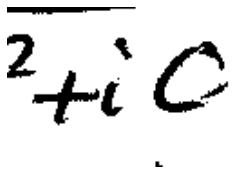
Diagram and typed description.



The meeting of the strokes in the third sketch and the direction of the loop in the fourth are unclear.

$$(\gamma_\mu p_\mu - m)K(p) = 1$$
$$K(p) = \frac{\gamma_\mu p_\mu + m}{\underline{p}^2 - m^2}$$
$$K(x, x') = \int K(p)e^{ip(x-x')}\, d^4p$$

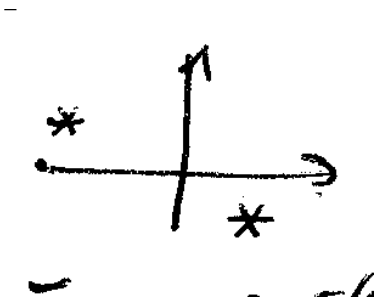
The pole is bypassed if we take $K(p) = \dfrac{\gamma_\mu p_\mu + m}{\underline{p}^2 - m^2 + i0}$



Original source details: GELFAND-P000608

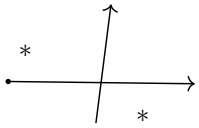
Uncertain reading. The last mark is open and is read as zero.

$$\underline{p}^2 = E^2 - \vec{p}^2.$$

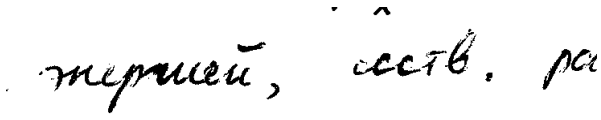


Original source details: GELFAND-P000608

Diagram and typed description.



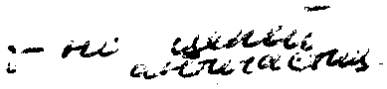
There are levels with neg. energy, nat. propagate backwards in time (pred. the existence of antiparticles).



Original source details: GELFAND-P000608

Uncertain reading. The abbreviation after "energy" tentatively reads "ество." (nat.).

There is a unique choice of bypass leading to a consistent theory: — nor have antiparticles



Uncertain reading. The last two lines are close together. The reading “nor have antiparticles” is tentative.

Original source details: GELFAND-P000608

GELFAND-P000609

Charge normalization.

e — is this the obs. electron charge?

Scattering: electrons exch. photons, force $\sim e^2$



Diagram and typed description.

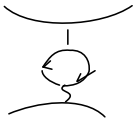


Original source details: GELFAND-P000609

Correction: $\sim e^4$.



Diagram and typed description.



The directions of the two arrow-like strokes on the loop are unclear. The visible gaps are retained.

Original source details: GELFAND-P000609

Obs. charge $e^2_{\text{phys}} < e^2$.

In strong interact. theory the reverse holds $g^2_{\text{phys}} > g^2$.

— · —

GELFAND-P000610

22.9.75. Gelfand Sem.



Diagram and typed description.



Yellow comment icon in the upper right margin of the digitized frame.

Original source details: GELFAND-P000610

Gelfand on Atiyah's res.

$\mathfrak{A}$ — all oper. in Hilb. sp., $\mathcal{K}$ — comp. oper.

$$\mathfrak{A}/\mathcal{K} = \mathfrak{B}.$$

When are $B_1, B_2 \in \mathfrak{B}$ unitarily equivalent?

This is [crossed out] Brown, Douglas, Fillmore (Lect. Notes, $N = \ldots$)

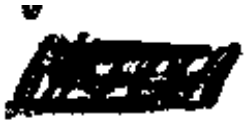


Original source details: GELFAND-P000610

Uncertain reading. An entry before the surnames is crossed out. It may read "Все" (all).

a) spectra coincide (ess.)

b) indices coincide in [crossed out] $\forall$ conn. comp. of the complement

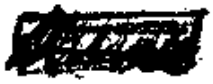


Original source details: GELFAND-P000610

Undeciphered passage. An entry before the quantifier is crossed out.

Thm. 2. [crossed out] Let X — be a compact space,

$$\underset{\substack{\uparrow \\ \text{set of embeddings of } C(X) \text{ in } \mathfrak{B}.}}{\mathrm{Ext}(X)} \qquad \cong K_1(X)$$



Original source details: GELFAND-P000610

Undeciphered passage. An entry before "Let" is crossed out.

—— · ——

Moser — Korteweg–de Vries

—— · ——

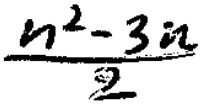
Kostant

$$\psi(x) = \prod_{n=1}^{\infty}(1 - x^n); \qquad \psi(x)^{-1} = \sum_{n=0}^{\infty} p(n)x^n$$

($p(n)$ — number of part. of n).

This is

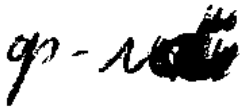
$$\psi(x) = \sum (-1)^n x^{\frac{n^2-3n}{2}}$$



Original source details: GELFAND-P000610

Uncertain reading. The exponent is written $(n^2 - 3n)/2$.

- “The most remarkable formula in all of mathematics” (Jacobi).



Uncertain reading. The heavily overwritten abbreviation after “remarkable” tentatively reads “ф-ла” (formula).

Original source details: GELFAND-P000610

$$(\psi(x))^3 = \sum_{n=0}^{\infty} (-1)^n (2n+1) x^{\frac{n(n+1)}{2}}$$

$$\eta(x) = x^{1/24} \cdot \psi(x).$$

$$\eta^{24}(x) = \sum \tau(n) x^n, \qquad \tau(n) \text{ — Ramanujan's function}$$

$\psi^{10}(x)$ exists too

Manin: if 6 different fns., obt. from different considerations

$$\eta(x)^{\square} = \sum \dots$$

Macdonald found powers for which it makes sense: These are the dim of compact Lie groups.

$$\psi(x)^{\dim K} = \sum_{\lambda} \text{[crossed out]}\ \chi_\lambda(a)\ \dim V_\lambda \cdot x^{(\text{depends on } \lambda)}$$



Undeciphered passage. An entry after the summation sign and before $\chi_\lambda(a)$ is crossed out.

Original source details: GELFAND-P000610

$a \in K$, $\chi_\lambda(a) = \pm 1$ or 0

Kostant: $\mathrm{Sp}(-\Delta + a)^{-s}$ etc.

Internal [ч-ие] G.: $\psi(x)^z$ — ? $\qquad (z \in \mathbb{C})$



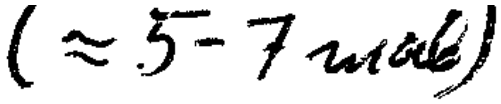
Uncertain reading. The middle abbreviation tentatively reads “ч-ие”.

Original source details: GELFAND-P000610

Polyakov. (continued).

References:

quant. mech.
1) Landau and Lifshitz (quant. mechanics). (≈ 5–7 chapters)
2) Dirac
3) Feynman (for schoolchildren)
4) Feynman. Path integrals



Uncertain reading. The word after "5–7" tentatively reads "глав" (chapters).

Original source details: GELFAND-P000610

Quant. field theory: 1st part of vol. IV Berestetskii, Lifshitz, Pitaevskii

GELFAND-P000611

$e(x)$ — field (bispinors)

$$\mathcal{L}_{\text{free}} = \bar{e}\, i\gamma_\mu \partial_\mu e - m\bar{e}e + \frac{1}{4e^2} F_{\mu\nu}^2$$

$$\mathcal{L}_{\text{int}} = \ldots \bar{e}\gamma_\mu e A_\mu \ldots \qquad \text{cont. 3-... terms}$$



Uncertain reading. The abbreviation after "3-" is uncertain.

Original source details: GELFAND-P000611

Localization of large int. symmetries — weak interaction.



Uncertain reading. "б-их вн." tentatively reads "large internal".

Original source details: GELFAND-P000611

$$\mathcal{L} = \bar{e}_L i\gamma_\mu \partial_\mu e_L + \bar{e}_R i\gamma_\mu \partial_\mu e_R + \bar{\nu}_L i\gamma_\mu \partial_\mu \nu_L +$$
$$+ \begin{pmatrix} e \to \mu \\ \nu_L \to \nu'_L \end{pmatrix} + m(\bar{e}_L e_R + \bar{e}_R e_L).$$

(all leptons) $\downarrow$

ν'_L — μ-meson neutrino

$$\mathcal{L} = \bar{e}_L i\gamma_\mu \partial_\mu e_L + \bar{\nu}_L i\gamma_\mu \partial_\mu \nu_L + \bar{e}_R i\gamma_\mu \partial_\mu e_R$$

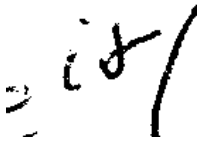
$$\begin{cases} \begin{pmatrix} e'_L \\ \nu'_L \end{pmatrix} = U \begin{pmatrix} e_L \\ \nu_L \end{pmatrix} \qquad SU_L(2) \\ e'_R = e_R \end{cases}$$

•

A new symmetry gr. $SU(2)$ arises, and also $U(1)$

$$F = SU_L(2) \times U_R(1) \times U_L(1) \quad \text{— overall lepton group.}$$

$$\begin{pmatrix} e_L \\ \nu_L \end{pmatrix} \overset{U_L(1)}{\longrightarrow} e^{i\gamma} \begin{pmatrix} e_L \\ \nu_L \end{pmatrix}$$



Original source details: GELFAND-P000611

Uncertain reading. The letter in the phase tentatively reads γ, possibly δ.

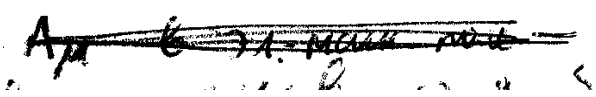
Localization



Original source details: GELFAND-P000611

Undeciphered passage. The crossed-out entry in a box in the left margin is illegible.

[Crossed out: ~~A_μ ... electromagn. field~~.]

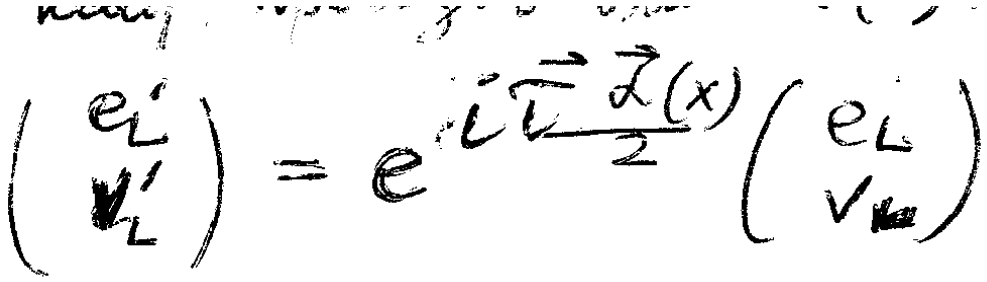


Original source details: GELFAND-P000611

Uncertain reading. A mark between A_μ and "electromagn. field" is obscured by the crossing-out.

Covar. derivative for $SU(2)$. $\bar{\tau}$ — Pauli matr. (3 in all)

$$\begin{pmatrix} e'_L \\ \nu'_L \end{pmatrix} = e^{i\frac{\bar{\tau}\vec{\alpha}(x)}{2}} \begin{pmatrix} e_L \\ \nu_L \end{pmatrix}, \qquad E' = e^{i\frac{\bar{\tau}\vec{\alpha}(x)}{2}} E$$

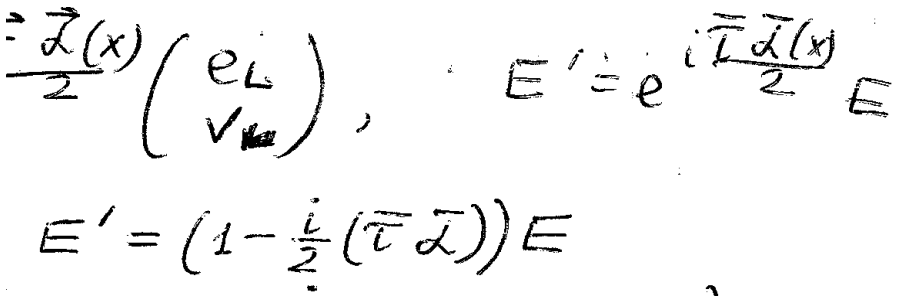


Original source details: GELFAND-P000611

Uncertain reading. The lower letter on the left and the lower subscript on the right are overwritten.

Infinitesim.:

$$E' = \left(1 - \frac{i}{2}(\bar{\tau}\bar{\alpha})\right) E$$



Original source details: GELFAND-P000611

Uncertain reading. The finite exponent is written with an implicit plus sign. A minus sign is visible in the infinitesimal expression.

$$\partial_\mu E' = \left(1 - \frac{i}{2}(\bar{\tau}\bar{\alpha})\right) \partial_\mu E - \frac{i}{2}(\bar{\tau}\partial_\mu\bar{\alpha})E$$

$$\boxed{\nabla_\mu E = \partial_\mu E + \frac{i}{2}(\vec{\tau}\vec{A}_\mu)E}$$

$$E' = e^{i\frac{\bar{\tau}\bar{\alpha}(x)}{2}} E$$

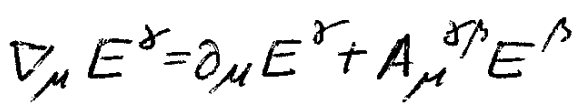
$$\nabla_\mu E' = e^{i\frac{\bar{\tau}\bar{\alpha}}{2}} \nabla_\mu E$$

Here 3 massless particles arise

GELFAND-P000612

one can write directly

$$\nabla_\mu E^\delta = \partial_\mu E^\delta + A_\mu^{\delta\beta} E^\beta$$

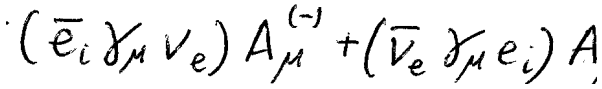


Original source details: GELFAND-P000612

Uncertain reading. The superscript tentatively reads δ, possibly γ.

and we look at how A transforms so that $\nabla_\mu E^\delta$ transforms like E^δ.

$$\mathcal{L}_{\text{int}} = (\bar{e}_i \gamma_\mu \nu_e) A_\mu^{(-)} + (\bar{\nu}_e \gamma_\mu e_i) A_\mu^{(+)} + (\bar{e}_L \gamma_\mu e_L - \bar{\nu}_L \gamma_\mu \nu_L) A_\mu^3$$



Uncertain reading. The electron index in the first two terms is written with a dotted i.

Original source details: GELFAND-P000612

where

$$A_\mu^\pm = A_\mu^1 \pm i A_\mu^2$$

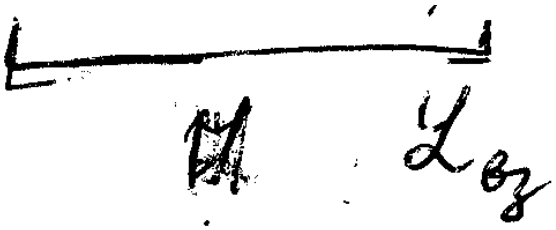
Example of weak interaction: $\mu^- \to e^-, \bar{\nu}_e, \nu_\mu$.

$$\mathcal{L} = \bar{E} \left(i \gamma_\mu \frac{\partial E}{\partial x_\mu} \right) + \underbrace{\bar{E} \gamma_\mu \vec{\tau} E \vec{A}_\mu}_{\mathcal{L}_{\text{int}}} + \bar{e}_R i \gamma_\mu \partial_\mu e_R$$



Uncertain reading. There is a separate short stroke with dots to the left of $\mathcal{L}$. Its meaning is uncertain.

Original source details: GELFAND-P000612



Undeciphered passage. An entry beneath the left side of the bracket is crossed out.

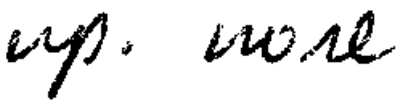
Original source details: GELFAND-P000612



Undeciphered passage. A separate crossed-out entry in the right margin is illegible.

Original source details: GELFAND-P000612

We also localize the r. field



Uncertain reading. The abbreviation “пр.” tentatively means “right”.

Original source details: GELFAND-P000612

$$\bar{E}\left(i\gamma_\mu \frac{\partial E}{\partial x_\mu}\right) + \bar{E}\gamma_\mu \vec{\tau} E \vec{A}_\mu + \bar{e}_R i\gamma_\mu \partial_\mu e_R$$
$$+ \bar{e}_R \gamma_\mu e_R B_\mu$$
$$A_\mu = A_\mu^3 \cos\theta + B_\mu \sin\theta$$
. . . . . _ . . . . .

Next time — ≈ in 3 weeks.

12 quarks: 4 flavors and 3 colors.



Diagram and typed description.



Yellow comment icon to the right of the quark entry in the digitized frame.

Original source details: GELFAND-P000612

—— · ——

GELFAND-P000613

29.9.75. Gelfand Sem.

Yu. I. Manin.

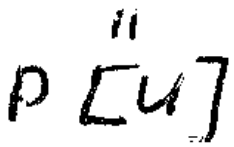
$$x\prod_{n=1}^{\infty}(1-x^n)^{24} = \sum_{n=1}^{\infty}\tau(n)x^n; \qquad \tau \text{ — Ramanujan's function.}$$

I. M. Gelfand. (and Dikii)

∞-dimensional Lie algebra, more remarkable than all the Cartan ones.

Elements — polynomials $P(u, u', \ldots, u^{(n)})$ in any number of variables.

$\overset{''}{P}[u]$, where $u = \{u, u', \ldots, u^{(n)}, \cdot\}$



Uncertain reading. Two strokes stand above the left square bracket.

Original source details: GELFAND-P000613

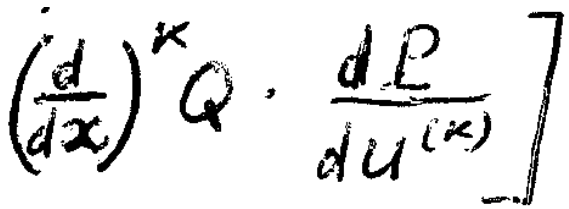
There is a notion of

$$\frac{d}{dx}P(u,u',\dots,u^{(n)}) = \sum_{i=0}^{\infty}\frac{\partial P}{\partial u^{(i)}}\cdot u^{(i+1)}$$

There is multiplication (algebr.)

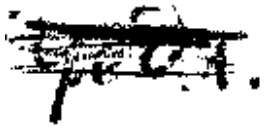
Unexpectedly, there is a notion of the commutator:

$$[P,Q] = \sum_{k=0}^{\infty}\left[\left(\frac{d}{dx}\right)^k P\cdot\frac{\partial Q}{\partial u^{(k)}} - \left(\frac{d}{dx}\right)^k Q\cdot\frac{dP}{du^{(k)}}\right]$$



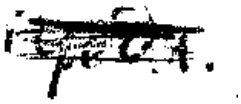
Uncertain reading. The mark after Q reads as a multiplication dot. The last fraction uses d.

Original source details: GELFAND-P000613



Uncertain reading. A short entry before the surname is crossed out.

Original source details: GELFAND-P000613



Undeciphered passage. The crossed-out entry cannot be read in full.

Original source details: GELFAND-P000613

Povzner. This alg. is connected with the eq. $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}$. Can be connected with $\forall$ eq.

$$\frac{\partial u}{\partial t} = a(u,u_x,\dots).$$

Consider fns. $F(u,u_x,\dots)$ and the operator

$$a[F(u,u_x,\dots)] = aF_u + \left(\frac{d}{dx}a\right)F_{u_x} + \frac{d^2a}{dx^2}F_{u_{xx}} + \dots$$

The operator is det. by the fn. a and an algebra is built from these oper.

$[a,b] = a[b] - b[a]$ — the same algebra.

— · —

Here the natural repr. of this algebra is indicated.



Uncertain reading. A mark to the left of “Here” is blacked out.

Original source details: GELFAND-P000613

Manin. 6 expressions for Ramanujan's fn. $\tau(n)$ (not obt. by a finite algorithm).

⊕ ① Definition ⇒

$$\tau(n) = \sum(-1)^k$$

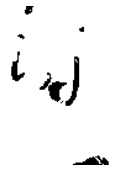
$$n-1 = n_1 + \ldots + n_k, \qquad n_i \geq 1$$

$$n_1 \leq n_2 \leq \ldots \leq n_k$$

in a row ≤ 24 equality signs.

⊕ ② Macdonald (Dyson)

$$\tau(n) = \frac{1}{1!2!3!4!} \sum \prod_{i<j} (u_i - u_j)$$



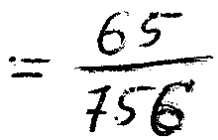
Uncertain reading. The index condition below the product tentatively reads $i < j$.

Original source details: GELFAND-P000613

$$10n = \sum_{i=1}^{5} u_i^2, \qquad \sum u_i = 0, \qquad u_i \equiv i \bmod 5$$

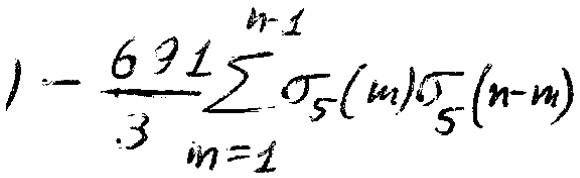
$\binom{+}{+}$ ③ Ramanujan

$$\tau(n) = \frac{65}{756}\sigma_{11}(n) + \frac{691}{756}\sigma_5(n) - \frac{691}{3}\sum_{m=1}^{n-1} \sigma_5(m)\sigma_5(n-m)$$



Uncertain reading. The last digit of the first denominator is overwritten and reads as 6.

Original source details: GELFAND-P000613



Uncertain reading. The argument in the final factor reads $n - m$.

Original source details: GELFAND-P000613

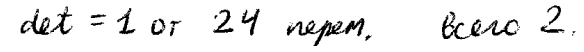
$\sigma_k(n)$ — $\sum$ of the kth powers of the divisors of n.

Corollary: $\tau(n) \equiv \sigma_{11}(n) \bmod 691$. (guessed by Ramanujan).

GELFAND-P000614

④ Int. quad. forms with det $= 1$ in 24 variables. 2 in all.

$\binom{+}{+}$ Let Q_{24} — one of them.



Uncertain reading. The line reads “2 in all”.

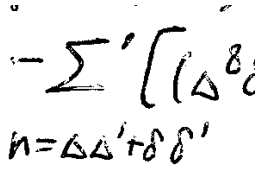
Original source details: GELFAND-P000614

$$\tau(n) = \alpha \left(\sum_{Q_{24}=n} 1 \right) - \beta \sigma_{11}(n)$$

(α, β computed from $n = 1$ and 2).

⑤ (Manin) $\binom{+}{+}$

$$\tau(n) = \sigma_{11}(n) - \sum_{n=\Delta\Delta'+\delta\delta'}{}' \left[(\Delta^8\delta^2 - \Delta^2\delta^8) + \frac{691}{6}(\Delta^6\delta^4 - \Delta^4\delta^6) \right],$$



Uncertain reading. There is a prime on the summation sign.

Original source details: GELFAND-P000614

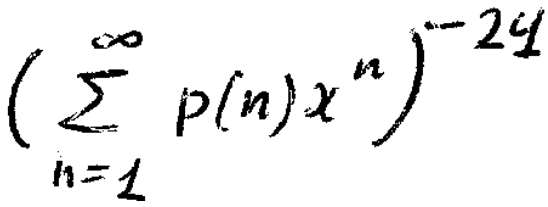
$$\Delta > \delta > 0, \quad \text{and } [\, \Delta' > \delta' > 0 \quad (\text{integers})$$
$$\text{or } \Delta \mid n, \quad \Delta' = \frac{n}{\Delta}, \quad \delta' = 0, \quad 0 < \frac{\delta}{\Delta} \le \frac{1}{2}\,].$$

⑥ $\begin{pmatrix} + \\ + \\ + \end{pmatrix}$

$$\sum p(n)x^n = \prod_{n=1}^{\infty} (1 - x^n)^{-1}$$

($p(n)$ — number of partitions of n into any number of summands).

$$\Rightarrow \quad \sum \tau(n)x^n = x \left(\sum_{n=1}^{\infty} p(n)x^n \right)^{-24}$$



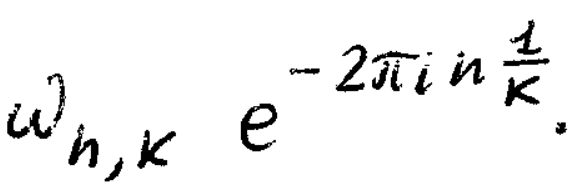
Uncertain reading. The lower limit of the inner sum is written $n = 1$.

Original source details: GELFAND-P000614

Using $p(n)$, find $\tau(n)$ recursively.

Hardy–Ramanujan–Rademacher formula

$$p(n) = \frac{1}{\sqrt{3}} \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} \sum_{\substack{h \bmod k \\ (h,k)=1}} \omega_{h,k} e^{-2\pi i n \frac{1}{k}} \cdot \frac{d}{dn} \frac{\operatorname{sh} \frac{\pi}{6k}\sqrt{24n-1}}{\frac{\pi}{6k}\sqrt{24n-1}}$$



Uncertain reading. The exponent is written with $1/k$.

Original source details: GELFAND-P000614

$\omega_{h,k}^{24} = 1$ there are also explicit formulas for $\omega_{h,k}$.

Hardy–Ramanuj. wrote [crossed out] sh $\to e^{\cdots}$.



Undeciphered passage. A short entry before "sh" is crossed out.

Original source details: GELFAND-P000614

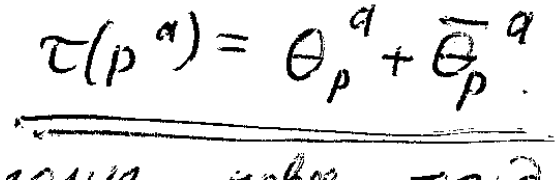
[Below sh $\to e^{\cdots}$: diverges.] but if one truncates at the $\sqrt{n}$ term, it approximates well. One has to stop at $\sqrt{n}$ terms. Each of these formulas is part of an infinite series of formulas.

⑦ $\binom{+}{+}$ $\underline{\underline{\tau(mn) = \tau(m)\tau(n)}}$, $(m,n) = 1$

(not apparent from the formulas).

$$\tau(p) = \theta_p + \bar{\theta}_p, \qquad \theta_p \bar{\theta}_p = p^{11} \quad \Rightarrow$$

$$\underline{\underline{\tau(p^a) = \theta_p^a + \bar{\theta}_p^a.}}$$

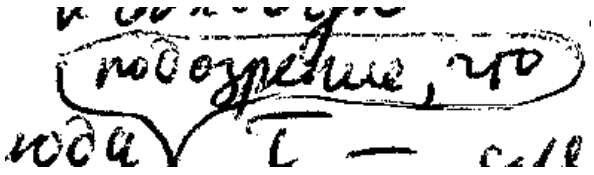


Uncertain reading. The doubly underlined formula is given as written.

Original source details: GELFAND-P000614

One can substitute in each formula and obtain new identities.

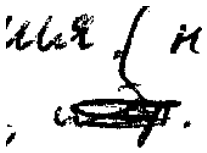
⑧ $\begin{pmatrix} + \\ + \\ + \\ + \end{pmatrix}$ From this [circled insertion: the suspicion that] τ — trace of a representation (much harder) — Serre, Deligne.



Uncertain reading. The circled insertion reads "the suspicion that".

Original source details: GELFAND-P000614

Grothendieck, Weil, Kuga, Sato, [crossed out] Rankin.



Undeciphered passage. The crossed-out entry before Rankin is illegible.

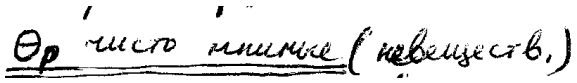
Original source details: GELFAND-P000614

An ℓ-adic repr.

$$\mathrm{Gal}_{\{\ell\}}(\bar{Q}/Q) \longrightarrow Q_\ell^2$$

(ramif. only at ℓ). The trace is taken of the Frobenius autom. gen. by p.

From this it was possible to prove that θ_p are purely imaginary (nonreal)



Uncertain reading. The manuscript reads "purely imaginary".

Original source details: GELFAND-P000614

$\longrightarrow$ The longest thm. in modern mathematics (Deligne's estimate: 2,000 pages).

GELFAND-P000615

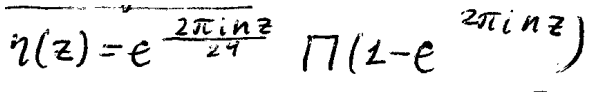
Levels of difficulty.

$+$	combinatorics
$++$	modular forms
$+++$	complex analysis
$++++$	alg. geom. and the like, étale cohom. and so on

—— · ——

How it is obtained.

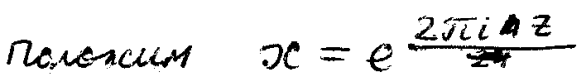
$$\eta(z) = e^{\frac{2\pi i n z}{24}} \prod (1 - e^{2\pi i n z})$$



Uncertain reading. The prefactor exponent is written $2\pi inz/24$.

Original source details: GELFAND-P000615

Set $x = e^{\frac{2\pi i n z}{24}}$.



Uncertain reading. The heavily inked mark between i and z tentatively reads n. The denominator is overwritten.

Original source details: GELFAND-P000615

The fn. behaves [crossed out] well under the substitution $z \to \dfrac{az+b}{cz+d}$, $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 1$.

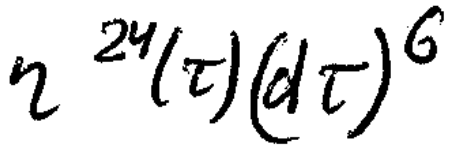


Undeciphered passage. Short strokes after the verb are crossed out.

Original source details: GELFAND-P000615

$$\eta(z+1) = e^{\pi i/12}\eta(z),$$
$$\eta(-1/z) = \sqrt{z/i}\,\eta(z).$$

or $\eta^{24}(\tau)(d\tau)^6$ — inv. under frac.-lin. subst..

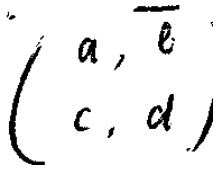


Uncertain reading. The exponent on $d\tau$ is written 6.

Original source details: GELFAND-P000615

Loss of inf. — $\sqrt[24]{\ldots} = \omega_{c,d}$ — the same ones.

$$\begin{pmatrix} a, & \bar{b} \\ c, & d \end{pmatrix}$$



Uncertain reading. In the matrix below "Loss of inf." there is a separate stroke over b.

Original source details: GELFAND-P000615

Fns. of this kind (modular) form a ring with 2 generators, and one must write them in terms of the generators, [crossed out ~~or~~] Identity in the ring ⇒ equality of Fourier coeffs. (⇒ ③, ④).



Uncertain reading. A word above the line is crossed out. It tentatively reads "or".

Original source details: GELFAND-P000615

Pure algebra:

$$\left\{\varphi(z) \,\middle|\, \sum_{n=0}^{\infty} a_n e^{2\pi i n z}\right\}, \qquad \exists k: \quad \varphi(z)dz^k \,\middle|\, \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \varphi(z)dz^k.$$

$\longleftarrow$ converg. series for $\operatorname{Im} z > 0$.

Thm.: the ring $\Phi = C[E_4, E_6]$.

$$\left.\begin{aligned} a_n|_{E_4} &= \sigma_3(n) \\ a_n|_{E_6} &= \sigma_5(n) \end{aligned}\right\| \qquad \sigma_3 \text{ — sums of cubes of divisors.}$$

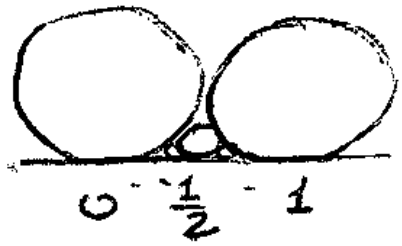
—— · ——

H. R. R. (formula 6).

$$\sum p(n) e^{2\pi i n z} = \frac{1}{\prod(1 - e^{2\pi i z n})}.$$

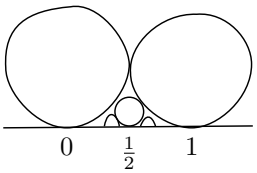
Sing. points on the boundary of convergence (poles of order ∞ at rat. points).

Rademacher's integr. contour: at $\forall$ rat. point, circles of a given radius.



Original source details: GELFAND-P000615

Diagram and typed description.



The exact shape of the short arcs near the small circle is uncertain.

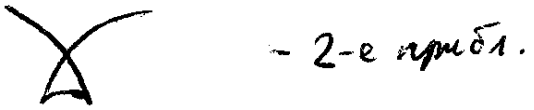


Original source details: GELFAND-P000615

Diagram and typed description.



— 1st approx.



Original source details: GELFAND-P000615

Diagram and typed description.



— 2nd approx.

etc.

Special case of Macdonald's formulas in the form transformed by Kostant:

$$\varphi(x) = \prod_{n=1}^{\infty}(1 - x^n), \qquad \varphi(x) = \sum_{\lambda \in D} \chi_\lambda(a) \dim V_\lambda \, x^{c(\lambda)}.$$



Undeciphered passage. A short entry beneath the product is heavily crossed out.

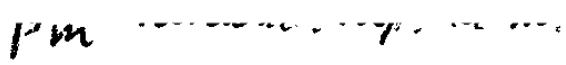
Original source details: GELFAND-P000615

[under $\lambda \in D$: highest weights of irred. reps.; under $\chi_\lambda(a)$: $0, \pm 1$.]

$$c(\lambda) = (\lambda + \rho, \lambda + \rho) - (\rho, \rho).$$

GELFAND-P000616

$6/\overline{X} - 75$ Gelfand Sem.



Undeciphered passage. Only the lower parts of a line of strokes are visible at right, level with the date.

Original source details: GELFAND-P000616

Leray.



Undeciphered passage. A hatched strip is visible under the name. Any letters beneath it are illegible.

Original source details: GELFAND-P000616

“5 years ago Arnold suggested that I understand one of Maslov’s books.”

“Now I am beginning to understand where this book leads.”



Uncertain reading. “мне” (to me) is overwritten. The next word tentatively reads “понять” (understand).

Original source details: GELFAND-P000616

Algebraic prelim. to Maslov’s work,
in part. Buslaev’s metaplectic group.

Maslov: $X = \mathbb{R}^\ell$, Buslaev: X — any set.

$$X = R^\ell: \quad S(X),\ S'(X),\ \mathcal{H}(X) \text{ — intermed. Hilb. sp..}$$

$$a\left(x, \frac{1}{\nu}\frac{\partial}{\partial x}\right), \qquad \frac{\nu}{i} \in \mathbb{R}_+.$$

Assume a — lin. [crossed out] in both arguments, to begin with.



Undeciphered passage. An entry after “lin.” is heavily crossed out.

Original source details: GELFAND-P000616

We seek a gr. of automorph. of $S'(X)$.

$$S: f' \longrightarrow Sf'.$$

$$SaS^{-1} \in A$$

(A — [crossed out] the class of oper. linear in x and $\partial/\partial x$.)



Undeciphered passage. An entry between A and “class of oper.” is crossed out.

Original source details: GELFAND-P000616

Example: $\overline{\mathcal{F}}$ $\overline{\mathcal{F}}^{-1}$ and $\overline{\mathcal{F}}$ in several coord..

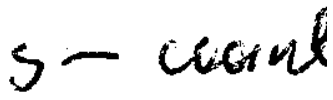
lin. change of var. in x-sp..



Original source details: GELFAND-P000616

Uncertain reading. The first bar over $\mathcal{F}$ has an arched edge. The next two are closer to horizontal.

S — symb. autom. A (inner).



Original source details: GELFAND-P000616

Uncertain reading. The first abbreviation tentatively reads “симв.” (symbol).

$$a : X \oplus X^* \longrightarrow \mathbb{C}$$

$$a(x, p)$$



Original source details: GELFAND-P000616

Uncertain reading. A short blackened stroke is visible between X^* and the line below.

gr. of autom. — sympl. group

$$\langle p', x\rangle - \langle p, x'\rangle.$$

$$X \oplus X^* = Z(\ell).$$

$G(\ell)$ — gr. of autom. in the sp. of distr. $\Rightarrow$ obtain its repr.

$$G(\ell) \longrightarrow \mathrm{Sp}(\ell), \qquad \text{kernel: } \dot{\mathbb{C}}$$



Original source details: GELFAND-P000616

Undeciphered passage. A short entry to the left of “kernel” cannot be read with confidence.

$$G(\ell)/\dot{\mathbb{C}} = \mathrm{Sp}(\ell)$$

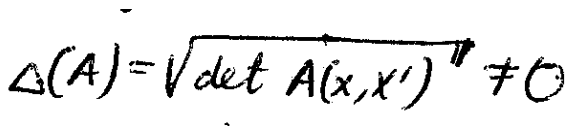
$$c \in \dot{\mathbb{C}} = \mathbb{C} \setminus 0$$

In particular, $G(\ell)$ — a Lie gr..

GELFAND-P000617

Many elem. are known — full and partial Fourier transf..

$$A : X \oplus X \longrightarrow \mathbb{R} \quad \text{real quadr.}, \qquad \Delta(A) = \sqrt{\det A(x, x')''} \neq 0$$



Original source details: GELFAND-P000617

Uncertain reading. Two strokes at the end of the radical tentatively read as $A(x, x')''$.

$$(S_A f')(x) = c\left(\frac{|\nu|}{2\pi i}\right)^{\ell/2} \Delta(A) \int_X e^{\nu A(x,x')} f'(x')\, d^\ell x'$$

$$(x,p) = s_A(x',p') \quad \Leftrightarrow \quad p = A_x,\ p' = -A_{x'}$$

— the corr. sympl. transformation.

S' — a unit. operator.



Original source details: GELFAND-P000617

Uncertain reading. The short stroke at the top of S tentatively reads as the prime in S'.

Unitarity $\Rightarrow$ metapl. group $M_p(\ell)$

I. Segal ; A. Weil ; K. Siegel

Math. phys. arbitr. field $\longrightarrow$ to understand Siegel's work



Original source details: GELFAND-P000617

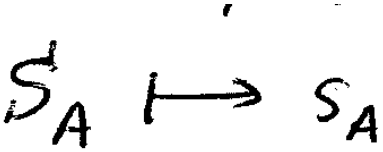
Uncertain reading. The note beneath Weil tentatively reads "arbitr. field". The arrow leads to "to understand Siegel's work".

$M_p(\ell) \ni$ [crossed out] $S'_A \longmapsto s_A$



Original source details: GELFAND-P000617

Undeciphered passage. The entry after the membership sign is heavily crossed out.



Original source details: GELFAND-P000617

Uncertain reading. The stroke at the first S_A tentatively reads as the prime in S'_A.

this is the formula with $|c| = 1$.

Take $c = \pm 1$

$$S_A \cdot S_{A'} = S_{A''}$$

[crossed out] $\underline{\underline{S_A S_{A'} S_{A''} = \pm E}}$

$\Uparrow$ $\overset{\text{sign}}{\sim}$ Maslov ind.

$$s_A s_{A'} s_{A''} = e$$



Original source details: GELFAND-P000617

Undeciphered passage. An entry to the left of the doubly underlined equality is crossed out.

Decompose:

$$A(x,x') = \langle Px, x\rangle - \langle Lx, x'\rangle + \langle Qx', x'\rangle, \qquad P, Q, L : X \longrightarrow X'.$$

. . . . . .



Uncertain reading. A crossed stroke is visible to the left of the row of dots.

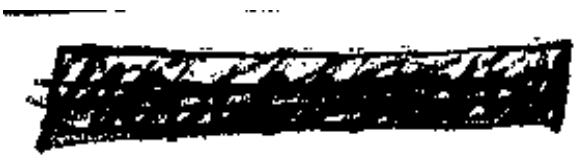
Original source details: GELFAND-P000617

GELFAND-P000618

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13.10.75. Gelfand Sem.

Fradkin. (Efim [crossed out] Samoilovich).



Undeciphered passage. An entry before "Samoilovich" is crossed out.

Original source details: GELFAND-P000618

What does one need to know?

1. Quant. mechanics: Landau
2. Field theory Bogoliubov, Shirkov
   Akhiezer and Berestetskii
   Pauli — old.

Fradkin, Proceedings of FIAN, (field theory and stat. from the funct. approach)
vol. 29 (old) ↓ briefly. S-matr. and so on..

3. Theory of group reps. Gelfand, Minlos, Shapiro



Uncertain reading. The middle surname tentatively reads "Minlos".

Original source details: GELFAND-P000618

4. Anticommuting vars. (Fermi vars.) Berezin; methods of second quant..

5. Abrikosov, Gor'kov, Dzyaloshinskii, "Methods of Field Theory in Stat. Physics" (book).

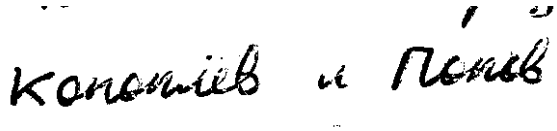


Original source details: GELFAND-P000618

Diagram and typed description.

The digitized frame's yellow icon obscures part of the name "Gor'kov" and nearby strokes.

6. "Gauge Fields" Konoplev and Popov.



Original source details: GELFAND-P000618

Uncertain reading. The surnames in the manuscript tentatively read "Коноплёв и Попов".

7. Reviews by Weinberg, Lee and others in

Perelomov. On completely int. Ham. systems. ($\exists n$ first int. in involution)

$$H = \frac{1}{2}\sum_{k=1}^{n} p_k^2 + U(q_1, \ldots, q_n)$$

G. Liouville's thm.: the syst. is then solved by quadratures.

$$H = \tfrac{1}{2}p^2 + \sin q$$

2nd int. — elliptic fn..

0th example — [crossed out] chain [crossed out] Toda.



Original source details: GELFAND-P000618

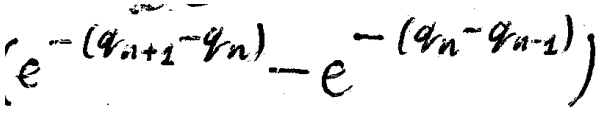
Undeciphered passage. An entry before "chain" is crossed out.



Original source details: GELFAND-P000618

Undeciphered passage. An entry before "Toda" is crossed out.

$$U = g^2 \sum_{k=1}^{n} \left(e^{-(q_{n+1}-q_n)} - e^{-(q_n-q_{n-1})}\right), \qquad q_{k+n} = q_k.$$



Original source details: GELFAND-P000618

Uncertain reading. The summation starts at $k = 1$. The exponents use $n+1, n, n-1$, and a minus separates the exponentials.

[crossed out] S. Manakov ξ 1974; Toda 1970 (Prog. Th. Ph.);



Original source details: GELFAND-P000618

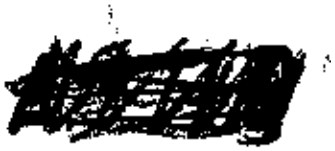
Undeciphered passage. An entry before “S. Manakov” is crossed out.



Original source details: GELFAND-P000618

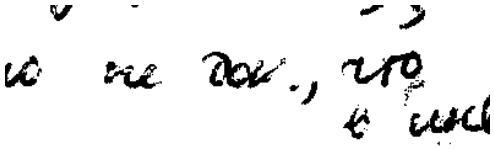
Uncertain reading. A heavily inked mark resembling ξ appears between “Manakov” and the date.

1973 Hénon, Flaschka [crossed out] 1974 $\longrightarrow$ there were int., but it was not proved that they were in invol..



Original source details: GELFAND-P000618

Undeciphered passage. An entry after “Flaschka” is crossed out.

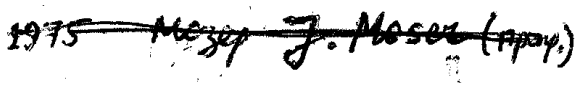


Original source details: GELFAND-P000618

Uncertain reading. The abbreviation after “not” tentatively reads “док.” (proved).

M. Kac, van Moerbeke 1975.

[crossed out ~~1975 Moser J. Moser (prepr.)~~]

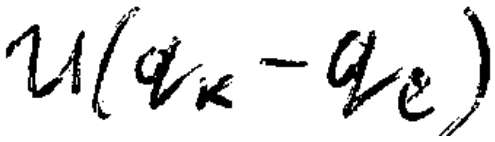


Original source details: GELFAND-P000618

Uncertain reading. The Moser reference is crossed out. The last abbreviation tentatively reads “prepr.”.

A:

$$U = g^2 \sum_{k<\ell} U(q_k - q_\ell).$$



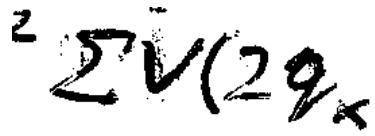
Original source details: GELFAND-P000618

Uncertain reading. The letter under the summation sign in A resembles U.

G. Problems: a) find the int. b) prove invol. c) solve the inv. problem d) express the int. by quadratures.

B.

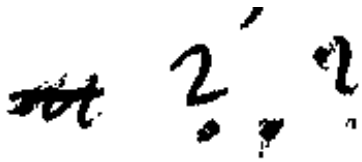
$$U = g^2 \sum_{k<\ell} [V(q_k - q_\ell) + \varepsilon V(q_k + q_\ell)] + g_1^2 \sum V(q_k) + g_2^2 \sum V(2q_k$$



Undeciphered passage. The closing parenthesis is not visible at the right edge of the frame.

Original source details: GELFAND-P000618

	$V(q)$	I_k	$\{I_k, I_\ell\} = 0$	I.P. (inv. problem)	$x(q)$
1	q^{-2}	M, OP	M, OP	?	q^{-1}
2	$a^2 \operatorname{sh}^{-2} aq$	CMR, OP	CMR, OP	?	$a \operatorname{cth} aq$
3	$a^2 \sin^{-2} aq$	M, OP	CMR, OP	?	$a \operatorname{ctg} aq$
4	$a^2 \wp(aq \mid \omega_1, \omega_2)$	C, OP	?, ?	?	$a \frac{\operatorname{cn} aq}{\operatorname{sn} aq}$
5	$q^{-2} + a^2 q^2$	P, P	ϕ, ϕ	?	



Undeciphered passage. In row 4 a short entry before the two question marks is crossed out.

Original source details: GELFAND-P000618



Uncertain reading. The involution column in row 5 has two looped letters, tentatively ϕ, ϕ.

Original source details: GELFAND-P000618

$\wp$ — Weierstrass fn.. 2, 3 — limiting cases of 4.



Undeciphered passage. Separate strokes before the note on the columns are crossed out.

Original source details: GELFAND-P000618

1 column — A

2 —"— — B

GELFAND-P000619

1975 J. Moser prepr.

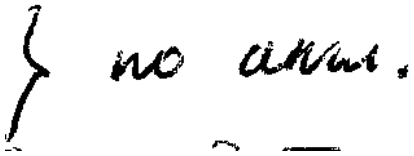
F. Calogero, M., R., (3 authors) Lett. Notes



Uncertain reading. The manuscript reads “Lett.” before “Notes”.

Original source details: GELFAND-P000619

	Calogero	
OP	Olshanetsky, Perelomov	} by name
P	Perelomov	



Uncertain reading. The words beside the brace tentatively read “по имен.” (by name).

Original source details: GELFAND-P000619

Coleman, Schroer — connection [crossed out ~~of the mod.~~] of the Thirring mod. with sin-Gordon.



Uncertain reading. An entry before “Thirring mod.” is crossed out. It tentatively reads “с мод.” (with the mod.).

Original source details: GELFAND-P000619

0 — inv. problem — Stieltjes.



Uncertain reading. The surname after “inv. problem” reads “Стильтьес” (Stieltjes).

Original source details: GELFAND-P000619

$$0. \qquad g^2 \sum_{k=1}^{n} e^{-(q_{k+1}-q_k)}$$

The inverse problem is solved nowhere (except case 0).

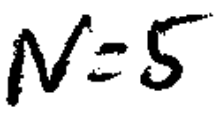
[left margin ‖ types 1–4.] The int. always have the form $I_k = \mathrm{Sp}(L^k)$, where the explicit form of L is written out (L-matrix).

$$L = P + iX, \qquad P_{k\ell} = p_k \delta_{k\ell}, \qquad X_{k\ell} = (1 - \delta_{k\ell}) x(q_k - q_\ell),$$

[under $P_{k\ell}$: diag. matr..] where $x(q)$ is written in the table.

The inv. problem here is to find matr. entries from the traces of powers. Idea: from Lax in KdV.

$$N = 5 \quad I_k = \mathrm{Sp}(N^k), \quad N = L^2 + \omega^2 Q^2, \qquad \text{where } Q_{ij} = q_j \delta_{ij}$$



Uncertain reading. $N = 5$ is written separately to the left of the formula.

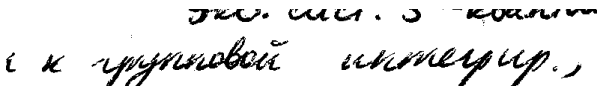
Original source details: GELFAND-P000619

Manakov — finite-dim. analogue of higher KdV. Treated the first two. [under Manakov: (dissert.)] Quant. syst. have also been treated and are also of interest.

$$p = -i\frac{\partial}{\partial q}.$$

Then $H =$ Laplace oper. on a symm. sp. (radial part). Equiv. to syst. 3 — the quantized one.

Gelfand's question: can everything be reduced to group integrability, but in a larger sp..



Original source details: GELFAND-P000619

Uncertain reading. The abbreviation after "group" tentatively reads "интегрир." (integrability).

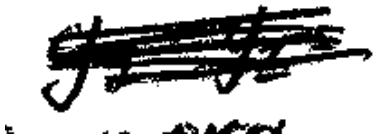
—— · ——

Connection with Dynkin diag.: $\{\alpha\}$ — roots, $q_\alpha = (\alpha, q)$.

$$U = \sum_{\alpha>0} g_\alpha^2 V(q_\alpha), \qquad g_\alpha = g_\beta, \quad \text{if } \alpha \text{ and } \beta \text{ have equal lengths}$$

$\alpha_{ij} = e_i - e_j$ — one obtains Moser's syst.. A — root syst. of type A_n. B — at diff. values of const. B, C, BC, D etc..

[crossed out] There are also all the exceptional groups.



Original source details: GELFAND-P000619

Undeciphered passage. A short entry before "There are" is crossed out.

It is unclear as yet whether the syst. for exceptional groups are completely integrable.

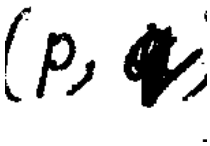
Fradkin. E. S. What is new in the last 20 years?

1. Consider an arbitr. system. It is given by a Lagr. or a Hamiltonian $H(p, q)$ (classical for now).

p, q can be Fermi and Bose (a mixture).

p, q — either Bose or Grassmann (Fermi)
$\downarrow$
$pq + qp = 1.$
(see Berezin).

There are also constraints $\varphi_k(p, q) = 0$.

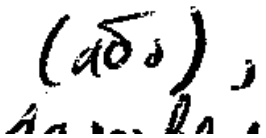


Original source details: GELFAND-P000619

Uncertain reading. The second argument is heavily overwritten and tentatively reads q.

(distinguish between Bose and Fermi vars.).

Example: gauge fields (electrodynamics (Ab.), Yang–Mills, gravit. field, Einstein eqs.), grad. inv. of Maxwell's eqs..



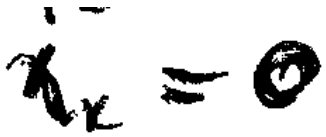
Original source details: GELFAND-P000619

Uncertain reading. The parenthesis after "electrodynamics" tentatively reads "аб." (Ab.).

and light is in 2 states, rather than 3, the extra one is eliminated by gauge inv. all fields with higher spins.

$\varphi_k = 0$ Bose constraints

$\chi_k = 0$ Fermi constraints



Original source details: GELFAND-P000619

Uncertain reading. The first sign in the last line is heavily overwritten and tentatively reads χ.

GELFAND-P000620

[top left, crossed out: $\prod(d\varphi, d\pi, d\psi, d\pi^{(\psi)})$.]



Original source details: GELFAND-P000620

Uncertain reading. Parts of a product of differentials are legible. The formula is crossed out with diagonal strokes.

Lagrangian $L(q, \dot{q})$, $p = \dfrac{\partial L}{\partial^r \dot{q}}$. Poisson bracket: ($r$ — denotes the right derivative).



Original source details: GELFAND-P000620

Uncertain reading. The heading uses the singular "bracket".

$$\{A, B\} = \frac{\partial A}{\partial^r q}\frac{\partial B}{\partial^\ell p} - (-1)^{n_A n_B}\frac{\partial B}{\partial^r q}\frac{\partial A}{\partial^\ell p}.$$

$n_A = 0$, if the number of Fermi vars. is even

$n_A = 1$ —"— odd

(in A).

Nondegeneracy: $\dot{q} = f(q, p)$ (can be expressed). Sometimes it cannot: when $\det \dfrac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} = 0$.

Then addit. constraints arise.

One can proceed in the same spirit and obtain (see Dirac's slim book). all the constraints. [crossed out] Close under the Poiss. bracket (to a Lie alg.).



Original source details: GELFAND-P000620

Undeciphered passage. A parenthesized entry after "all the constraints" is crossed out.

ψ_k — addit. conditions obtained in this way.

Dirac: [circled important] second-class constraints, between which $\{\varphi_k, \psi_j\} \neq 0$ taking the addit. conditions into account. Discard constraints belonging to the center, etc.

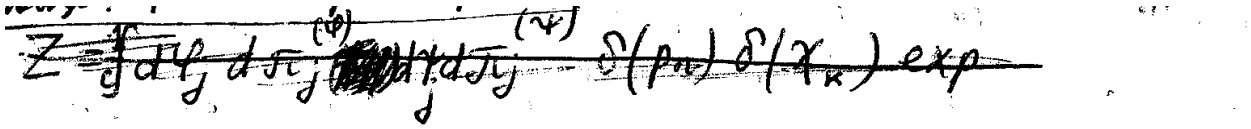


Original source details: GELFAND-P000620

Uncertain reading. The first abbreviation tentatively reads “Выбрас.” (discard).

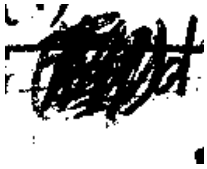
Generating fn. for all Green’s fns.

[crossed out] ~~$Z = \frac{1}{C}\int d\varphi_j\, d\pi_j^{(\varphi)} \ldots d\psi_j\, d\pi_j^{(\psi)}\, \delta(\varphi_k)\delta(\chi_k)$ exp~~



Original source details: GELFAND-P000620

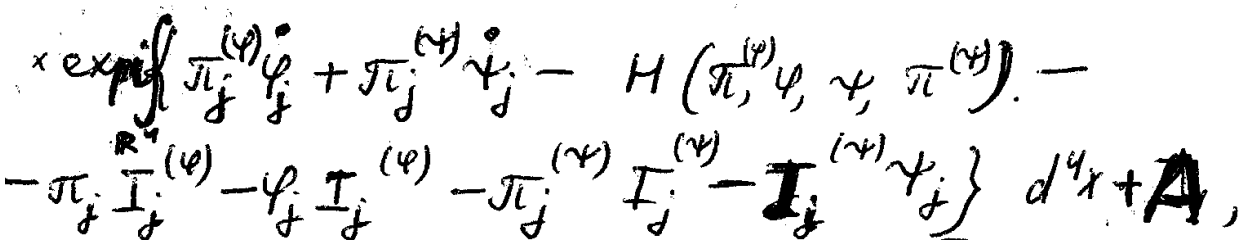
Uncertain reading. The whole preceding line is crossed out. The legible parts are typeset.



Original source details: GELFAND-P000620

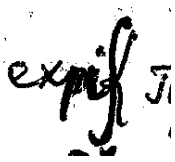
Undeciphered passage. A measure factor in the crossed-out line is obscured by heavy ink.

$$Z = \frac{1}{C}\int \prod_x d\varphi_j(x)\, d\pi_j^{(\varphi)}(x)\, d\psi_j(x)\, d\pi_j^{(\psi)}(x)\, \delta(\varphi_k)\delta(\chi_k)\times$$
$$\times \exp i \int_{\mathbb{R}^4} \Big\{ \pi_j^{(\varphi)}\dot{\varphi}_j + \pi_j^{(\psi)}\dot{\psi}_j - H(\pi^{(\varphi)}, \varphi, \psi, \pi^{(\psi)}) -$$
$$- \pi_j \bar{I}_j^{(\varphi)} - \varphi_j I_j^{(\varphi)} - \pi_j^{(\psi)} I_j^{(\psi)} - \bar{I}_j^{(\psi)}\psi_j \Big\}\, d^4x + A,$$



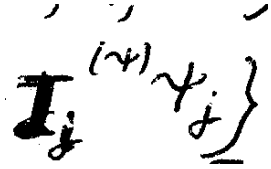
Original source details: GELFAND-P000620

Uncertain reading. A minus is visible at the end of the H line and at the start of the next. The closing brace precedes $d^4x + A$.



Original source details: GELFAND-P000620

Uncertain reading. The overwritten sign between exp and the integral tentatively reads i.

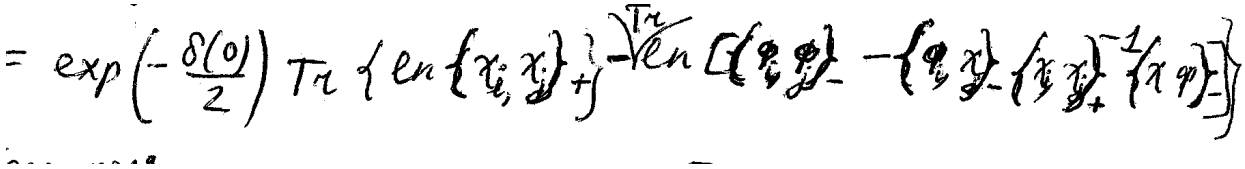


Original source details: GELFAND-P000620

Uncertain reading. The subscript on the final I looks like j. The upper stroke merges with the letter.

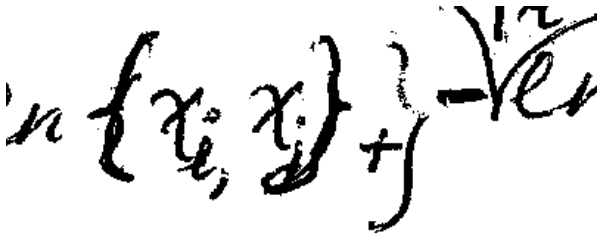
where

$$A = \exp\left(-\frac{\delta(0)}{2}\right) \operatorname{Tr}\{\ln\{\chi_i, \chi_j\}_+\} \\ - \operatorname{Tr}\ln\left[\{\varphi_i, \varphi_j\}_- - \{\varphi_i, \chi_j\}_-\{\chi_i, \chi_j\}_+^{-1}\{\chi\varphi\}_-\right]\Big\}.$$



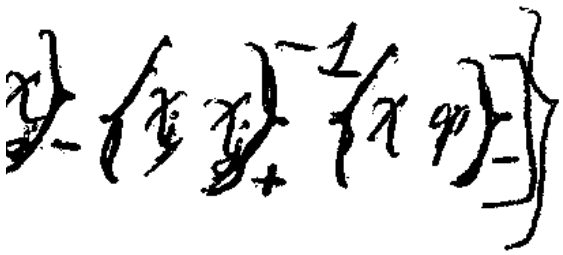
Original source details: GELFAND-P000620

Uncertain reading. The grouping after $\delta(0)/2$ and at the end follows the manuscript.



Original source details: GELFAND-P000620

Uncertain reading. An additional closing brace follows the first bracket with a plus sign. The second Tr is inserted above the line.



Original source details: GELFAND-P000620

Uncertain reading. The last bracket contains χ and φ without explicit indices.

φ — Bose fields
ψ — Fermi fields

Tr of matrices

(general formula for all quantum theories).

φ, χ — constraints (only noncommuting ones).

I — "sources" φ and ψ — elem. of the conj. space.

One obtains an analogue of the Fourier transform.

A — a term associated with the constraints.

$\delta(0)$ then cancels (in the S-matrix)

$\{\chi, \chi\}_\pm$ — Poisson brackets (Bose and Fermi)

—— · ——

$20/\overline{X} - 75.$ Gelfand Sem.

I. M. Gelfand:

$$L = -\frac{d^2}{dx^2} + u(x), \qquad A = \frac{d^3}{dx^3} + \dots, \qquad [L, A] = K(u).$$

$A = L^{3/2}$ with the nondifferential terms cut off.

1-zone potential: $K(u) = 0$. The point (hypothesis): for u the operators L and A commute.

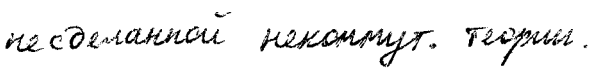
Higher eqs.. $A = L^{5/2}, L^{7/2}, \dots$ almost commute — the remaining oper. is multiplication by a fn.

[crossed out] Hypothesis: this is a piece of an unworked-out noncommut. theory.



Original source details: GELFAND-P000621

Undeciphered passage. A short entry before "Hypothesis" is crossed out.



Original source details: GELFAND-P000621

Uncertain reading. The adjective before "noncommut. theory" reads "несделанной" (unworked-out).

Krichever: in fact started from this.

[crossed out] Krichever. Igor.



Original source details: GELFAND-P000621

Undeciphered passage. An entry to the left of the name is crossed out.

Consider the operator eq. $[L_n, L_m] = 0$, where

$$L_n = \sum_{i=0}^{n} u_i(x) \frac{d^i}{dx^i}, \qquad u_n(x) \equiv c_0, \quad u_{n-1} \equiv c_1$$

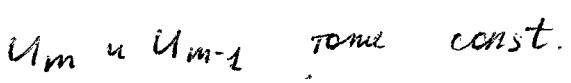
$$L_m = \sum_{i=0}^{m} v_i(x) \frac{d^i}{dx^i},$$

[crossed out ~~$u_m(x) \equiv$~~] u_m and u_{m-1} are also const.



Original source details: GELFAND-P000621

Uncertain reading. The crossed-out expression tentatively reads $u_m(x) \equiv$.



Original source details: GELFAND-P000621

Uncertain reading. The two constant coefficients in this line are written with u.

(operators of this kind are always considered).

Write the syst. of eqs.:

$$0 = \sum_{k,\ell} u_k C_k^{k+\ell-s} v_\ell^{(k+\ell-s)} - \sum_{k,\ell} v_k C_k^{\ell+k-s} u_\ell^{(\ell+k-s)}$$
$$0 \le s \le m+n-3.$$

Number of eqs. $=$ number of unknowns.

$$mu'_{n-2} - nv'_{m-2} = 0 \quad \Rightarrow \quad v_{m-2} = \frac{m}{n} u_{n-2} + h_2 \qquad (h_2 = \text{const}).$$

It is easy to check that the first [above the line $(m-1)$] eqs. (for $n-1 \le s \le m+n-3$) allow one to express all v_i as polynomials in u_i (and their derivatives). Also [crossed out] constants of integration arise.



Undeciphered passage. A short entry after "Also" is crossed out.

Original source details: GELFAND-P000621

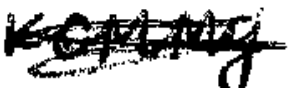
Substituting them in the eqs. $0 \le s \le n-2$, we obtain eqs. for the [crossed out] coeffs. of L_n.



Undeciphered passage. An entry before "coeffs." is crossed out.

Original source details: GELFAND-P000621

for $L = -\dfrac{d^2}{dx^2} + u(x)$ one obtains for $m = 2k+1$ a set of oper., [crossed out] whose commutator is multiplication by a fn.



Undeciphered passage. A word before "whose" is crossed out.

Original source details: GELFAND-P000621

For u one eq. remains, the higher KdV eq.. The family [crossed out] depends on the const. of integration.



Undeciphered passage. An entry between "The family" and "depends" is crossed out.

Original source details: GELFAND-P000621

Theorem (A. B. Shabat). If [crossed out] for L_n there exists an oper. L_m:



Undeciphered passage. Part of the hypothesis before "for" is crossed out.

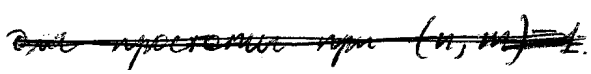
Original source details: GELFAND-P000621

$$[L_n, L_m] = 0, \quad \text{then} \quad \exists \text{ polynomial } P(x,y): \qquad P(L_n, L_m) = 0$$

(commutative).

GELFAND-P000622

Proof. [crossed out ~~all simple when $(n,m)=1$.~~]



Uncertain reading. The crossed-out phrase tentatively reads "all simple when $(n,m)=1$".

Original source details: GELFAND-P000622

Let y — a solution of the eq.:

$$L_n y = Ey, \qquad y_j(x,E) \text{ — basis of solns. (entire fns. of } E\text{).}$$

$$y_i^{(j)}(x_0, E) = \delta_{ij}$$

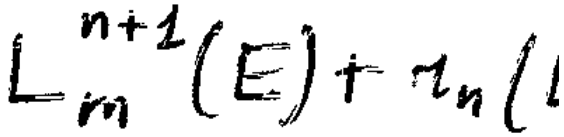


Uncertain reading. The equals sign meets the overwritten parenthesis.

Original source details: GELFAND-P000622

L_m gives a lin. oper. in the sp. of solutions of $L_n y = Ey$ $(\forall E)$. Its char. polynomial

$$0 = L_m^{n+1}(E) + r_n(E) L_m^n(E) + \ldots + r_0(E).$$



Uncertain reading. The first exponent is written $n+1$.

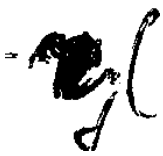
Original source details: GELFAND-P000622

Coeffs. — entire fns. of E. Prove that they are polynomials in E. This follows from the asymptotics of $y_i(E)$ in E (as $E \to \infty$).

$$y_i(x,E) \sim \sum_{j=0}^{n-1} a_{ij} e^{\varepsilon_j k x}, \qquad \varepsilon_j^n = 1, \quad k^n = E.$$

Hence the growth order of $r_j(E) \leq$ a polyn. in $k \Rightarrow$ they are themselves polynomials.

$$r_j(E) y = r_j(L_n) y$$



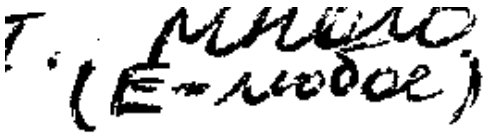
Uncertain reading. The r on the right is overwritten.

Original source details: GELFAND-P000622

Hence the char. polynomial

$$P(L_n, L_m)y = 0.$$

But there are sufficiently many solutions (E — arbitrary).



Uncertain reading. Below “many” the manuscript reads “E arbitrary”.

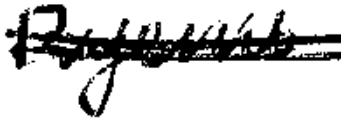
Original source details: GELFAND-P000622

[circled inf.-dimensional sp.]

$$\Rightarrow P(L_n, L_m) = 0. \qquad \triangleright$$

I. Bernstein: for polynomial coeffs. it was in F. A., vol. 1, no. 4.

[crossed out ~~Let $\psi_i(x, E)$ — eigenfn..~~]



Uncertain reading. The opening word is crossed out and tentatively reads “Пусть” (let).

Original source details: GELFAND-P000622



Uncertain reading. The crossed-out line contains $\psi_i(x, E)$ and “соб.” (eigenfn.).

Original source details: GELFAND-P000622

Let $\psi_i(x, E)$ be an eigenfn. for the pair of op.:

$$L_n\psi_i(x, E) = E\psi_i(x, E)$$
$$L_m\psi_i(x, E) = E_i\psi_i(x, E).$$

Then consider the surf. [left, crossed out ~~$\psi(x, P)$~~.]



Uncertain reading. The entry under the crossing-out tentatively reads $\psi(x, P)$.

Original source details: GELFAND-P000622

$P(x, y) = 0$, x, y — eigenvalues of L_n and L_m.

$\Rightarrow E_i$ alg. fns. of E,

$$\psi|_{x_0} = 1 \text{ (norm.)}.$$

$\Psi(x, P)$ — a fn. on this surf. [crossed out ~~BAKER~~.]



Uncertain reading. The hatched entry tentatively reads “BAKER”. Some letters are obscured.

Original source details: GELFAND-P000622

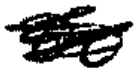
$$P \in X.$$

—— · ——

Consider $\forall$ nonsing. alg. curve X with marked point $P_0 \in X$.

The following properties uniquely determine the fn. $\Psi(x, P)$, $x \in (a, b)$, $P \in X$. (Akhiezer).

1°. $\psi(x, P)$ is meromorphic on [crossed out ~~x_0~~] $X \setminus P_0$, has polar divisor D with $\deg D = g$, and D is nonspecial and independent of x.



Uncertain reading. The entry crossed out before X tentatively reads x_0.

Original source details: GELFAND-P000622

2°.

$$\psi(x, P) = e^{k(x-x_0)} \underbrace{\left(1 + \sum_{i=1}^{\infty} \frac{\xi_i(x)}{k^i}\right)}_{\text{regular fn. near } P_0}, \qquad \frac{1}{k} = z \text{ — loc. parameter near } x_0.$$



Uncertain reading. The overwritten letter in the numerator tentatively reads ξ.

Original source details: GELFAND-P000622

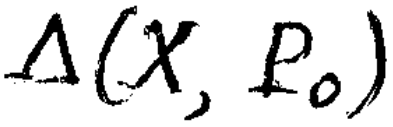


Uncertain reading. The note on the right reads "near x_0". The label under the bracket reads "near P_0".

Original source details: GELFAND-P000622

GELFAND-P000623

Introduce the ring $\Delta(X, P_0)$ of fns. rat. on the curve X with a single sing. at P_0.



Uncertain reading. The first sign resembles Δ with a thin lower stroke. Below, the ring is written with Λ.

Original source details: GELFAND-P000623

Theorem. $\forall E \in \Lambda(X, P_0)$, $\exists!$ diff. oper. L, such that

$$L\Psi(x, P) = E(P)\psi(x, P)$$

[crossed out] Here if

$$E(P) = c_n k^n + c_{n-1} k^{n-1} + \ldots + c_0 + \ldots$$

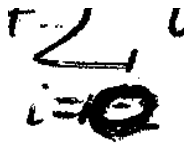


Undeciphered passage. An entry before "Here if" is crossed out. It may read "Д-во" (proof).

Original source details: GELFAND-P000623

($1/k$ — loc. param. near P_0), then

$$L = c_n \frac{d^n}{dx^n} + c_{n-1} \frac{d^{n-1}}{dx^{n-1}} + \sum_{i=0}^{n-2} u_i(x) \frac{d^i}{dx^i},$$



Uncertain reading. The lower limit is overwritten and reads $i = 0$.

Original source details: GELFAND-P000623

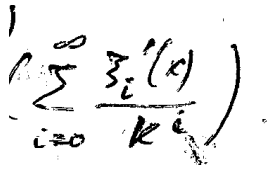
where u_i — certain fns..

Proof.

$$\psi(x, P) = e^{k(x-x_0)} \left(1 + \sum_{i=1}^{\infty} \frac{\xi_i(x)}{k^i} \right)$$

Differentiate in x: the poles are the same (the same divisor), and near P_0 (apply L) obtain

$$\frac{d}{dx}\psi(x, P) = k\Psi + e^{k(x-x_0)} \left(\sum_{i=0}^{\infty} \frac{\xi_i'(x)}{k^i} \right).$$



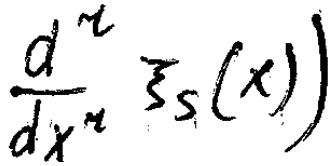
Uncertain reading. The lower limit in the differentiated sum is written $i = 0$.

Original source details: GELFAND-P000623

i.e. one obtains a new pole at P_0.

Seek L by requiring the sing. parts to coincide for $L\psi$ and $E\psi$. We obtain:

$$u_i(x) = Q\left(c_j, \frac{d^r}{dx^r}\xi_s(x) \right), \qquad \begin{array}{l} 0 \le s \le i-1, \\ 0 \le r \le i-s, \end{array}$$



Uncertain reading. The derivative order in the numerator and denominator reads r.

Original source details: GELFAND-P000623

i.e. u_i is found explicitly from the asymptotics.

Mapping $E \longrightarrow L$.



Uncertain reading. The letter at the end of the arrow is overwritten. It tentatively reads L.

Original source details: GELFAND-P000623

This is a ring homomorphism

$$\varphi : \Lambda(X, P_0) \longrightarrow \mathcal{L} \qquad \text{(diff. oper.)}$$

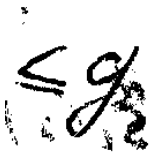
The image is a commut. ring. [left] Let n — the least possible, Let $\mu_1, \ldots, \mu_{n-1}$ — be numbers such that $\exists \psi$ with a pole of ord. $s + \mu_s n$.

$$\mu_1 + \ldots + \mu_{n-1} = g.$$

At a point in gen. position — g algebraically independent commuting operators.

Def. A diff. op. L_n is called algebraic of genus g, if it arises in this construction.

Theorem. L_n — alg. of genus $\leq g \Longleftrightarrow \exists$ operators $L^{(s)}$, $1 \leq s \leq n-1$, such that



Uncertain reading. There is a separate dark trace below g. An index cannot be read with confidence.

Original source details: GELFAND-P000623

$$[L_n, L^{(s)}] = 0, \quad \text{and} \quad [L^{(s)}, L^{(1)}] = 0, \qquad \text{ord}\, L^{(s)} = s + \mu_s n,$$



Uncertain reading. The second superscript in the commutator reads (1).

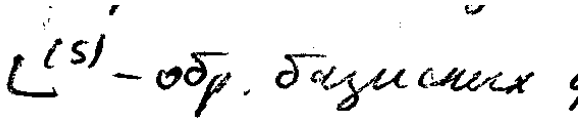
Original source details: GELFAND-P000623

where μ_i are such that $\mu_1 + \ldots + \mu_{n-1} = g$.

GELFAND-P000624

Remark. If op. commute with a pair, they commute with one another.

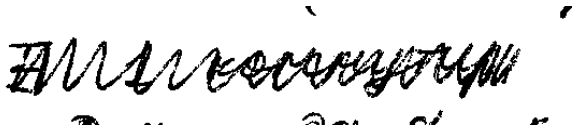
Proof: a) necessity is obvious ($L^{(s)}$ — “обр.” of basis fns.) from the preceding.



Uncertain reading. The abbreviation reads “обр.”.

Original source details: GELFAND-P000624

b) sufficiency — [crossed out]



Undeciphered passage. The entry before “construct” is heavily crossed out.

Original source details: GELFAND-P000624

from the eigenvalues construct fns. on the curve; suff. for genus g.



Uncertain reading. A separate closing mark appears after g.

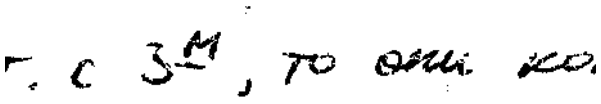
Original source details: GELFAND-P000624

Limanov: if 2 oper. with polyn. coeffs. commute with a third, then they commute with one another (oper. in 1 variable).



Uncertain reading. The surname tentatively reads “Лиманов” (Limanov).

Original source details: GELFAND-P000624



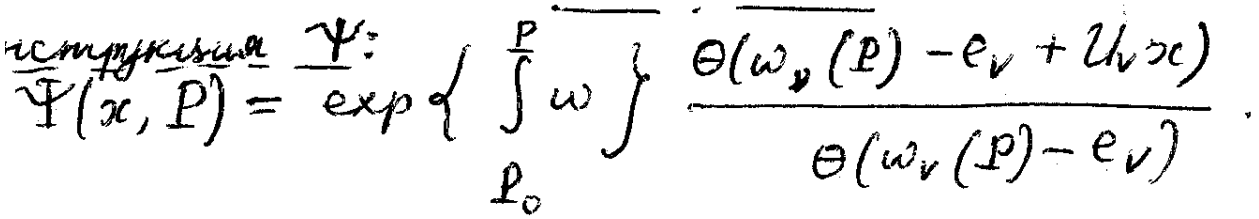
Uncertain reading. “With a third” is written as 3 with a raised Cyrillic “м” and a short stroke.

Original source details: GELFAND-P000624

—·—

Construction of Ψ:

$$\Psi(x,P)=\exp\left\{\int_{P_0}^{P}\omega\right\}\frac{\Theta\big(\omega_\nu(P)-e_\nu+U_\nu x\big)}{\Theta\big(\omega_\nu(P)-e_\nu\big)}.$$

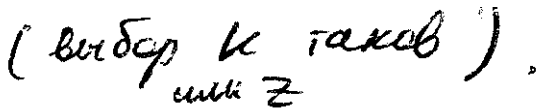


Original source details: GELFAND-P000624

Uncertain reading. The two strokes and dot above the formula form a divider. No factor x is written in the exponent.

$$\omega_2=\frac{dz}{z^2}\qquad\text{(the choice of } k \text{ is such)}.$$

[Beneath k: or z.]

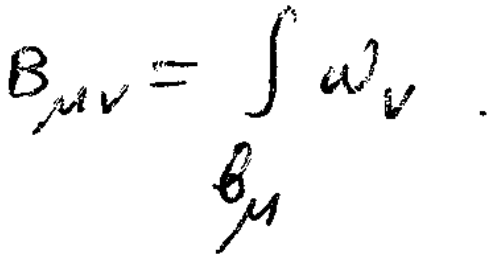


Uncertain reading. “or z” is written below the line.

Original source details: GELFAND-P000624

$$\omega_\nu(P) = \int_{P_0}^{P} \omega_\nu, \qquad \omega_\nu \text{ — Ab. diff. on the curve.}$$

$$\omega_\nu(P) = \int_{P_0}^{P} \omega_\nu, \qquad \int_{a_i} \omega_\nu = \delta_{i\nu}, \qquad B_{\mu\nu} = \int_{b_\mu} \omega_\nu.$$



Uncertain reading. The curved stroke at right tentatively belongs to the letter ω.

Original source details: GELFAND-P000624

$$\Theta(u_1, \dots, u_n) = \sum_{m_1,\dots,m_n} \exp\left\{2\pi i \text{ [crossed out]} \sum_{\mu,\nu} B_{\mu\nu} m_\mu m_\nu + \sum_\nu m_\nu u_\nu\right\}.$$



Undeciphered passage. The mark after $2\pi i$ is heavily crossed out.

Original source details: GELFAND-P000624

—·—

GELFAND-P000625

27/X-75. Gelfand Sem.

Dubrovin.

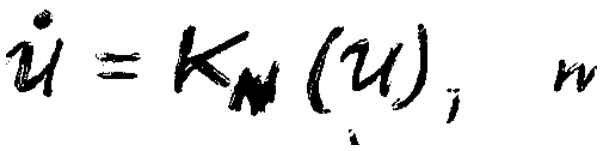
$\forall$ oper. L $\exists$ a series of eqs. of KdV type.

$$L = \frac{d}{dx} + U(x), \qquad U(x) \text{ — matr. } n \times n. \qquad (u_i^j(x)) \tag{1}$$

The eq. will be

$$\dot{U} = K_N(U); \qquad m = 1, 2, \dots. \tag{2}$$

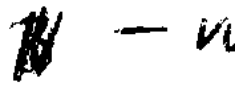
[Below $K_N(U)$:] nonlin. diff. op. in x.



Uncertain reading. The overwritten index on K tentatively reads N. To the right is $m = 1, 2, \dots$.

Original source details: GELFAND-P000625

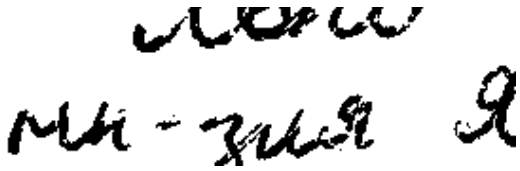
$K_m(U) = 0$ — a completely int. Ham. syst. which is solved explicitly. It admits a commut. repr. of Lax type on matrices $\Lambda' = [\Lambda, Q]$. (matr. entries of Λ — polynomials in $U, U', \ldots$).



Uncertain reading. The symbol before the dash is overwritten. It tentatively reads Λ.

Original source details: GELFAND-P000625

Eigenvalues — integrals of ord. eqs.. It is solved explicitly!! Introduce an extra par. E, alg. geom. arises: Jacobi varieties of Riemann surfaces.



Uncertain reading. The abbreviation before "Jacobi" tentatively reads "мн-зия" (varieties).

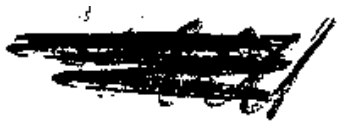
Original source details: GELFAND-P000625

Consider the spectral problem $L\Psi = EA\Psi$, $L = \dfrac{d}{dx} + U(x)$; [Upper right: uncertain entry] [crossed out].



Uncertain reading. Slanted strokes are visible above the crossed-out entry. No reading is secure.

Original source details: GELFAND-P000625



Undeciphered passage. The entry to the right of the spectral equation is heavily crossed out.

Original source details: GELFAND-P000625

$$A = (a_i \delta_i^j), \qquad a_i \neq a_j.$$

Let next $U_i^i = 0$ (this can always be achieved).

From it we construct eqs. preserving the spectrum of this problem.

$$Q = U - EA.$$

Consider the eq. $\lambda' = [\lambda, Q]$ and its soln. as a formal series

$$\lambda_B = B + \frac{\lambda_{1,B}}{E} + \frac{\lambda_{2,B}}{E^2} + \cdots, \qquad \lambda_B = \lambda_B(x, E). \tag{3}$$

B — const. diag. matrix, $= b_i \delta_i^j$; $\lambda_{j,B}$ — polyn. in the matr. entries of [crossed out] U.



Uncertain reading. A short entry before U is crossed out.

Original source details: GELFAND-P000625

Consider the eq.

$$\dot{U} = [A, \lambda_{N+1,B}] \tag{4}$$

or their lin. comb..

This is an eq. of KdV type. All these eqs. leave the spectrum of L, unchanged, since there is a Lax-type repr.

$$\dot{L} = [L, M_N] \tag{5}$$

Now seek all stat. solns.

$$K_N(U) = 0. \tag{6}$$

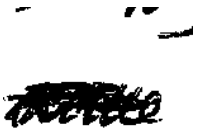
Hence $[L, M_N] = 0$ for this U.

Lemma. The eq. $K_N(U) = 0$ [crossed out] admits a Lax-type repr. [crossed out].



Original source details: GELFAND-P000625

Uncertain reading. The right parenthesis in $K_N(U)$ touches the sign, which reads as an equality.



Original source details: GELFAND-P000625

Undeciphered passage. A word immediately after the zero is heavily crossed out.

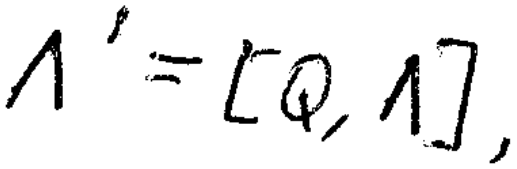


Original source details: GELFAND-P000625

Uncertain reading. A short entry after “Lax” is crossed out.

$$\Lambda' = [Q, \Lambda], \qquad \text{where } \Lambda \text{ — a polynomial in } E \tag{7}$$

(and in $U, U', \ldots$).



Original source details: GELFAND-P000625

Uncertain reading. In (7), Q comes first in the commutator.

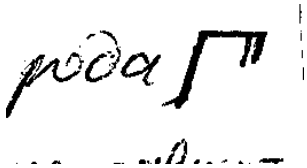


Original source details: GELFAND-P000625

Uncertain reading. The equation number is overwritten and tentatively reads (7).

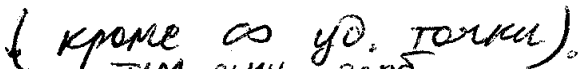
Proof. The eq. $L\Psi = EA\Psi$ has a basis of solns. $y_1, \dots, y_n$, M_N determines there (in the soln. space) a lin. transf. — a matrix, which we denote by Λ. We then have $[L, \Lambda] = 0$, i.e. $\Lambda' = [Q, \Lambda]$. ▷.

Finite-gap lin. operator. $\mathcal{L}$ — a lin. diff. operator, depending on par. E [inserted: (polynomially)] and the problem considered is $\mathcal{L}_E\Psi = 0$. $\mathcal{L}$ is called finite-gap, if the problem $\mathcal{L}_E\Psi = 0$ has an eigenfn., merom. in E on a Riemann surf. of finite genus Γ covering the E-plane (except for ∞ the distant point). (Γ is independent of x).



Original source details: GELFAND-P000626

Uncertain reading. Two short strokes descend at the right of the top bar of Γ.



Original source details: GELFAND-P000626

Uncertain reading. After ∞ the manuscript reads “уд. точки” (distant point).

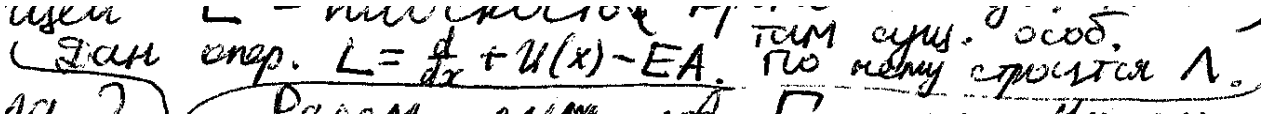
[Vertically in the left margin: Γ is called the spec. of the oper.]



Original source details: GELFAND-P000626

Uncertain reading. The underlined entry tentatively reads “наз. спек. оп-ра”. Additional strokes are visible below the line.

Lemma 2. [inserted: Given oper. $L = \frac{d}{dx} + U(x) - EA$. From it we construct Λ. [above the insertion: there is an essent. sing. there]]



Original source details: GELFAND-P000626

Uncertain reading. The insertion begins “Given oper.” Above “From it” is “there is an essent. sing. there”.

Consider the Riem. surf. Γ, defined by the eq. $R(z, E) = 0$ ($R(z, E) = \det(z \cdot 1 - \Lambda(E))$). We then have L — finite-gap with spectrum Γ.

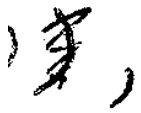
Proof.



Original source details: GELFAND-P000626

Uncertain reading. The initial letter is overwritten. There is a separate mark to the left of the heading.

Construct an eigenfn. Ψ simultaneously for L and Λ. ($L\Psi = 0$ [crossed out], $\Lambda\Psi = z\Psi$, z — the same z, as above).



Uncertain reading. A sign resembling a closing parenthesis is crossed out after the zero.

Original source details: GELFAND-P000626

Remark 1. $U(x+T) = U(x) \Longrightarrow \Psi$ [crossed out] [inserted: turns out] to be a Bloch fn..



Undeciphered passage. The word before "Bloch" is crossed out. "получ." (turns out) is inserted above it.

Original source details: GELFAND-P000626

(eigenfn. of translation by a period). (One can show that $[\Lambda, \widehat{T}] = 0$ [crossed out], and deduce the Bloch property from this).



Uncertain reading. A sign resembling a parenthesis is crossed out after the zero.

Original source details: GELFAND-P000626

Remark 2. $[L, M_N] = 0 \Longrightarrow \Psi$ — an eigenfn. of this pair.

—·—

Consider the eq. $\Lambda' = [Q, \Lambda]$. Coeffs. of the polynomial $R(z, E)$ — integrals. One can verify that the eqs. are Hamiltonian, the integr. are in involution.



Original source details: GELFAND-P000626

Uncertain reading. The final word is abbreviated "инволют.".

$$\sum_{i,j} I_{ij} \text{ [crossed out] } z^i E^j = R(z, E).$$



Uncertain reading. A separate vertical stroke stands to the left of the summation.

Original source details: GELFAND-P000626



Uncertain reading. A short sign is crossed out between I_{ij} and z^i.

Original source details: GELFAND-P000626

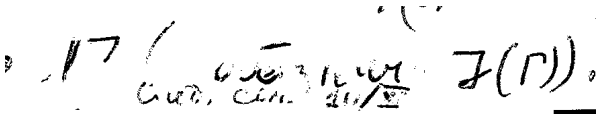
We then have $I_{ij} = I^0_{ij}$ — integrals.

Problem: determine the finite-gap oper. L with the same spectrum Γ.

Theorem. All finite-gap oper. with given spectrum Γ — the space of a principal [inserted: π-]bundle, where

$$\pi = (\text{diag. matr.})/(\text{scalar matr.})$$

over the Jacobian variety of the complete Γ ([faint entry] $J(\Gamma)$).

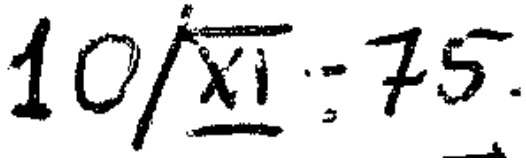


Original source details: GELFAND-P000626

Undeciphered passage. There is a separate angular stroke near Γ. Before $J(\Gamma)$ is a faint two-line entry. Parts of “см.” and of a fraction bar are visible in the lower line.

GELFAND-P000627

10/XI-75. Gelfand Sem.



Original source details: GELFAND-P000627

Uncertain reading. The month has a bar above and below. Two small marks are visible near the dash.



Original source details: GELFAND-P000627

Undeciphered passage. There is a hatched strip beneath the heading. No letters can be read with confidence.

Talk for Polyakov.

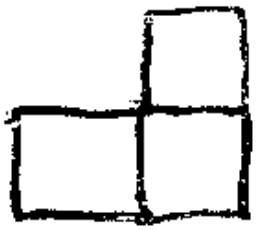
G. Torsion for schoolchildren.



Original source details: GELFAND-P000627

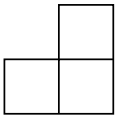
Uncertain reading. The doubly underlined initial “Г.” stands in the left margin before “Torsion”.

An ant lives on a cube.



Original source details: GELFAND-P000627

Diagram and typed description.



Three faces unfolded into the plane, with no labels.

[To the right of the figure:] go round a closed path in the plane corresp. path on the cube: we will arrive at another point (torsion) and turn (curvature).

It is unclear what the basic geom. things look like on polyhedra (n-dimensional case). Char. classes and so on. [crossed out]



Undeciphered passage. A short entry after “and so on” is heavily crossed out.

Original source details: GELFAND-P000627

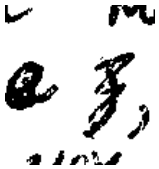
Problem: make a general theory of relativity on lattices.

A list of what one needs to know from physics (for a sensible mathematician).

1. Feynman, Hibbs «Quant. Mech. and Path Ints.».

2. Feynman’s lectures for schoolchildren.

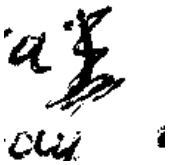
3. Quantum mechanics: Landau, Lifshitz (perturbation theory, quasiclassical approximation [crossed out], Lie groups, basic concepts of scattering theory).



Uncertain reading. A sign resembling a parenthesis is crossed out after “approximation”. A comma is visible to its right.

Original source details: GELFAND-P000627

4. Relativ. quant. mech. — the beginning of Berestetskii, Lifshitz, Pitaevskii; second quantization (Dirac’s book [crossed out]; Feynman «Stat. Mechanics»).



Uncertain reading. A sign resembling a parenthesis is crossed out after “Dirac”. A semicolon is visible to its right.

Original source details: GELFAND-P000627

Stat. physics — Landau and Lifshitz.

GELFAND-P000628

What mathematics should physicists know?

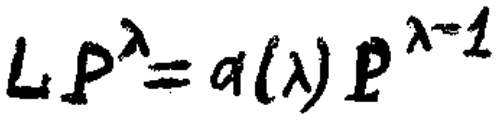
1. Group reprs.

2. Pseudodiff. operators.

3. Fourier–Hörmander integral oper..

4. P^λ (Bernstein polynomials).

$$LP^\lambda = a(\lambda)P^{\lambda-1}.$$



Uncertain reading. The coefficient on the right reads $a(\lambda)$.

Original source details: GELFAND-P000628

5. Riemann–Roch theorem.

6. Relation of sing. to top. cycles.

Polyakov.



Undeciphered passage. There is a heavily hatched strip below the surname.

Original source details: GELFAND-P000628

[Right:] TMF, the next-to-last issue; art. on the Yang–Mills field — worked out in detail. Yang, Mills — trans. in the coll. «Elem. Particles and Compensating Fields» (1964).



Uncertain reading. The abbreviation before “в сб.” (in the collection) reads “перев.” (translation).

Original source details: GELFAND-P000628

There are quarks $q \simeq \begin{pmatrix} u \\ d \\ s \end{pmatrix}$. quark spin $\frac{1}{2}$, $B \sim$ spin $\frac{1}{2}$ (baryon), M — spin 0 (meson).

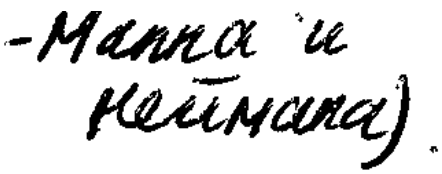
$$M \sim \bar{q}q \quad \text{(quark and antiquark)},$$
$$B \sim qqq.$$

Where from: Grouping into multiplets (8 at a time). (the idea of Gell-Mann and Ne’eman).



Uncertain reading. The entry reads “мультиплеты” (multiplets).

Original source details: GELFAND-P000628



Uncertain reading. The second surname is written “Неймана”.

Original source details: GELFAND-P000628

$\Rightarrow$ Symmetries: the most suitable gr. $SU(3)$, reprs. $8, 10, 27$ (various).

Lowest representations? [crossed out] scalar, stand., and tens. products.

Uncertain reading. There is a short crossed-out mark before “scalar”.



Original source details: GELFAND-P000628

Let these be real particles! remarkable idea:

$$3^* \times 3 = 1 + 8,$$
$$3 \times 3 \times 3 = 1 + 8 + 8 + 10.$$

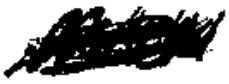
$\Rightarrow$ one cannot build mesons as 3 quarks; there are 8 of them, and $3 \times 3 = 3^* + 6$.

We scatter:

$$\frac{\sigma(BB)}{\sigma(BM)} = \frac{3}{2} \qquad \text{(cross sections)}.$$

— that is how it comes out (accur. 10%). High en. — scattering of a [crossed out] particle on a particle as of a part. on an antiparticle.

Undeciphered passage. A short word before "particle" is heavily crossed out.



Original source details: GELFAND-P000628

$qqqq\bar{q}$ — absent in experiment (called exotic).

Many particles — speeds of rot. of quarks (like the electron in the hydrogen atom). but: unstable (resonance). Trans. to the ground state.

Uncertain reading. The abbreviation before "вращ." (rotation) reads "скор.".



Original source details: GELFAND-P000628

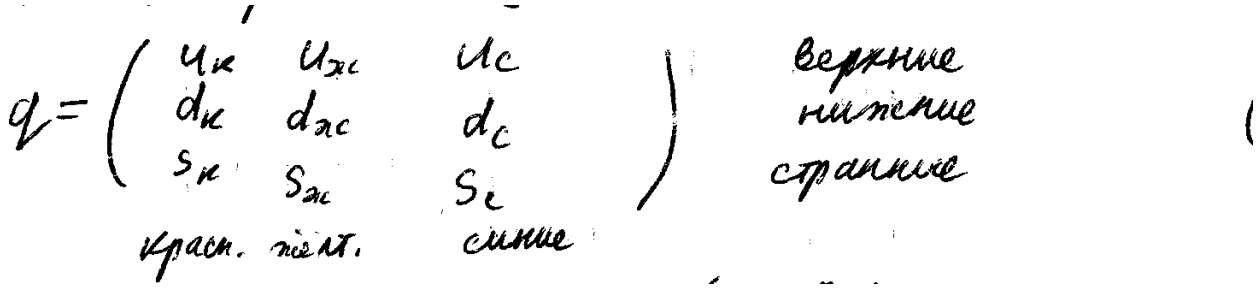
Baryon wave fn.:

$$\Delta^{++} \simeq u(x_1)u(x_2)u(x_3),$$

but antisymm. is needed.

Propose 3 kinds and 3 colors.

$$q = \begin{pmatrix} u_{\text{к}} & u_{\text{ж}} & u_{\text{с}} \\ d_{\text{к}} & d_{\text{ж}} & d_{\text{с}} \\ s_{\text{к}} & s_{\text{ж}} & s_{\text{с}} \\ \text{red} & \text{yellow} & \text{blue} \end{pmatrix} \begin{matrix} \text{up} \\ \text{down} \\ \text{strange} \end{matrix} \qquad \text{(Gell-Mann)}.$$



Original source details: GELFAND-P000628

Uncertain reading. The matrix retains the source indices "к", "ж", "с". The color names are written under the corresponding columns.

In experiment quarks do not exist (90%). $\mathcal{L}$ cannot be invar. under

$$SU(3) \otimes SU_c(3) \qquad \text{color (change of color)}.$$

GELFAND-P000629

Assume that only colorless part. have masses that are not very large, colored ones — ∞.

$$M \sim \overline{q}_{\text{к}} q_{\text{к}} + \overline{q}_{\text{жс}} \overline{q}_{\text{жс}} + \overline{q}_{\text{с}} q_{\text{с}}.$$



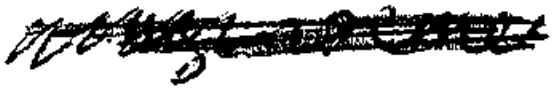
Uncertain reading. Both qs in the middle term are barred. Both subscripts tentatively read "жс".

Original source details: GELFAND-P000629

The Lagrangian must prohibit the appearance of colored particles (they must have inf. masses). Let the interact. be carried by colored Yang–Mills fields (gluons).

$SU(3)$ — approx. symmetry. (20–30%). Let $SU_c(3)$ — be the exact group. (Possibly there is a fourth kind (charmed), but there are certainly 3 colors).

Of mesons (see the fmla.) there should be 9. But there are more. Hence a fourth quark is necessary. (4 years ago). [crossed out] $SU(4) \otimes SU_c(3)$.



Undeciphered passage. The entry after "4 years ago" is heavily crossed out.

Original source details: GELFAND-P000629

$$\mathcal{L} = \overline{q}(i\gamma_\mu \nabla_\mu - m)q + \frac{1}{4e^2} \operatorname{Sp}(F_{\mu\nu}^2).$$

There are 8 Yang–Mills fields. (Gluons — from glu — glue — stick quarks together). They are massless. 8 colored gluons.

Conserved currents $\Longleftrightarrow$ presence of symmetry. We want to show that if the gr. is nonabelian $\Rightarrow$ interact. energy $\sim$ [crossed out] R (dist.) ($\Rightarrow$ the string joining the quarks does not break) (in the abelian case $\sim R^{-1}$, Maxwell eqs.).



Uncertain reading. A short entry with a horizontal stroke is crossed out before R.

Original source details: GELFAND-P000629

$$\mathcal{L} = \frac{1}{4e^2} \operatorname{tr}(F_{\mu\nu}^2).$$

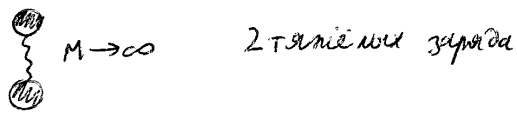


Diagram and typed description.



$M \to \infty$
2 heavy charges

Two shaded circles are joined by a wavy line.

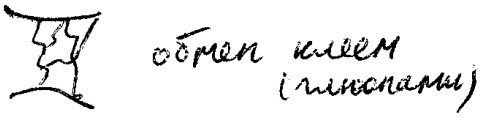
Original source details: GELFAND-P000629

G. Classical vers.: noncommut. electrical engineering.



Original source details: GELFAND-P000629

Uncertain reading. The abbreviation after “Классич.” (classical) tentatively reads “вар.” (version).



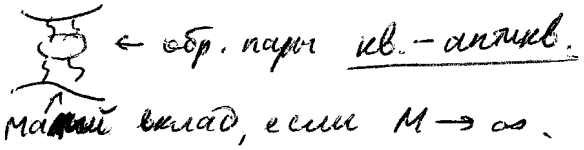
Original source details: GELFAND-P000629

Diagram and typed description.



exchange of glue
(gluons)

Two boundaries and connecting strokes. The exact joins in the middle are uncertain.



Original source details: GELFAND-P000629

Diagram and typed description.



← formation of a pair
q.–antiquark

Small contribution if $M \to \infty$

An oval between two boundaries. Arrows lead to the oval and to the lower line.



Original source details: GELFAND-P000629

Uncertain reading. The first word beneath the sketch is overwritten and tentatively reads “Малый” (small).

$$\frac{\int DA_\mu\, e^{i\int \mathcal{L}\, d^4x}\ \mathrm{tr}\, e^{i\oint A_\mu\, dx_\mu}}{\int DA_\mu\, e^{i\int \mathcal{L}\, d^4x}} = \left\langle \mathrm{tr}\, e^{i\oint_C A_\mu\, dx_\mu} \right\rangle = e^{iW(C)}.$$



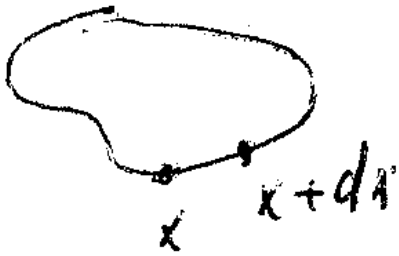
Original source details: GELFAND-P000629

Uncertain reading. The last character after d^4 in the numerator is overwritten. It tentatively reads x.



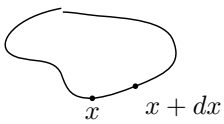
Original source details: GELFAND-P000629

Uncertain reading. The x after d in the numerator contour integral is overwritten. No subscript C is visible on that integral.



Original source details: GELFAND-P000629

Diagram and typed description.

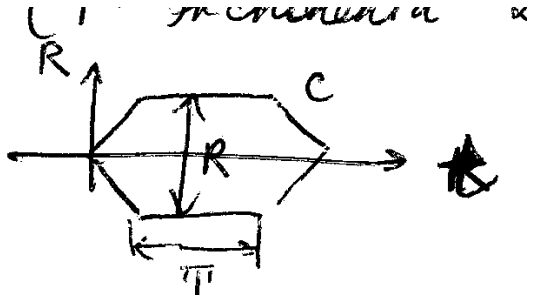


Two points are marked on the contour. A short stroke follows $x + dx$. No index can be read with confidence.

$$\operatorname{tr} \prod_x (1 + iA_\mu(x)\, dx_\mu)$$

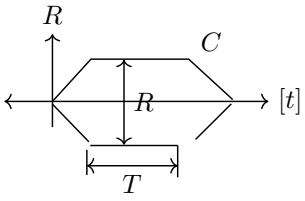
— contour integral. (this ordering). (T — Dyson exponential).

Assertion:



Original source details: GELFAND-P000629

Diagram and typed description.



Contour C, the R axis, and spans R and T. There is a gap at the right of the lower side of the contour.

Uncertain reading. The overwritten horizontal axis label tentatively reads t.



Original source details: GELFAND-P000629

Consider such a contour. Here $W(C) \xrightarrow[T\to\infty]{} E(R)T$, where $E(R)$ — the interaction energy of the charges.

GELFAND-P000630

Explanation: if the syst. is in a stat. state, the trans. amplitude in time T: $F = e^{-iET}$ (E — energy).

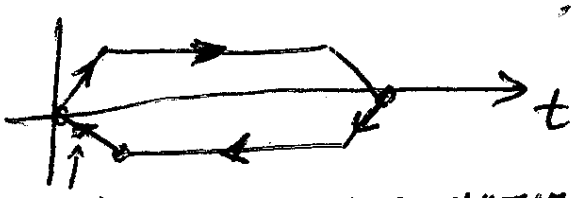


Original source details: GELFAND-P000630

Uncertain reading. The exponent is written $-iET$. A separate stroke is visible above the first F.

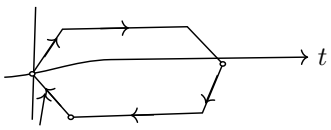
On the other hand

$$F = e^{-iET} = \sum_{\text{(paths)}} e^{iS}.$$



Original source details: GELFAND-P000630

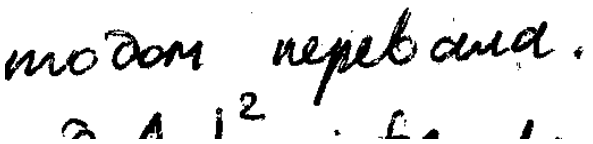
Diagram and typed description.



Arrows point right on the upper side and left on the lower side. A separate arrow points up toward the start.

[Beneath the figure:] creation of a part.–antipart. pair (superheavy); we hold it for a long time.

Classical action $S = \int_{(x)}^{(x')} A_\mu \, dx_\mu$. In the el.-magn. field. Antipart. $\sim$ a part. going backward in time. Hence $W(C) \sim S_C$ (area inside the contour). $\sim RT \to E \sim R$. It is calculated by the saddle-point method.



Uncertain reading. The last word is written “перевала” (saddle point).

Original source details: GELFAND-P000630

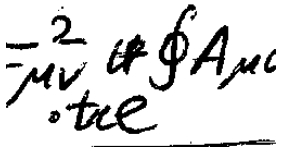
$$\int DA_\mu \, e^{i\int (\partial_\mu A_\nu - \partial_\nu A_\mu)^2} e^{i \oint A_\mu \, dx_\mu} = e^{-\oint\oint dx \, dx' \, D(x-x')},$$

where $D = \dfrac{1}{(x-x')^2 + i0}$ (Green’s fn. of the d’Alembert eq.).

$$W = \text{const} \cdot T + \frac{e^2 T}{R} + \dots$$

Next time:

$$\frac{\int DA \, e^{\frac{1}{4e^2}\int F_{\mu\nu}^2} \text{[crossed out]} \cdot \operatorname{tr} e^{\oint A_\mu \, dx_\mu}}{N}$$ — we will calculate.



Uncertain reading. A short sign is crossed out before the contour integral. The insertion below reads $\cdot \operatorname{tr} e$.

Original source details: GELFAND-P000630

non-Gaussian, since $F_{\mu\nu}$ has the quadr. term $[A_\mu, A_\nu]$.

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

(In the el.-magn. field the commutator is 0). El.-dyn. with a compact gauge group — nonuniqueness of extremals already arises. Quarks in Yang–Mills have yet to be worked out.



Uncertain reading. The last word reads “сделана”, with the feminine ending.

Original source details: GELFAND-P000630

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GELFAND-P000631

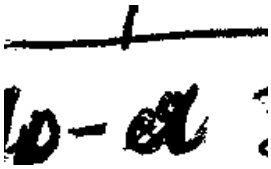
9/II-76. Gelfand Sem.



Uncertain reading. The digit of the day is heavily overwritten and reads “9”.

Original source details: GELFAND-P000631

Krichever. Zakharov–Shabat eqs.



Uncertain reading. The ending of the abbreviation “Ур-я” (eqs.) is heavily overwritten.

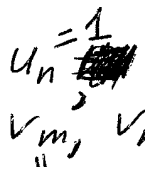
Original source details: GELFAND-P000631

$$\left[L_1 - \frac{\partial}{\partial y}, L_2 - \frac{\partial}{\partial t}\right] = 0, \tag{1}$$

$$L_1 = \sum_{i=0}^{n} u_i(x,y,t)\frac{\partial^i}{\partial x^i},$$

$$L_2 = \sum_{i=0}^{m} v_i(x,y,t)\frac{\partial^i}{\partial x^i},$$

$$\left.\begin{matrix} u_n = 1, & u_{n-1} \\ v_m = 1, & v_{m-1} \end{matrix}\right\} = \text{const}.$$



Undeciphered passage. An entry after u_n is crossed out, with $= 1$ above it. A 1 is inserted below v_m.

Original source details: GELFAND-P000631

Consider the formal series

$$\Psi = e^{kx+k^n y}\left(\sum_{i=0}^{\infty} \xi_i(x,y,t)\frac{1}{k^i}\right), \qquad \xi_0 = 1. \tag{2}$$

Let $\left(L - \dfrac{\partial}{\partial y}\right)\Psi = 0$, where $L = L_1$. The coeffs. are computed recursively.

$$\sum_{i=0}^{n} u_i \sum_{\ell=0}^{i} C_i^{\ell}\, \xi_{\ell-s}^{(i-\ell)} = \xi_{n-s} + \frac{\partial}{\partial y}\xi_{-s}, \quad s = n, n-1, \ldots \text{ down to } -\infty. \tag{3}$$

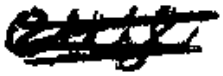
$\xi_{-s} = 0$ for $s > 0$.



Uncertain reading. The sign before 0 is heavily overwritten. It tentatively reads $>$.

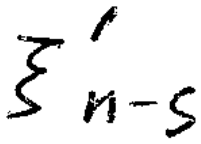
Original source details: GELFAND-P000631

[crossed out] determines $\xi'_{n-s} \Rightarrow$ const. of integr. are needed.



Undeciphered passage. A short word before "determines" is crossed out.

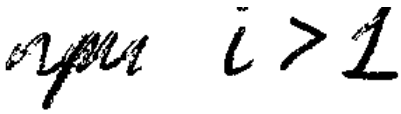
Original source details: GELFAND-P000631



Uncertain reading. The subscript of ξ' tentatively reads $n - s$.

Original source details: GELFAND-P000631

$\xi_0 = 1, \quad \xi_i(0, y, t) = 0$ for $i > 1$ — normalization.



Uncertain reading. The bound in the normalization is visibly written $i > 1$.

Original source details: GELFAND-P000631

Conversely, $\forall$ series $\exists$ an oper., for which everything holds to accuracy k^{-1}.

Algebraic operator. — if $\exists$ an anal. continuation in $1/k$ to a compact alg. curve (merom. on the complement of the point, [below the line: where there is an essent. sing.]).

G. Zakharov–Shabat eq.

$$\frac{\partial K_2}{\partial y} - \frac{\partial K_1}{\partial t} = [K_1, K_2],$$

K_1, K_2 — diff. operators. K_1, K_2 — matr. in y and t or oper. in other variables. There are many ways leading to this eq.. y, t — parameters. [Left: crossed out]



Uncertain reading. The short crossed-out entry tentatively reads "Ha" (on).

Original source details: GELFAND-P000631

Put on an equal footing x, y, t. Then it is written in the form (1).

Problem for schoolchildren: find oper. of the form

$$-\frac{d^2}{dx^2} + u(x) = L_1, \qquad \frac{d^3}{dx^3} + v_1\frac{d}{dx} + v_2 = L_2,$$

which commute. This is done via ellipt. fns. or ellipt. curves. If we have only $\frac{d^n}{dx^n}$, then

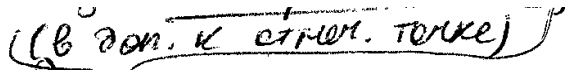
$$\frac{d^n}{dx^n}\Psi - \frac{\partial \Psi}{\partial y} = 0 \quad \Rightarrow \quad \Psi = e^{kx + k^n y}.$$

More precisely: the method of Fourier integral oper..

$$e^{kx+k^n y}\left(1+\frac{\xi_1(x,y,t)}{k}+\dots\right).$$

Kr. Associate to each oper. L — the formal series (2). The recurrence relation (3) recovers it uniquely up to normalization.

Definition. L is called algebraic, if the formal series (2) has a merom. continuation [inserted: (in the complement of the marked point)] to a compact alg. curve, with poles independent of x, y, t.



Original source details: GELFAND-P000632

Uncertain reading. "In the complement of the marked point" is inserted above the line.

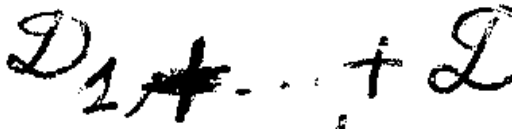
$L \longrightarrow$ [crossed out] $\mathcal{R}$



Original source details: GELFAND-P000632

Undeciphered passage. An entry after the arrow and before $\mathcal{R}$ is heavily crossed out.

$\mathcal{R}$ — alg. curve with marked point P_0 and set of poles $D_1 + \dots + D_g$, g — genus of the curve — these are called scattering data for L.



Original source details: GELFAND-P000632

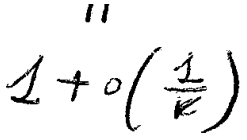
Uncertain reading. The first sign after D_1 is overwritten. It tentatively reads $+$.

P_0 — point where there is an essent. sing.. $1/k$ — loc. param. near P_0.

G. An asympt. expansion as a rule does not converge. The def. assumes convergence. K. To an alg. operator $L \rightsquigarrow$ alg. curve with a marked point and a set of poles. G. C^∞ — Taylor fmla. asympt. $\Rightarrow$ analytic — converg.. Analogue of quasianalyticity?

K. Thm. (Akhiezer) $\forall$ nonsing. alg. curve $\mathcal{R}$ with marked point P_0 and a fixed loc. param. near P_0 and a marked set of poles $\exists$ a fn. determined uniquely up to proportionality $\Psi(x,y,P)$ with the given set of poles and

$$\underbrace{\Psi(x,y,P)e^{-kx-k^n y}}_{1+o(1/k)} \quad \text{— reg. near } P_0.$$



Original source details: GELFAND-P000632

Uncertain reading. Below the bracket is $1 + o(1/k)$, with a small o.



Diagram and typed description.



Unlabelled rectangle.

Original source details: GELFAND-P000632

$e^{\zeta(P)x}$ [arrow to ζ:] Weierstrass (Kamke).

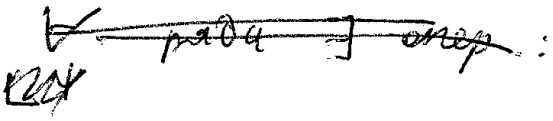
Proof: explicit fmla. (will be presented later). Thm. The eq. is algebraic $\Rightarrow \exists$ a fn. $\Psi \Rightarrow$ we know all the $\xi \Rightarrow$ we find all the u_i (uniquely) — why do they exist? From [crossed out] Ψ one finds the u_i!!

Uncertain reading. A short sign is crossed out before Ψ.



Original source details: GELFAND-P000632

[Lower right, crossed out: $\forall$ series $\exists$ an oper.:]



Undeciphered passage. A second short crossed-out entry stands beneath the left end of the cancelled line.

Original source details: GELFAND-P000632

GELFAND-P000633

$\forall$ series $\exists$ an oper.:

$$L\Psi - \frac{\partial}{\partial y}\Psi = \text{[crossed out]} \left(e^{kx+k^n y} o(1/k)\right).$$

(first $n-2$ eqs.)



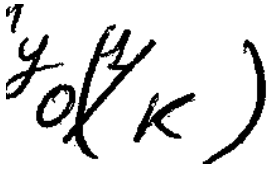
Uncertain reading. The first line reads “ряда” (series).

Original source details: GELFAND-P000633



Uncertain reading. A short entry after the equals sign is heavily crossed out.

Original source details: GELFAND-P000633



Uncertain reading. The argument reads $1/k$. The small round letter before it tentatively reads o.

Original source details: GELFAND-P000633

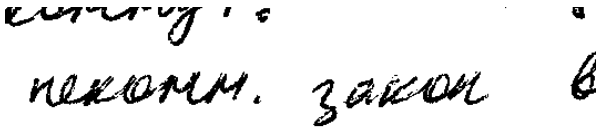
$L\Psi - \dfrac{\partial \Psi}{\partial y}$ meets all the requirements of Akhiezer's thm. $\Rightarrow$ it $\equiv 0$.

Drinfeld.

An attempt at algebraizing Krichever's previous talk

Description of commut. subrings in the ring of diff. operators.

G. The goal is a noncommut. reciprocity law.



Uncertain reading. The abbreviation before "law" reads "некомм." (noncommutative).

Original source details: GELFAND-P000633

B — [crossed out] a comm. ring with 1, equipped with a deriv. [crossed out] $b \mapsto b'$ [crossed out]. ($\mathbb{C}$-algebra)



Undeciphered passage. The entry after B and the dash is heavily crossed out.

Original source details: GELFAND-P000633



Uncertain reading. A short sign is crossed out after "дифф." (deriv.).

Original source details: GELFAND-P000633



Uncertain reading. A short sign is crossed out after b'.

Original source details: GELFAND-P000633

$$B[\mathcal{D}], \qquad \underbrace{\sum b_i \mathcal{D}^i}_{\text{polyn.}}, \qquad \underbrace{Db - bD = b'}_{\text{multiplication rule}} .$$

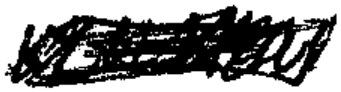
(b' — derivative in B)

X — a compact Riem. surf. over $\mathbb{C}$, P — a fixed point. Denote by A [crossed out] the ring of merom. fns. on X [crossed out] with a pole only at P.



Uncertain reading. A sign after A is heavily crossed out.

Original source details: GELFAND-P000633



Original source details: GELFAND-P000633

Undeciphered passage. The word after X is heavily crossed out.

We describe the homomorphisms $A \to B[\mathcal{D}]$. (in fact there are no other comm. subrings — Shabat's thm.)

G. The real problem: find all extensions of Gelfand–Kirillov skew fields.

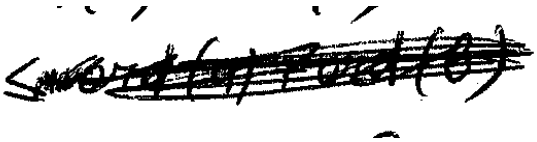
Polynomiality gets in the way.

$$\sum_{i=0}^{k} b_i D^i, \qquad \text{ellipticity}: \quad \exists b_k^{-1}.$$

Req.: the image of $\forall$ nonzero elem. of A is elliptic.

Rank. $a \neq 0, \quad a \in A$.

$$\left.\begin{array}{c} \operatorname{ord}(a) \text{ — order of the image of } a \\ \operatorname{ord}(ab) = \operatorname{ord}(a) + \operatorname{ord}(b) \\ \operatorname{ord}(a+b) \leq \text{[crossed out]} \max(\operatorname{ord}(a), \operatorname{ord}(b)) \\ \operatorname{ord}\lambda = 0 \quad \text{for } \lambda \in \mathbb{C} \end{array}\right] \quad \text{valuation}.$$



Original source details: GELFAND-P000633

Uncertain reading. An entry after the inequality sign is crossed out. It tentatively reads $\operatorname{ord}(a) + \operatorname{ord}(b)$.

$\deg a$ — order of the pole of a at the point P.

Then $\operatorname{ord}(a) = n \deg(a)$. (clear for polynomials)

n — rank of the homomorphism.

Krichever — $n = 1$ (coprimality of the orders of the oper.).

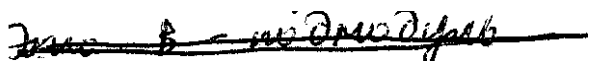
—·—

$M = B[\mathcal{D}]$, M has the structure of a B-module; $\varphi : A \to B[\mathcal{D}]$.

GELFAND-P000634

M_n — oper. of order $\leq n$.

[crossed out: This is a B-submodule.]



Original source details: GELFAND-P000634

Uncertain reading. The cancelled line tentatively reads "This is a B-submodule".

M — this is an $A \otimes B$-module, i.e. an A-module and B-module and mult. by A and B commute.

$$bx = bx \quad \text{(mult. on the left)},$$
$$ax = x\varphi(a),$$
$$a \in A \Rightarrow aM_i \subset M_{i+\deg a \cdot n}.$$

$\nabla : M \to M$ — mult. on the left by D.

$$\nabla(x) = Dx.$$

∇ commutes with the action of A, and $\nabla b - b\nabla = b'$.

Conversely, if there is an $A \otimes B$-mod. M with a filtr. and all these relations, then $\exists$ a homom. $\varphi : A \to B[\mathcal{D}]$ of rank n.

We can forget φ and consider the problem of $A \otimes B$-modules.

$\to$ Thus, M — a module over $A \otimes B$, equipped [crossed out] with the filtration

$$0 = M_{-1} \subset M_0 \subset M_1 \subset M_2 \subset \cdots$$



Undeciphered passage. A short entry before "filtration" is crossed out.

Original source details: GELFAND-P000634

[Under A: merom. fns. on X with a single pole. Under B: a ring ($\mathbb{C}$-alg.) with deriv. $b \mapsto b'$.] and with the operator $\nabla : M \to M$, such that

1) $\nabla a = a\nabla$.

2) $\nabla b - b\nabla = b'$.

3) $\nabla(M_i) \subset M_{i+1}$, $\nabla : M_i/M_{i-1} \xrightarrow{\sim} M_{i+1}/M_i$ for $i \geqslant 0$.



Uncertain reading. A separate slanted stroke lies below $>$. The bound reads $i \geqslant 0$.

Original source details: GELFAND-P000634

4) $bM_i \subset M_i$, M_i/M_{i-1} [crossed out] a free B-module with one generator.



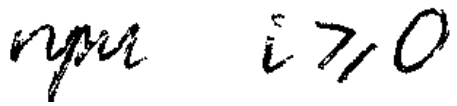
Uncertain reading. A sign resembling an arrow is crossed out before "free".

Original source details: GELFAND-P000634

5) $aM_i \subset M_{i+n\deg a}$,

$$a : M_i/M_{i-1} \xrightarrow{\sim} M_{i+n\deg a}/M_{i+n\deg a-1} \qquad \text{for } i \geqslant 0.$$

Uncertain reading. Item 5 repeats the sign with a separate stroke. The bound reads $i \geqslant 0$.



Original source details: GELFAND-P000634

$\sim$ the data of a sheaf over $X \otimes B$ with certain filtrations.

$\nabla \sim$ a connection with a sing. at ∞.

—·—

GELFAND-P000635

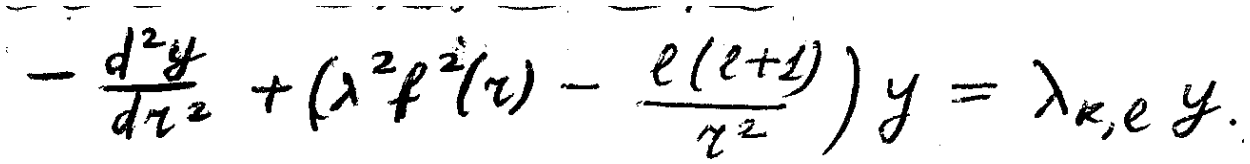
23/II-76. Gelfand Sem.

S. P. Novikov on the work of M. Brodsky. (Minsk)

Some cases of explicitly solvable problems unrelated to KdV.

Problem of vibrations of an elastic sphere.

$$-\frac{d^2y}{dr^2} + \left(\lambda^2 f^2(r) - \frac{\ell(\ell+1)}{r^2}\right) y = \lambda_{k,\ell} y.$$



Original source details: GELFAND-P000635

Uncertain reading. The right-hand side is written $\lambda_{k,\ell} y$. A minus sign precedes the fraction in parentheses.

Inverse problem — recover the par. of the elastic medium from the set of frequencies $\lambda_{k,\ell}$. The problem is overdetermined. Find when it is solvable and how.

Let $rf(r) = u(r)$ be a monotone fn..

Consider

$$m(\tau) = \int_{H(\tau)}^{1} \left(\tau^{-2} f^2(r) - \frac{1}{r^2}\right)^{1/2} dr \qquad \text{(Bohr quant. rule)}.$$

($H(\tau)$ — a zero of the expression under the radical)

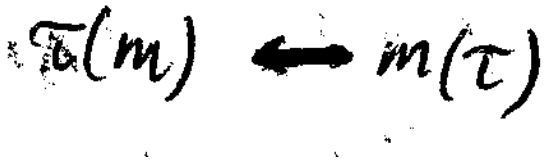
Consider

$$\tau^*_{k\ell} = \frac{\sqrt{\ell(\ell+1)}}{\lambda_{k\ell}}, \qquad m_{k\ell} = \frac{k + \frac{1}{4} + C}{\sqrt{\ell(\ell+1)}}.$$

$\tau^*(m_{k\ell}) = \tau^*_{k\ell}$ establishes a one-to-one corresp.

$m_{k\ell}$ are dense everywhere; nec. cond. — $\tau_{k\ell}$ dense.

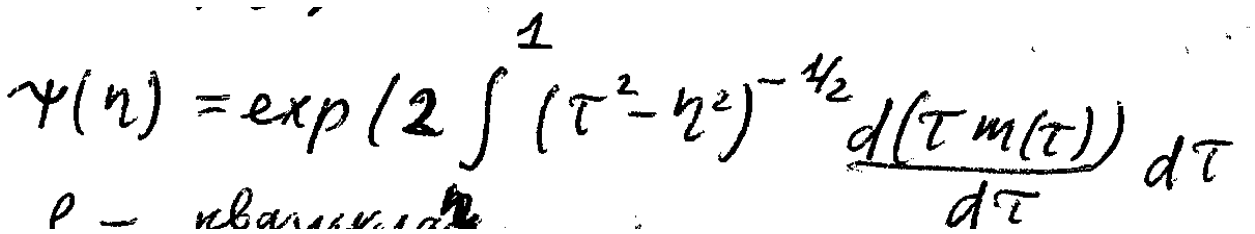
$$(rf(r))^{-1} = \psi(\eta). \qquad \tau(m) \longleftrightarrow m(\tau)$$



Uncertain reading. The arrow between the expressions has heavy strokes at both ends.

Original source details: GELFAND-P000635

$$\psi(\eta) = \exp\Big(2\int^{1} (\tau^2 - \eta^2)^{-1/2} \frac{d(\tau m(\tau))}{d\tau}\, d\tau$$



Original source details: GELFAND-P000635

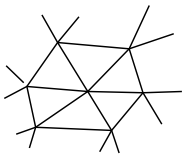
Uncertain reading. The upper limit of the integral is written 1. The lower limit and the closing parenthesis are not visible.

Large ℓ — quasiclassics.

Fuchs. There is a triang. manifold.



Diagram and typed description.



A triangular mesh with edges extending beyond it.

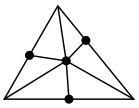
Original source details: GELFAND-P000635

Consider $\sum$ of all simplices of dim. 1 with coeff. 1 mod 2. It is not a cycle.
For the first barycentr. subdivision 0.



Original source details: GELFAND-P000635

Diagram and typed description.



A triangle with three internal lines and four marked points.

all the centers, 0 mod 2 (even).
Stief. class — is Poincaré dual to it.



Original source details: GELFAND-P000635

Uncertain reading. The entry reads “Штиф. класс” (Stief. class).

$$\sum \text{number} \underset{\text{mod } 2}{=} \sum (-1)^i \text{ of numbers} = \text{Euler char.}.$$

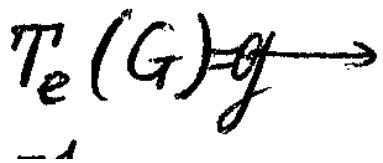
Losik (Mark).
Connection. $E \to X$ vector bundle with fiber $\mathbb{R}^n$; $p : P \to X$ assoc. principal bundle.
$x \in X$, E_x — fiber of E at x.
$P_x \ni z$, $z = (e_{i,x})$, $i = 1, \dots, n$, z — a basis in E_x.
Consider $G = GL(n, \mathbb{R})$ and define its action on P.

$$A \in G \rightsquigarrow R_A : P \to P, \qquad R_A(z) = zA = \left(\sum_{j=1}^{n} e_{j,x} A_i^j \right), \quad A = (A_i^j), \quad i, j = 1, \dots, n.$$

This action is smooth.
GELFAND-P000636

1. $(zA)B = z(AB)$.
2. G preserves the fibers and acts simply transitively.

$z \in P_x$, V_z — vertical tangent space ($\subset T_z P$). $V_z = T_z(P_x)$.
define the map $\ell_z : G \to P_x$, $\ell_z(A) = zA$
and the induced map $\alpha_z : T_e(G) = \mathfrak{g} \to V_z$ (isomorphism).



Original source details: GELFAND-P000636

Uncertain reading. In $T_e(G) = \mathfrak{g} \to V_z$, the lettering of $\mathfrak{g}$ overlaps the strokes of the equals sign and arrow.

$\beta_z = \alpha_z^{-1} : V_z \to \mathfrak{g}$ (a linear form on V_z with values in $\mathfrak{g}$).

A connection on a principal bundle — a 1-form on P with values in $\mathfrak{g}$, such that

1. $\omega|_{V_z} = \beta_z$,

2. $R_A^* \omega = \operatorname{ad} A^{-1} \omega$ (invariance condition).

For the tangent bundle — an affine connection.

Consider the equation $\omega = 0$ at the point z,

$$H_z = \{\xi \in T_z P : \omega(\xi) = 0\}.$$

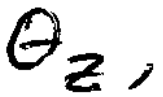
H_z is called the horizontal space. $T_z = H_z \oplus V_z$. H_z is preserved under the action of G.

—·—

A bundle will be constructed whose sections are in one-to-one correspondence with connections on P.

Consider $T_z^* \otimes \mathfrak{g}$ — 1-forms at z with values in $\mathfrak{g}$.

Define $\Theta_z \subset T_z^* \otimes \mathfrak{g}$ as the subspace of those Θ_z, such that $\theta_z|_{V_z} = 0$.



Uncertain reading. After "those", the symbol resembles uppercase Θ_z. The restriction condition reads lowercase θ_z.

Original source details: GELFAND-P000636

$$C_z \subset T_z^* \otimes \mathfrak{g}, \qquad C_z \ni \omega_z : \omega_z|_{V_z} = \beta_z.$$
$$C_z - C_z \subset \Theta_z, \qquad \omega_{1,z} - \omega_{2,z} \in \Theta_z, \qquad \omega_z + \Theta_z \in C_z.$$



Uncertain reading. The last relation has the sign $\in$ before C_z.

Original source details: GELFAND-P000636

i.e. C_z — a plane parallel to Θ_z, which does not pass through 0.

This gives a bundle of affine spaces («the difference of two Christoffel symbols is a tensor»).

Over P consider the bundle with fiber $T_z^* \otimes \mathfrak{g}$.

There are maps:

$$\widetilde{r}_{z,A} : T_z^* \otimes \mathfrak{g} \to T_{zA}^* \otimes \mathfrak{g},$$

induced by the maps $R_{A^{-1}}^* : T_z^* \to T_{zA}^*$ and $\operatorname{ad} A^{-1} : \mathfrak{g} \to \mathfrak{g}$. [At right: Tensor product of maps.]

GELFAND-P000637

C_z goes to the corresponding plane at zA under this map.

$$C(P) = \bigcup_{z \in P} C_z$$

— a natural smooth bundle over P.

G acts on $C(P)$.

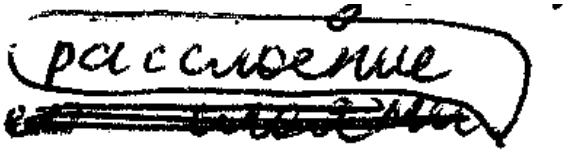
[Crossed out: C?] Invariant sections $\longleftrightarrow$ connections.



Uncertain reading. The crossed-out entry before “Invariant” begins with a mark resembling C.

Original source details: GELFAND-P000637

$C(P)/G = B(P)$, [“bundle” inserted and circled, with an arrow to π] $\pi : B(P) \to X$.



Undeciphered passage. The words below the circled “bundle” are heavily crossed out.

Original source details: GELFAND-P000637

$C(P)$ — a principal bundle with base P, there is a canonical connection there.

Gabrielov. [Crossed out: The connection γ? gives an isomorphism $T^*X \otimes \mathfrak{g} \to B(P)$.]



Uncertain reading. The cancelled symbol after “The connection” resembles γ.

Original source details: GELFAND-P000637

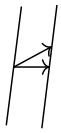
(on $\forall$ fiber — an affine transformation)

1 conn. chosen $\to$ another conn. determines a transformation of the original space.

($\Longleftrightarrow$ the difference of connections is a tensor)



Diagram and typed description.



Two slanted lines and two arrows pointing right in the margin.

Original source details: GELFAND-P000637

$$\lambda \in T_x X \rightsquigarrow E_x \to E_x$$

i.e. a form with values in homomorphisms.

$T^*X \otimes \mathrm{Hom}(E, E)$.

—·—

GELFAND-P000638

1/III-76. Gelfand Sem.



Original source details: GELFAND-P000638

Diagram and typed description.



Hatched strip with a wavy upper edge below the heading.

Drinfeld. (Volodya).

$\overline{X}$ — a smooth proj. curve over $\mathbb{C}$, P — a fixed point on X,

$A \sim$ the ring of rat. fns. on X with a pole at P (reg. on $X \setminus P$).



Original source details: GELFAND-P000638

Undeciphered passage. A separate word below the definition of the ring is obscured by heavy hatching.

G. Example. $y^2 = x(x-1)(x-2)$ (elliptic curve).

$$P = \infty, \qquad A = \mathbb{C}[x,y]/(y^2 - x(x-1)(x-2)).$$



Original source details: GELFAND-P000638

Uncertain reading. The field symbol in the first quotient was overwritten. Tentatively $\mathbb{C}$.

There is no fn. with a pole of order 1. — since [crossed out] it is a torus.



Original source details: GELFAND-P000638

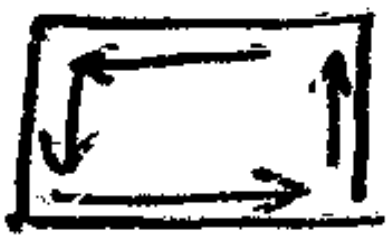
Undeciphered passage. A short word between “since” and “it is a torus” is heavily crossed out.

doubly per.



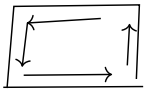
Original source details: GELFAND-P000638

Uncertain reading. The abbreviation reads “двоякопер.” (doubly periodic).



Original source details: GELFAND-P000638

Diagram and typed description.



$\int = 0$, residue $= 0$.

Four arrows run counterclockwise along the contour.

Pole of order 2 — Weierstrass $\wp$-fn. (see Hurwitz, Courant).

[crossed out] $y^2 = x(x-1)$ of genus 0 — Euler subst..

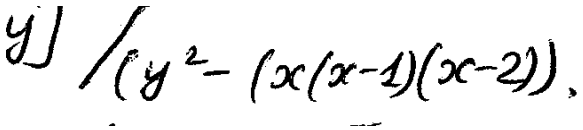


Original source details: GELFAND-P000638

Undeciphered passage. A short word before the equation is heavily crossed out.

A for a curve of genus 0 — polynomials, gen. z.

A for an ell. curve — $\mathbb{C}[x,y]/(y^2 - (x(x-1)(x-2))$.



Original source details: GELFAND-P000638

Uncertain reading. The manuscript lacks a further closing parenthesis after the product.

x has a pole of order 2. It goes to the Weierstrass fn. $x = \wp(z)$,

y —"— order 3. $y = \wp'(z)$.

The ring A uniquely determines the curve and the point.

How to describe these rings?

—·—

Krichever.

almost $\forall$ points P least pole order $g+1$.

Sometimes it can be lower — at Weierstrass points.

$$y^2 = \prod_{i=1}^{2g+1} (z - z_i)$$

(hyperelliptic curve) — at a branch point there is a fn. with a pole of order 2.

n — least possible pole mult. at $P \Rightarrow$ there are poles of mult. $2n, 3n, \ldots$.

Let μ_s be such that $\mu_s n + s$ — pole multiplicity, $s = 1, \ldots, n-1$.

(μ_s — the least).

Then $\mu_1 + \cdots + \mu_{n-1} = g$. ‖ all generators are among them. (Weierstrass theorem).

GELFAND-P000639

P in general position — the ring A has g generators — with poles of $g+1, g+2, \ldots, 2g+1$ order.



Original source details: GELFAND-P000639

Uncertain reading. The manuscript gives g before "generators" and lists orders from $g+1$ to $2g+1$.

B — an algebra over $\mathbb{C}$ with a derivation $\mathcal{D}$.
There is a ring $B[\mathcal{D}] = \{\sum b_i D^i\}$, $\mathcal{D}b - b\mathcal{D} = b'$.
An elem. is called elliptic if its lead. coeff. is invertible.
We describe homs. $A \to B[\mathcal{D}]$, such that the image of $\forall$ elem. is elliptic.
The rank of such a homomorphism is a (nat.) number n, such that

$$\forall a \in A, \quad \operatorname{ord} \varphi(a) = n \cdot \deg a.$$

$\deg a$ — pole order of a [crossed out] at the point P.



Undeciphered passage. The entry after a is covered by heavy cross-out strokes.

Original source details: GELFAND-P000639

(G. There are diff. opers. of orders 2 and 3 that commute with each other.)
[crossed out: M?] homoms. $\longleftrightarrow$ modules with additional structures.



Uncertain reading. The crossed-out mark resembles M.

Original source details: GELFAND-P000639

Krichever thm.
The leading coefficient of the homomorphism $\varphi : A \to B[\mathcal{D}]$.
Let z — a local coordinate at P. Let $a = z^{-k} + \cdots$ (Laurent).
[crossed out: Lead.?] $\varphi(a) = b^k D^{nk} + \cdots$,



Uncertain reading. The crossed-out entry tentatively reads “Стар” (lead.).

Original source details: GELFAND-P000639

where $\underset{\uparrow}{b}$ is independent of k and of the choice of loc. parameter.
B
(there are poles of $\forall$ orders from a certain order onward. One has to divide)
b is called the leading coeff. of the homomorphism φ.
def. [crossed out: This is?] $\varphi_1 \sim \varphi_2$, if $\exists b \in B^*$, $\varphi_1(a) = b\varphi_2(a)b^{-1}$.



Uncertain reading. The crossed-out entry above the start of the definition tentatively reads “Это” (this is).

Original source details: GELFAND-P000639

$\frac{d}{dx} \sim \frac{b'}{b} + \frac{d}{dx}$. Among germs, b'/b — any fn..

Krichever thm. B [crossed out] — germs of smooth fns. on the line.



Undeciphered passage. The formula after B is crossed out. Delimiters are visible, and the reading remains unresolved.

Original source details: GELFAND-P000639

homom. φ of rank 1 (up to equiv.!) $\longleftrightarrow$ the leading coeff. and a point on the Jacobian of the curve (Jacobian variety), J_X.

S. Gelf. 1-dimensional vect. bundles of degree 0 ($\sim$ divisor up to equivalence)

$$\begin{matrix} Q_1, \dots, Q_\ell \\ n_1, \dots, n_\ell \\ \text{integers} \end{matrix} \qquad \text{degree } n_1 + \dots + n_\ell = 0.$$

equivalence: if they differ by the divisor of a (meromorphic) fn..

Equiv. — bundles given by global merom. sections.

GELFAND-P000640

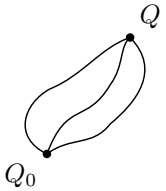
G. Analytic def. of the Jacobian variety: on $\overline{X}$ there exist differentials of the first kind (without sing.) there are g of them.

$y^2 = x(x-a)(x-b)$,

$\| \frac{dx}{y}$ has no sing. (at $x = 0, a, b$ there are none, since x is no loc. param.).



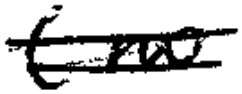
Diagram and typed description.



Three paths between Q_0 and Q.

Original source details: GELFAND-P000640

$z_i = \int_{Q_0}^{Q} \omega_i$. [crossed out: This is?] ($\int$ over homol. classes. Basis — g numbers)



Uncertain reading. The crossed-out entry after the integral tentatively reads “Это” (this is).

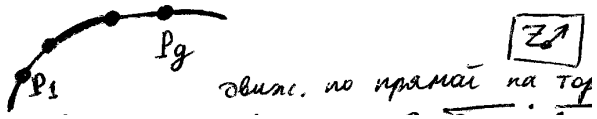
Original source details: GELFAND-P000640

the form determines a set of int. over periods.

up to periods — $\mathbb{C}^g/\underline{\text{lattice}}$. $n_1\alpha_1 + \cdots + n_{2g}\alpha_{2g}$.

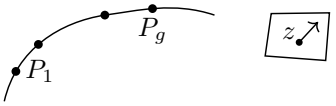
Only some tori arise.

A divisor is a lin. comb. of points. Points of the torus can be added.



Original source details: GELFAND-P000640

Diagram and typed description.



Points on the curve and an arrow from z in a rectangle.

motion along a straight line on the torus $\to$ motion of $P_1, \ldots, P_g$ [crossed out] along the curve.



Original source details: GELFAND-P000640

Undeciphered passage. The word between P_g and "along the curve" is heavily crossed out.

—·—

Giving a hom. from A into the ring $B[\mathcal{D}]$ is equiv. to giving the following.

M — a module over $A \otimes B$, $\nabla : M \to M$, $\nabla(am) = a\nabla(m)$, $\nabla(bm) = b\nabla(m) + b'm$.

$$0 = M_{-1} \subset M_0 \subset M_1 \subset M_2 \subset \cdots, \qquad \dim_B M_i/M_{i+1} = 1 \quad \text{for } i \geqslant 0.$$



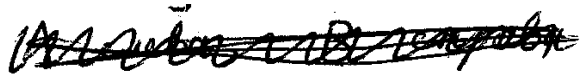
Original source details: GELFAND-P000640

Uncertain reading. The manuscript writes M_i/M_{i+1} in the dimension statement.

$$\begin{aligned} BM_i &\subset M_i, \\ \nabla M_i &\subset M_{i+1}, \qquad \nabla_* : M_i/M_{i-1} \xrightarrow{\sim} M_{i+1}/M_i, \\ aM_i &\subset M_{i+\deg a}, \\ a : M_i/M_{i-1} &\xrightarrow{\sim} M_{i+\deg a}/M_{i+\deg a-1}. \end{aligned}$$

—

$\underline{\varphi : A \to B[\mathcal{D}]}$, $\Rightarrow B[\mathcal{D}] = M$. [crossed out: $A \otimes B$-module?]

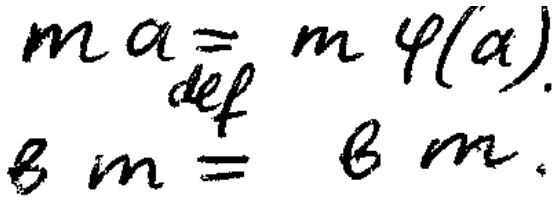


Original source details: GELFAND-P000640

Uncertain reading. The crossed-out line tentatively reads $A \otimes B$-module.

Mult. by B on the left and by A on the right,

$$\begin{array}{rcl} ma & = & m\varphi(a). \\ & \overset{\text{def}}{} & \\ bm & = & bm. \end{array}$$



Uncertain reading. The mark def is written between the two equals signs.

Original source details: GELFAND-P000640

$\nabla : M \to M$ mult. by $\mathcal{D}$ on the left.

Filtr. — order.

last cond. — ellipticity.

—·—

How, from M to construct φ?

[crossed out: M?] $M_{-1} = 0$,



Uncertain reading. The crossed-out mark resembles M.

Original source details: GELFAND-P000640

$\nabla : M_0 \xrightarrow{\sim} M_1/M_0$. in M_0 there is a generator e (over B).

put $D = \nabla(e)$.

$\varphi(a)$ is clear since there is an A-module, etc..

—·—

Next step: translate into the language of sheaves.

[crossed out: on the curve] $M_{-1} = 0 \subset M_0 \subset M_1 \subset \cdots$



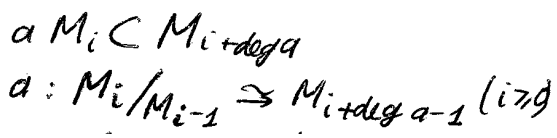
Uncertain reading. “On the curve” is crossed out before the filtration.

Original source details: GELFAND-P000640

$\sim$ a vect. bundle [inserted: 1-dimensional] on X with 0 cohom. (0-dimensional and 1-dimensional) of the sheaf of sections.

[At right]

$$aM_i \subset M_{i+\deg a}, \qquad a : M_i/M_{i-1} \xrightarrow{\sim} M_{i+\deg a-1} \quad (i \geqslant 0).$$



Uncertain reading. Only one filtration term is written at the target of the arrow in the right margin.

Original source details: GELFAND-P000640

GELFAND-P000641

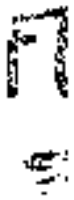


Diagram and typed description.

A three-sided mark at upper right with several detached strokes below.

Original source details: GELFAND-P000641

L — a 1-dimensional vect. bundle over $\overline{X}$, $H^0(X, \underline{L}) = H^1(X, \underline{L}) = 0$.

($H^0(X, \underline{L}) = 0$ and $\deg L = g - 1$).

$M = H^0(X - P, \underline{L})$ (algebraic sects., holom. with a single pole).



Uncertain reading. The abbreviation after “algebraic sects.” reads “голом.” (holom.).

Original source details: GELFAND-P000641

M_i [crossed out: $-$?] with a pole of order $i + 1$.



Uncertain reading. A short horizontal mark after M_i is crossed out.

Original source details: GELFAND-P000641

$M_{-1} = 0$ — first condition, ($H^0 = 0$).

$$\dim M_i/M_{i-1} = 1, \qquad M_0 = H^0(X, \underset{\substack{\uparrow \\ \text{with a pole at } P}}{\underline{L}(P)}).$$

$$0 \to \underline{L} \to \underline{L}(P) \to \underline{L}(P)/\underline{L} \to 0.$$

exact cohom. seq. $\Rightarrow \dim_{\mathbb{C}} M_i/M_{i-1} = 1$.

$=$

Conversely, given a seq. of modules, construct a sheaf

$$0 = M_{-1} \subset M_0 \subset M_1 \subset M_2 \subset \cdots, \qquad \dim_{\mathbb{C}} M_i/M_{i-1} = 1.$$

M — a module over A.

$$aM_i \subset M_{i+\deg a}$$
$$a_* : M_i/M_{i-1} \xrightarrow{\sim} M_{i+\deg a}/M_{i+\deg a-1}, \qquad i \geqslant 0.$$



Uncertain reading. The subscript on a in the map resembles an asterisk.

Original source details: GELFAND-P000641

We give sects. of the sheaf on Zariski open sets.

The sheaf is quasi-coherent. It suffices to give it on open affine subsets.

$a_0 \in A$, $a_0 \neq$ const $\|$ cover

$$X = (X \setminus P) \cup (X \setminus \text{zeros of } a_0).$$

Sects. over $X \setminus P$, A-module — M.

Over $X \setminus$ zeros of a_0 — m/a_0^k, $m \in M_{k \cdot \deg a_0 - 1}$.

Next time: Drinfeld 20 min., Shnirelman.



Original source details: GELFAND-P000641

Undeciphered passage. Only isolated upper strokes are discernible below the last line.

GELFAND-P000642

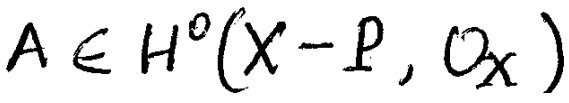
. —·—

15/III-76 Gelfand Sem. Drinfeld.

$\overline{X}$ — a smooth proj. curve over $\mathbb{C}$

$P \in \overline{X}$,

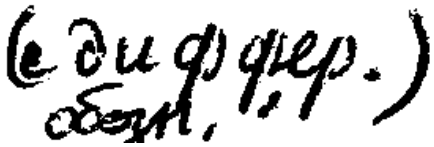
$A \in H^0(X - P, \mathcal{O}_X)$ — rat. fns. on $\overline{X}$ with a pole at P.



Uncertain reading. The sign between A and H^0 is $\in$.

Original source details: GELFAND-P000642

B — an algebra over $\mathbb{C}$ (with a deriv. [below: notation?]), z — a loc. coord. on $\overline{X}$ near P.



Original source details: GELFAND-P000642

Uncertain reading. Below "with a deriv." is an abbreviation resembling "обозн." (notation), with two short strokes.

We seek homomorphisms $\varphi : A \to B[\mathcal{D}]$ of rank 1,

i.e. $\forall a \in A$, $a = z^{-m} + \cdots \Rightarrow \varphi(a) = b^m D^m + \cdots$

b — the leading coefficient of φ.

They are considered up to conjugation by inv. elements of B.

[crossed out: B?] Giving φ is then equivalent to giving an $A \otimes B$-module M



Original source details: GELFAND-P000642

Uncertain reading. The mark crossed out before "Giving" resembles B.

[crossed out: e?] and a map $\nabla : M \to M$, such that



Original source details: GELFAND-P000642

Uncertain reading. The short cancelled mark before "and a map" resembles "e".

$\nabla(am) = a\nabla(m)$, [crossed out: and?] $\nabla(bm) = b\nabla(m) + b'm$, $M = \bigcup M_i$,
(filtration)



Original source details: GELFAND-P000642

Uncertain reading. The mark crossed out between the formulas for the map resembles "и" (and).

$BM_i \subset M_i$, [crossed out: BM_i?] $M_{-1} = 0$, $M_i/M_{i-1} = B$ for $i \geqslant 0$.



Original source details: GELFAND-P000642

Uncertain reading. The crossed-out entry resembles BM_i.

$$\forall a \in A, \qquad aM_i \subset M_{i+\deg a},$$
$$a : M_i/M_{i-1} \xrightarrow{\sim} M_{i+\deg a}/M_{i+\deg a-1},$$
$$\nabla : M_i/M_{i-1} \xrightarrow{\sim} M_{i+1}/M_i, \qquad i \geqslant 0.$$

G. corollary: $\exists$ an oper. whose powers, except for the 1st — are diff. operators (2nd, 3rd etc.).

$\overline{X}$ — an alg. curve of genus g, P — a point.

n — the least possible pole multiplicity at P (the sole pole).

μ_i — the least numbers for which there is a pole of mult. $i + \mu_i n$, $i = 1, \ldots, n-1$.

Then $\sum \mu_i = g$.

GELFAND-P000643

[crossed out: There exist fn...?]



Original source details: GELFAND-P000643

Uncertain reading. The crossed-out opening tentatively reads “Сущ. ф...” (there exist fn...).

(There are g omitted pole multiplicities*)

[Arrow from the asterisk] this [crossed out] also conversely implies the statement formulated by Krichever.



Original source details: GELFAND-P000643

Undeciphered passage. The word after “this” is heavily crossed out.

D. Into the language of sheaves.

Everything except ∇, $\sim$ specifying a vect. bundle L on $\overline{X}$, for which $H^0(X, \underline{L}) = H^1(X, \underline{L}) = 0$.

$\sim$ specifying a 1-dimensional bundle L over $X \otimes B$,

$$H^0(X \otimes B, \underline{L}) = H^1(X \otimes B, \underline{L}) = 0.$$

$$X = U_1 \cup \cdots \cup U_k, \qquad U_i = \text{Spec}\, R_i.$$

Glue the aff. schemes Spec [crossed out] $R_i \otimes B$.



Original source details: GELFAND-P000643

Undeciphered passage. Marks between Spec and the expression on the right are heavily crossed out.

Intuitively: a germ of a smooth family of vect. bundles.

$$\nabla : \underline{L} \longrightarrow \underset{\text{with a first-order sing. at } P}{\underline{L}(P)} \qquad \underline{L} \longrightarrow \underline{L}(P)/\underline{L} \longrightarrow 0.$$

[At right] $H^0(\underline{L}(P)/\underline{L})$ is free (the restriction to P is trivial).

$\nabla b - b\nabla = b'$, ∇ — O_X-linear.

leading coeff. — ?

Trivialize near P, Let e — a basis.

$$\frac{\nabla(e)}{e} = \frac{1}{b_0 z} + \cdots \quad \Rightarrow \quad b_0 \text{ — leading coeff..}$$

Krichever thm. Homom. of rank 1 up to conjugation by inv. elements of $B \Longleftrightarrow$ a point on the Jacobian of $\overline{X}$ in general position.

1. Given φ. $\Rightarrow$ a 1-dimensional bundle is given over $X \otimes B$. We construct a point on the Jacobian.

Let $\mathbb{C} \leftarrow B$ and the corresp. embedding $\overline{X} \subset X \otimes B$, since there is a natural map $B \to \mathbb{C}$ (value at 0).

Restrict L to $\overline{X}$, i.e. obtain a vect. bundle on $\overline{X}$.

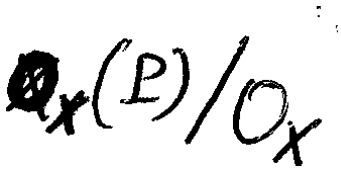
2. Let there be given L_0 — a bundle over $\overline{X}$ and let there be given B. We construct everything else.

J — the Jacobian of $\overline{X}$, its points are vect. bundles of the same degree.

Tangent space to the Jacob. $\cong H^1(X, O_X)$ (the Jacobian itself $H^1(X, O_X^*)$. $\Rightarrow$ the tangent space is obtained by removing $*$).

Consider the exact seq.

$$0 \longrightarrow O_X \longrightarrow O_X(P) \longrightarrow O_X(P)/O_X \longrightarrow 0.$$



Original source details: GELFAND-P000643

Uncertain reading. The first letter of the third term has been overwritten. Tentatively $O_X(P)/O_X$.



Original source details: GELFAND-P000643

Undeciphered passage. A crossed-out entry is visible below the third term of the sequence.

[Upward arrow] a sheaf over a point.

Consider

$$\frac{1}{[\text{crossed out}]z} \in H^0(O_X(P)/O_X) \xrightarrow[\text{boundary hom.}]{} H^1(X, O_X).$$



Undeciphered passage. The mark before z in the denominator is obscured by ink.

Original source details: GELFAND-P000643

(z — a loc. parameter).

The image of $1/z$ in the tangent space to the Jacobian is denoted by $\eta_P \in H^1(X, O_X)$.

GELFAND-P000644

ℓ_0 — the corresp. point on the Jacobian. As we vary x we get a smooth path on the Jacobian (a point of the Jacobian with values in B).

$$\underline{\ell'(x) = \frac{\eta_P}{b_0(x)}}$$

(a straight line with changing parameter).



Uncertain reading. The looped sign in both entries resembles ℓ or e.

Original source details: GELFAND-P000644

Again: curve X, point P, loc. param..

$\Rightarrow$ a line in the tangent space to the Jacobian.

$x \in \overline{X}$ [crossed out] $x \mapsto \overline{x - P}$ — divisor.



Undeciphered passage. A group of looped signs is crossed out after the membership expression.

Original source details: GELFAND-P000644

We obtain a map from $\overline{X}$ into the Jacobian J.

The curve has at P a tangent space. The image — a 1-dimensional space tangent to the Jacobian.

—·—

A. Polyakov. Gauge fields. Quantization of the Yang–Mills eqs.

Lagrangian

$$S = \int \mathcal{L}(x) d^4x, \qquad \mathcal{L}(x) = \frac{1}{4} \operatorname{tr} F_{\mu\nu}^2.$$

For the group SU(2) we have $F_{\mu\nu}$ — an elem. of the Lie algebra.

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu].$$

[Outside the right rule]



Uncertain reading. The isolated mark resembles “ɜ” or “3”.

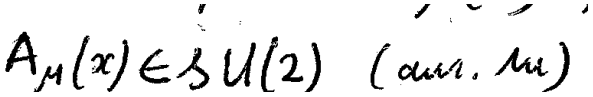
Original source details: GELFAND-P000644

A_μ, $\mu = 0, 1, 2, 3$ — anti-Hermitian with trace 0 (2×2) (“traceless”).

The fields A_μ — dynamical variables.

Elem. fields — fns. $A_\mu(x)$, $\mu = 0, 1, 2, 3$.

$$A_\mu(x) \in \mathrm{SU}(2) \quad \text{(Lie alg.)}.$$



Uncertain reading. The line reads $SU(2)$, followed by “Lie alg.” in parentheses.

Original source details: GELFAND-P000644

$$F_{\mu\nu}^2 = \sum_{\mu,\nu} F_{\mu\nu} F_{\mu\nu}.$$

Undeciphered passage. Several signs to the right of the sum are crossed out.



Original source details: GELFAND-P000644

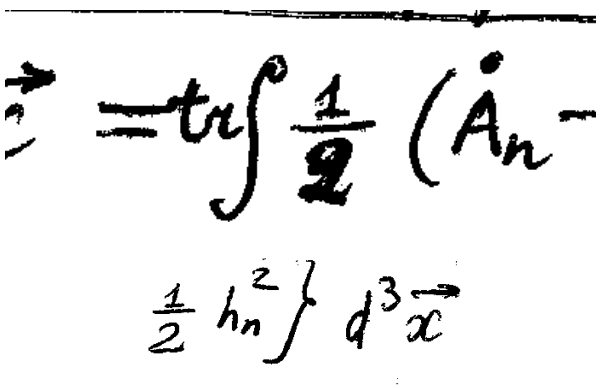
A_μ is called the vect. potential,

$F_{\mu\nu}$ —"— the field-strength tensor.

Passage to the Hamiltonian formalism. $n = 1, 2, 3.$

$$L = \int \mathcal{L} d^3\vec{x} = \text{tr} \int \frac{1}{2}(\dot{A}_n - \nabla_n A_0)^2 - \frac{1}{2} h_n^2 \Big\} d^3\vec{x}$$

(we single out time)



Uncertain reading. The manuscript has a closing brace. No opening brace is visible after the integral sign.

Original source details: GELFAND-P000644

Here $\nabla_n A_0 = \partial_n A_0 + g[A_n, A_0]$ (and we call it the covariant derivative).

$$h_n = \frac{1}{2}\varepsilon_{nm\ell}(\partial_m A_\ell - \partial_\ell A_m + g[A_m, A_\ell]),$$
$$F_{0n} = \partial_0 A_n - \partial_n A_0 + g[A_0, A_n].$$

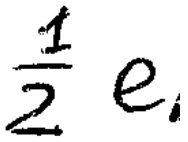
$$\nabla_\mu X \underset{\text{def}}{=} \partial_\mu X + g[A_\mu, X].$$

GELFAND-P000645

L does not contain $\dot{A}_0$, $e_n = \dfrac{\partial \mathcal{L}}{\partial \dot{A}_n} = \dot{A}_n - \nabla_n A_0$ — canon. momentum.

Hamiltonian.

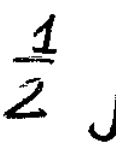
$$\mathcal{H} = e_n \dot{A}_n - \mathcal{L} = \frac{1}{2} e_n (e_n + \nabla_n A_0) + \frac{1}{2} h_n^2.$$



Uncertain reading. The manuscript has $\frac{1}{2}$ before $e_n(e_n + \nabla_n A_0)$.

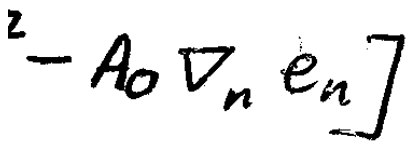
Original source details: GELFAND-P000645

$$H = \frac{1}{2} \int d^3\vec{x}\ \text{tr}[e_n^2 + h_n^2 - A_0 \nabla_n e_n].$$



Uncertain reading. The prefactor of the integral reads one half.

Original source details: GELFAND-P000645



Original source details: GELFAND-P000645

Uncertain reading. A minus sign stands before $A_0 \nabla_n e_n$, with no separate numerical factor.

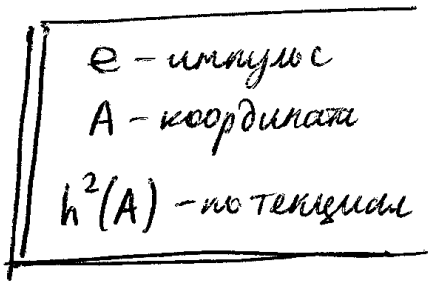
$$-\frac{\delta H}{\delta A_0} = \dot{e}_0 = 0.$$

Hamilton eqs. $\leftrightarrow$ Lagrange eqs..

Hence let us impose the cond.

$$\nabla_n e_n = 0 \qquad (*)$$

$$\widehat{H} = \frac{1}{2}\int d^3\vec{x}\,[e_n^2 + h_n^2].$$



Original source details: GELFAND-P000645

Diagram and typed description.

e — momentum
A — coordinates
$h^2(A)$ — potential

A note in the right margin with an open-sided frame.

Will the cond. hold at the next instant $(*)$?

Denote $\nabla_n e_n = \rho(\vec{x})$.

This is the condition $\{\rho(x), H\} = 0$, where

$$\{A, B\} = \int d^3\vec{x} \sum_n \left(\frac{\delta A}{\delta e_n} \cdot \frac{\delta B}{\delta A_n} - \frac{\delta A}{\delta A_n}\frac{\delta B}{\delta e_n} \right).$$

$$\begin{cases} \dot{A}_n = \dfrac{\delta H}{\delta e_n}, \\ \dot{e}_n = -\dfrac{\delta H}{\delta A_n}, \\ \nabla_n e_n = 0. \end{cases}$$

[At right] These eqs. $\sim$ the Lagrange eqs. This generalizes the usual result..

(generalization of Maxwell's eqs. from $U(1)$ to any [crossed out] compact group — and even a semisimple one).



Original source details: GELFAND-P000645

Undeciphered passage. A word between "any" and "compact" is crossed out. Lower loops are visible.

—·—

$\exists$ a comb. of Y.–M. fields on which the potential energy does not depend h^2.

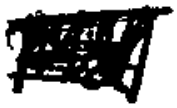
$$h_n = \frac{1}{2}\varepsilon_{nm\ell}F_{m\ell},$$
$$F_{m\ell} = \partial_m A_\ell - \partial_\ell A_m + g[A_m, A_\ell].$$

$$\widetilde{A}_n(x) = \Omega^+ A_n \Omega + \frac{1}{g}\Omega^+\partial_n\Omega(\vec{x}),$$

$\Omega(x)$ — arbitrary matrix, $\Omega(x) \in SU(2)$.

$\widetilde{h}_n = \Omega^+ h_n \Omega$ — checked directly.

$\Rightarrow$ the pot. en. [crossed out] invar. (Sp h^2 does not change).



Undeciphered passage. A short entry before "invar." is heavily crossed out.

Original source details: GELFAND-P000645

Introduce the field a_n. consider fields with cond. $\partial_n a_n = 0$.

Set

$$\begin{cases} A_n(x) = \Omega^+(x) a_n \Omega(x) + \frac{1}{g}\Omega^+\partial_n\Omega(x), \\ \partial_n a_n = 0. \end{cases}$$

[At right] $\|$ a new set of variables.

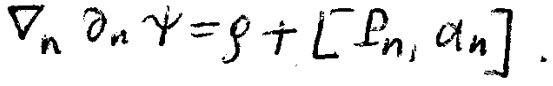
GELFAND-P000646

Canon. momentum split into two parts. P_n, $\rho(x)$, $\partial_n P_n = 0$.

$$e_n = P_n + \partial_n\Psi, \qquad H = \frac{1}{2}\int (P_n^2 + h_n^2(a) + (\partial_n\Psi)^2)d^3\vec{x}.$$

To find Ψ we use that $\nabla_n e_n \underset{\text{def}}{=} \rho(\vec{x})$.

$$\nabla_n P_n + \nabla_n\partial_n\Psi = \rho,$$
$$\nabla_n\partial_n\Psi = \rho + [P_n, a_n].$$



Uncertain reading. The first formula has $[P_n, a_n]$ with no factor g.

Original source details: GELFAND-P000646

$(\partial_n \Psi)^2$ is called Coulomb energy.

—·—

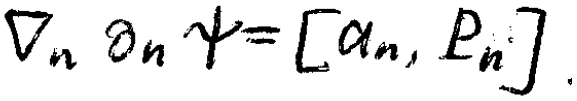
$$H = \int (e_n^2 + h_n^2(A)),$$
$$A_n \longrightarrow \Omega^+ a_n \Omega + \Omega^+ \partial_n \Omega,$$
$$\partial_n a_n = 0.$$

$$\begin{cases} H = \int (P_n^2 + (\partial_n \Psi)^2 + h^2), \\ \nabla_n \partial_n \Psi = [a_n, P_n]. \end{cases} \qquad \text{(Schwinger)}$$



Original source details: GELFAND-P000646

Uncertain reading. The parenthesis at right reads “швингер” (Schwinger).



Original source details: GELFAND-P000646

Uncertain reading. Here the bracket reads $[a_n, P_n]$.

Here $h_n(a) = \frac{1}{2}\varepsilon_{nm\ell}(\partial_m a_\ell - \partial_\ell a_m + g[a_m, a_\ell])$. Ψ is found from the last eq..

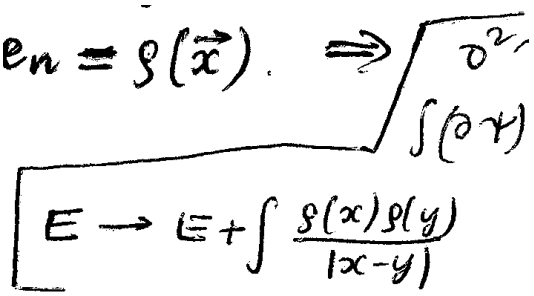
$$\nabla_n e_n = \rho(\vec{x}) \quad \Rightarrow \quad \begin{cases} \partial^2 \Psi = \rho(x) \quad \text{(in the scalar case)}, \\ \int (\partial \Psi)^2 dx = \int \dfrac{\rho(x)\rho(y)}{|x-y|}\, dx\, dy. \end{cases}$$

— Coulomb interaction.



Original source details: GELFAND-P000646

Uncertain reading. Above the equality the entry reads “в скал. случае” (in the scalar case).



Original source details: GELFAND-P000646

Diagram and typed description.

$$E \to E + \int \frac{\rho(x)\rho(y)}{|x-y|}$$

A bent line connects the energy expression to the preceding formulas.

Non-Ab. case. First by perturbation theory (first Khriplovich, then Gross, etc.).

Classical case: $g = 0 \Rightarrow$ the problem is Abelian.

We obtain

$$\partial^2 \Psi + g[a_n, \partial_n \Psi] = g[a_n, P_n] + \rho(x).$$

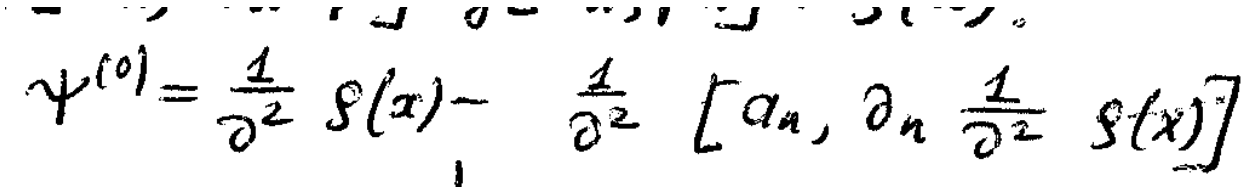


Uncertain reading. The right-hand side reads $g[a_n, P_n] + \rho(x)$.

Original source details: GELFAND-P000646

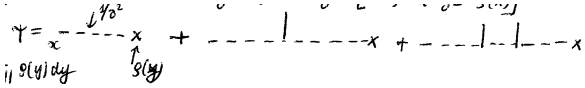
Approximations:

$$\Psi^{(0)} = \frac{1}{\partial^2}\rho(x) - \frac{1}{\partial^2}\left[a_n, \partial_n \frac{1}{\partial^2}\rho(x)\right].$$



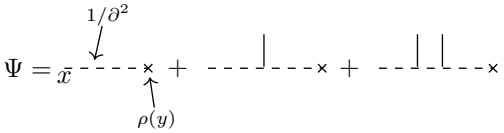
Original source details: GELFAND-P000646

Uncertain reading. The second term has prefactor $-1/\partial^2$. No separate factor g is visible.



Original source details: GELFAND-P000646

Diagram and typed description.



Three dashed diagrams with zero, one and two vertical branches.

[Below the first diagram] $\int \frac{1}{|x-y|}\rho(y)dy$.

Branch — a_n.

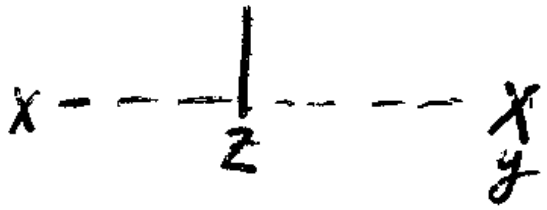
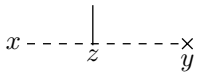


Diagram and typed description.

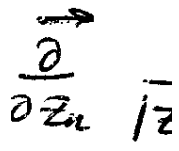


The lower diagram has a branch at z and a cross above y.

Original source details: GELFAND-P000646

[At right]

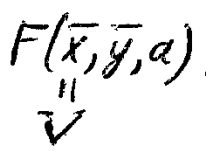
$$= \int \frac{1}{|x-z|} a_n(z) \frac{\vec{\partial}}{\partial z_n} \frac{1}{|z-y|} \rho(y)\, d\vec{y}\, d\vec{z}.$$



Uncertain reading. An arrow is written over the derivative sign in the numerator.

Original source details: GELFAND-P000646

$$\mathcal{H} \to \mathcal{H} + \int \rho(\bar{x}) F(\bar{x}, \bar{y}, a) \rho(\bar{y})\, d\bar{x}\, d\bar{y}.$$
$$\Big\Vert$$
$$V$$



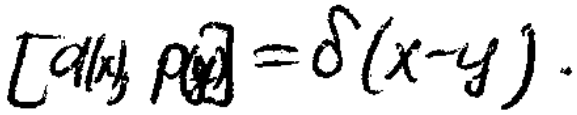
Uncertain reading. Below the arguments of F are two short strokes and a sign resembling V.

Original source details: GELFAND-P000646

Previously $F = \dfrac{1}{|x-y|}$. now F — an operator.

GELFAND-P000647

Passage to quant. mech. — a, p — opers., $[a(x)\ p(y)] = \delta(x-y)$.

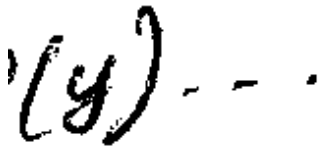


Uncertain reading. The second entry in the bracket reads $p(y)$. The marks between entries are crowded, and a separate comma is indistinct.

Original source details: GELFAND-P000647

$$\delta E = \langle V \rangle = \int \rho(x) \langle F(x, y, a) \rangle \rho(y) \dots$$

(averaging)



Uncertain reading. Three short strokes follow $\rho(y)$. The integration measure is unwritten.

Original source details: GELFAND-P000647

5 April — continuation.

GELFAND-P000648

22/III-76. Gelfand Sem.

G. A prob. distrib. arising from biology.

$$P_n = e^{-x_0 - \alpha n} x_0 \frac{(x_0 + \alpha n)^{n-1}}{n!}, \qquad 0 \leqslant \alpha < 1.$$



Uncertain reading. The initial letter appears to be uppercase P.

Original source details: GELFAND-P000648

For $\alpha = 0$ — Poisson distrib..

Exercise. check that the sum equals 1.

“This is the only case where math. makes sense”. There is one more case in the literature.



Uncertain reading. The end of the sentence reads “в лит-ре есть” (there is one in the literature).

Original source details: GELFAND-P000648

Roiter.

Λ — a finite-dimensional algebra over the field k.

There may be finitely or infinitely many indecomp. reps..

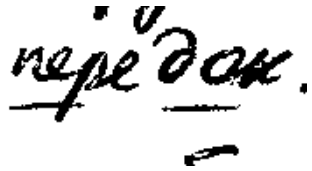
1968 Roiter: if there are infinitely many, there are indecomp. reps. of arbitrarily large dimension. Gabriel — a simpler proof.

(Maurice) Auslander: 1) generalized to Artinian rings.

2) if Λ has infinitely many indecomp. reps., there are infinite chains of modules embedded in one another.

$$A_1 \to A_2 \to A_3 \to \ldots \to A_n \to \ldots$$

Roiter reproved this with additional information.



Uncertain reading. The abbreviation before "with additional information" reads "передок." (reproved).

Original source details: GELFAND-P000648

Λ — a finite-dimensional algebra, A — an indecomp. module over Λ.

$\ell(A)$ — length of the mod. A — max. length

$$0 \subset A_1 \subset A_2 \subset \cdots \subset A_n = A.$$

$\bar{\ell}(A)$ — max. length [inserted with arrow: of such a] chain of indecomp. modules.

We have $\ell(A) \geqslant \bar{\ell}(A)$.

Theorem $\forall \Lambda$ and $\forall$ nat. N, $\exists n$:

$$\bar{\ell}(A) \leqslant n \Longrightarrow \ell(A) \leqslant N.$$

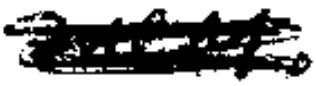


Uncertain reading. The arrow goes from $\bar{\ell}(A) \leqslant n$ to $\ell(A) \leqslant N$.

Original source details: GELFAND-P000648

—

There is an extended Dynkin diagram with n [crossed out] vertices.



Undeciphered passage. The word between n and "vertices" is heavily crossed out.

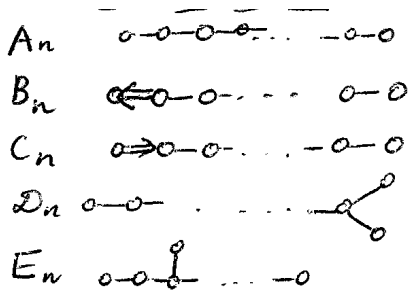
Original source details: GELFAND-P000648

It has an associated quadr. form

$$f(x_1, \dots, x_n) = \sum_{i=1}^{n} a_i x_i^2 + \cdots$$

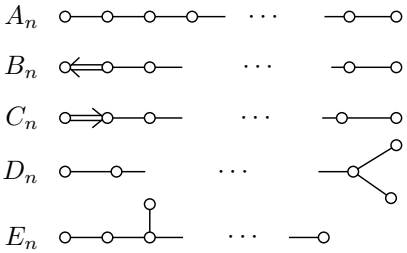
[At left] Kirillov.

Dynkin diagram one of the pictures.

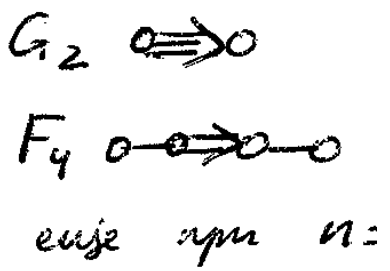


Original source details: GELFAND-P000648

Diagram and typed description.

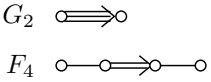


Left column with the diagrams A_n, B_n, C_n, D_n, E_n.



Original source details: GELFAND-P000648

Diagram and typed description.



Right column with the diagrams G_2, F_4.

[At right] and also for $n = 6, 7, 8$.

$\longrightarrow$

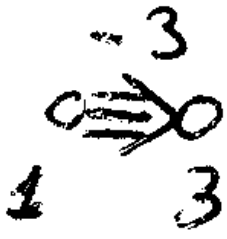
[At left] Roiter.

Dynkin diag. — an integer-valued quadr. form

$$\sum a_i x_i^2 + \sum_{i<j} b_{ij} x_i x_j.$$

additional conditions 1) positive definiteness. 2) $a_i > 0$. 3) $b_{ij} \leqslant 0$. 4) $a_i \mid b_{ij}$ (divides).

[At right] || one obtains precisely the Dynkin diagrams.



Original source details: GELFAND-P000648

Diagram and typed description.



The lower diagram has 1 and 3 below the circles. The stroke before the upper 3 resembles a minus sign.

|| Extended diag. — if the condition is replaced by nonnegative definiteness.

GELFAND-P000649

A reflection is a change of one coord. of the vector that leaves the quadr. form unchanged. (lin. transf.).

Generate the group by all the refl.

Simple root — $(0, 0, \dots, 1, \dots, 0)$.

Root: a vector obtained from a simple one by several reflections.

Given a vector $(x_1, \dots, x_n)$. When is it a root?

If all the strokes are single, then root $\iff$ the quadr. form on it equals 1.



Original source details: GELFAND-P000649

Uncertain reading. The abbreviation before “are single” reads “чёрт.” (strokes).

[Crossed out: If they are not all single] If they are not all single?



Original source details: GELFAND-P000649

Uncertain reading. The first “If they are not all single” is crossed out with several strokes.

Thm.

$$(x_1, \dots, x_n) \text{ — root} \quad \iff \quad \forall i: \ f(x) \mid a_i x_i.$$

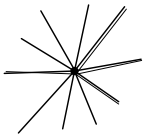
Exercise (for first-year students).

$$\sum_{i=1}^{n} a_i x_i^2 - \sum_{i<j} b_{ij} x_i x_j, \qquad \frac{b_{ij}}{a_i} \in \mathbb{Z}, \quad a_i > 0.$$



Original source details: GELFAND-P000649

Diagram and typed description.



Rays at the left of the following formula. Some strokes are doubled.

$$x_0^2 + \dots + x_n^2 - x_0 x_1 - x_0 x_2 - \dots - x_0 x_n. \quad \text{unclear.}$$

Refl. — a lin. transf. that leaves $f(x)$ unchanged and changes only one coord.

[Crossed out: $x \to \bar{x}$ and пр...в., similar.]



Original source details: GELFAND-P000649

Undeciphered passage. The interior letters of the crossed-out abbreviation are only partly legible.

describe all the roots! (obtained by refl. from standard vectors) (ВОЛК→КОЗА).

[At right, column]

ВОЛК (wolf)
ПОЛК (regiment)
ПОЛА (coat flap)
ПОЗА (pose)
КОЗА (goat)

— · —

Kirillov on Berezin's talk.

$$x_i x_k = x_k x_i, \qquad x_1, \dots, x_p$$
$$\xi_i \xi_k = -\xi_k \xi_i, \qquad \xi_1, \dots, \xi_q$$

Supermanifold: Berezin, Leites.

[In the left margin] Following Rudakov.

Consider the manifold M^p and the bundle $E \to M^p$, whose fibre is a Grassmann algebra (with q generators) (the struct. is specific to $\forall$ point).

Morphisms are described in a more complicated way.

A fn. on a superman. is a smooth sect. of the bundle.

Integration rules. w.r.t. x as usual, and

$$\int d\xi = 0, \qquad \int \xi \, d\xi = 1.$$

Reps. of a 64-dimensional Lie superalgebra describe elem. particles better than reps. of $SU(3)$.

Change of var.:

$$\begin{cases} x \to x + \xi_1 \xi_2, \\ \xi_1 \to \xi_1 + x \xi_2, \\ \xi_2 \to \xi_2 + \xi_1 \xi_2 \xi_3 \end{cases}$$

$$\int P(\xi_1, \xi_2, \xi_3) \, d\xi_1 \, d\xi_2 \, d\xi_3 = \text{coeff. of the highest term.}$$

Lie superalgebra: $\mathbb{Z}_2$-graded, and

$$[x, y] = -(-1)^{\deg x \cdot \deg y} [y, x].$$

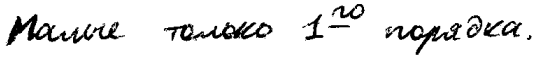
Jacobi id. $[x, \cdot]$ — a homogeneous derivation.

more conv. def.: a superalg. is obtained from a graded assoc. algebra by the rule

$$[x; y] = xy - (-1)^{\deg x \cdot \deg y} yx.$$

Lie thm. — Berezin and Kac from Kyiv.

G.'s remark. $p(t + \xi) = p(t) + p'(t)\xi$. Small quantities only of first order.



Uncertain reading. The first word tentatively reads "Малые" (small quantities).

Original source details: GELFAND-P000649

GELFAND-P000650

Fns. on a superman. form a superalgebra.

There is a Poisson bracket.

Find deformations in the direction of the Poisson bracket.

Sinai. Theory of phase transitions — spectra of matrices.

It was possible to calc. one of the interesting examples.



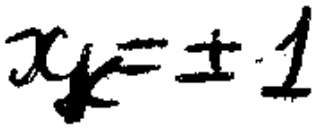
Uncertain reading. The abbreviation after "of" tentatively reads "инт." (interesting).

Original source details: GELFAND-P000650

Dimension of the space 2^n.

$$a_{ij} \qquad a(\underbrace{x_1, \dots, x_n}_{i}, \underbrace{y_1, \dots, y_n}_{j})$$
$$= e^{\beta(J_1 \sum_{i=1}^{n} x_i x_{i-1} + J_2 \sum_{i=1}^{n} x_i y_i)}.$$

$x_k = \pm 1$, $x_0 = x_n$.



Uncertain reading. The subscript on x is heavily overwritten and resembles k.

Original source details: GELFAND-P000650

Physical problem: one needs to study $\frac{e^{-\beta H}}{Z}$. Here [crossed out]



Undeciphered passage. A short passage after "Here" is heavily crossed out.

Original source details: GELFAND-P000650

$$\frac{\ln Z}{R^2} \underset{R \to \infty}{\longrightarrow} f(\beta).$$

The fns. $f(\beta)$ can have singularities.

Study $f(\beta)$ — [crossed out]

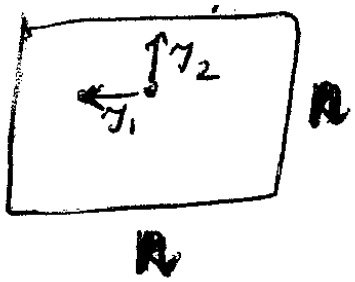


Undeciphered passage. A short word before "there is" is heavily crossed out.

Original source details: GELFAND-P000650

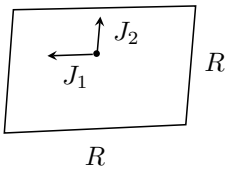
there is a transfer matrix (transfer-matrix).

Matrix with positive entries.



Original source details: GELFAND-P000650

Diagram and typed description.



The arrows point left and up. The labels read as J_1 and J_2.

$$Z = \sum e^{-\beta \sum_{x',x''} H(x',x'')}, \qquad Z = \operatorname{tr} A^R.$$



Original source details: GELFAND-P000650

Uncertain reading. The superscript is heavily overwritten and reads as R.

A^R — transition matrix from layer to layer.

(Onsager's 2-dimensional problem).

Onsager computed all the eigenvalues of this matrix:

Anticomm. variables help here: there are anticomm. matrices A_i, A_j, such that

$$A = e^{\sum c_{ij} A_i A_j}$$

and the integrals can be calculated.

[In the right margin] ‖ see Huang "Stat. Mech.".

Lieb similarly calculated the entropy of square ice.

— · —

GELFAND-P000651

29/III-76. Gelfand Sem.

Krichever: problem: solve the eq. $[L_1, L_2] = 0$. If ord L_1 and ord L_2 are coprime, the answer is a curve and a bundle (1-dimensional).

The orders are not coprime — multidimensional sheaves arise.

E.g. 4th and 6th order — not done. Reduced to 2-dimensional bundles.

$$\left.\begin{aligned} L_1 y = E_1 y \\ L_2 y = E_2 y \end{aligned}\right\} \quad P(E_1, E_2) = 0 \quad \text{— curve.}$$



Original source details: GELFAND-P000651

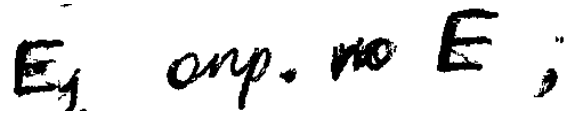
Undeciphered passage. A heavily crossed-out passage below the brace of the simultaneous equations.

Joint solution — a 2-dimensional bundle over the curve.

Coprime — the basis is 1-dimensional. (from the asymptotics).

$(m, n) = 1 \Longrightarrow$ the number of common eigenfns. equals n. (and the eigenvalues are distinct)

E_1 is determ. by E; y a uniquely determ. fn. on a Riemann surf.



Original source details: GELFAND-P000651

Uncertain reading. The subscript on the first E tentatively reads 1. The word “по” (by) is heavily overwritten.

Inverse problem: construct commuting diff. ops. from the bundle — not done.

Drinfeld.

$d = (m, n) \Longrightarrow$ specifying a pair of comm. ops. $\Longleftrightarrow$ specifying $(d-1)$ arbitrary fns.

$B^{-1}LB$ can be solved — 1 arbitrary fn. remains — the leading coeff.

Orders 4 and 6 — if square and cube — already 2 arbitrary fns.

Shnirelman. M — an n-dimensional compact analytic Riemannian manifold,

Δ — Laplace op. on M;

$$\Delta u_k + \lambda_k^2 u_k = 0; \qquad (1)$$

def. of Δ:



Original source details: GELFAND-P000651

Diagram and typed description.

 sphere of rad. τ;

Circle with a dot at the centre.

$$\frac{\text{avg. over sphere of rad. } \tau \text{ — value at centre}}{\tau^2} \longrightarrow \Delta f(x).$$

Growth as k: $\lambda_k \sim ck^{1/n}$, [crossed out: $a \ldots$]



Original source details: GELFAND-P000651

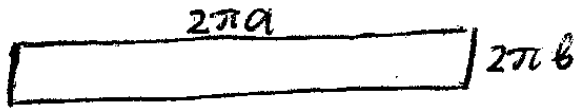
Undeciphered passage. A sign resembling a is visible at the start of the cancellation.

Let $a > 0$. How is the following distributed $a \cdot \lambda_k \bmod 2\pi$

(There are often arithmetic progressions).

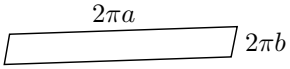
— subseq.

Torus.



Original source details: GELFAND-P000651

Diagram and typed description.



Rectangle under the heading “Torus”.

$$\Delta u = \lambda^2 u.$$

$\sin \alpha x \cdot \sin \beta y$ [crossed out: cos . . .]



Undeciphered passage. The start of the crossed-out entry resembles cos. The remaining signs are illegible.

Original source details: GELFAND-P000651

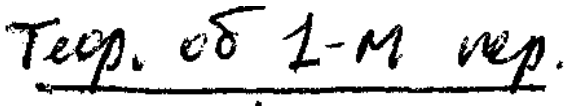
or $\cos \alpha x \cdot \sin \beta y$ etc.

$$\lambda^2 = \frac{n^2}{a^2} + \frac{m^2}{b^2}, \qquad \lambda = \sqrt{\frac{n^2}{a^2} + \frac{m^2}{b^2}}.$$

Series — for different n and m (for fixed).

Which 2 indices — unclear.

Thm. for 1 dim.



Uncertain reading. The last abbreviation tentatively reads “мер.” (dimension).

Original source details: GELFAND-P000651

$$-y'' + u(x)y = \lambda y \quad \Longrightarrow \quad \text{on}$$



Diagram and typed description.

⟝

Horizontal line with two end ticks.

Original source details: GELFAND-P000651

$$\sum_{i=0}^{n-1} c_i y_i \quad \text{has} \leq n \text{ zeros} \quad \Longrightarrow \quad \text{behaves like a polyn. of degree 1.}$$

Hence 2 dim. — a polynomial in n variables.



Uncertain reading. The abbreviation after 2 tentatively reads “мер.” (dimension).

Original source details: GELFAND-P000651

Lowest — positive everywhere.

1st — like linear fns.

Next — a curve of 2nd order.

follow the lin. combin.



Diagram and typed description.



Two ovals. The sloping line in the left one runs from an interior point to the upper boundary. In the right one it runs from the left boundary to an interior point.

Original source details: GELFAND-P000651

Rayleigh — theory of sound (2 volumes).

Arnold: transfer Harnack's thm. on the number of ovals (unknown for lin. combs.).
[In the right margin] ‖ For the nth eigenfn. the zeros divide the region $\leq$ into n parts.

GELFAND-P000652

Consider

$$\Phi(\lambda, a) = \lim_{\varepsilon \to 0} \varepsilon \left(\sum \delta((\lambda - a\lambda_k) \bmod 2\pi) e^{-\varepsilon \lambda_k} - \sum_k \frac{e^{-\varepsilon \lambda_k}}{2\pi} \right).$$

$$\mu_k = a\lambda_k \bmod 2\pi.$$

μ — param. on the circle.



Uncertain reading. The heavily retraced letter tentatively reads μ. The following words read "param. on the circle".

Original source details: GELFAND-P000652

$$\Phi(\mu, a) = \lim_{\varepsilon \to 0} \varepsilon \left(\sum_{k=1}^{\infty} \delta(\mu - \mu_k) e^{-\varepsilon \lambda_k} - \sum e^{-\varepsilon \lambda_k} / 2\pi \right).$$

[Arrow to the subtracted term] subtract so that the avg. value is 0. constant.

$\Phi^{\varepsilon}(\mu, a)$ — pre-limit expression.

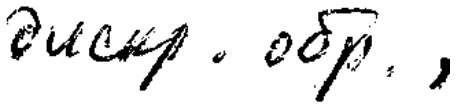
Theorem. Let $\gamma_1, \gamma_2, \ldots$ — closed geodesics on M, $\ell(\gamma_1), \ell(\gamma_2), \ldots$ — their lengths. Let невд. расп.,



Uncertain reading. The first Russian abbreviation resembles "невд.". The next reads "расп.".

Original source details: GELFAND-P000652

in a discr. manner,



Uncertain reading. The Russian abbreviation tentatively reads "дискр." (discr.).

Original source details: GELFAND-P000652

the lengths are all distinct and incommensurable. (general pos.).

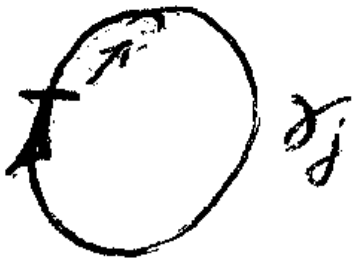
$$\Phi(\mu, a) \begin{cases} \equiv 0 & \text{if } a \neq \frac{p}{q} \ell(\gamma_j), \\ \not\equiv 0 & \text{[crossed out] if } a = \frac{p}{q} \ell(\gamma_j). \end{cases} \qquad (p, q \in \mathbb{Z})$$



Undeciphered passage. The word before the second "if" is heavily crossed out.

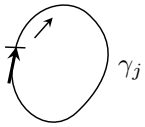
Original source details: GELFAND-P000652

[In the left margin]



Original source details: GELFAND-P000652

Diagram and typed description.



Oval and arrows. The heavy strokes on the left edge overlap.

consider the geodesic flow. ($\in SM$).

S — a plane transv. to the closed geod. (dim. $2n-2$).

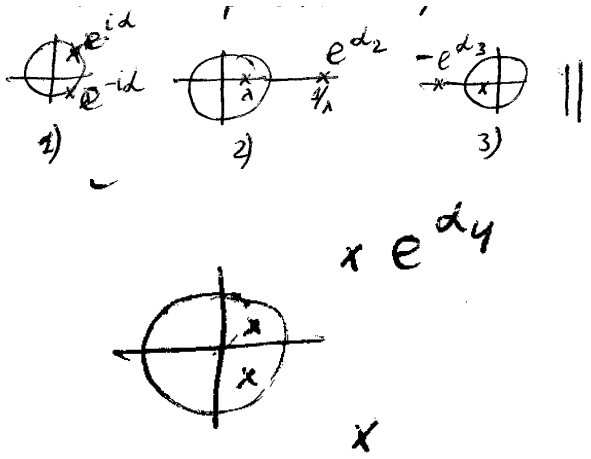
return map $\Psi : S \to S$.

$S \cap \gamma_j$ — fixed point.

$$d\Psi\big|_{S\cap\gamma_j} = \mathcal{M} \quad \text{— monodromy op.} \quad (\underline{\text{monodromy matrix}}).$$

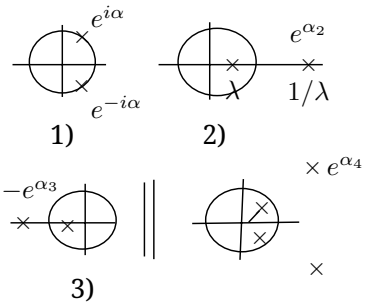
This transf. is symplectic, since the flow is Hamiltonian.

The eigenvalues fall into pairs and quadruples.



Original source details: GELFAND-P000652

Diagram and typed description.



Three numbered sketches and an unnumbered fourth. No legible label is visible at the inner cross in the third sketch.

[At the right of the fourth sketch] — this only in 3-dimensional space.
Consider the Fourier coeffs.:

$$\varphi_n(a) = \int_0^{2\pi} \varphi(\mu, a) e^{-i\mu n}\, d\mu.$$

$$1) \quad \varphi_n(a) = \frac{c}{\sin \frac{n\alpha_1}{2}},$$
$$2) \quad \varphi_n(a) = \frac{c}{\operatorname{sh} \frac{n\alpha_2}{2}},$$
$$3) \quad \varphi_n(a) = (-1)^n \frac{c}{\operatorname{sh} \frac{n\alpha}{2}}.$$

c depends on the dimension.
[A long enclosing line runs from the three formulas to the entry at right.]
1) α_1 is poorly approx. by rationals (required) $\Longrightarrow$



Uncertain reading. The word before “approx.” reads “плохо” (poorly).

Original source details: GELFAND-P000652



Uncertain reading. The parenthesis tentatively reads “треб.” (required).

Original source details: GELFAND-P000652

$$\text{A)} \quad \varphi(\mu, a) = \sum_{n=0}^{\infty} \delta\left(\mu - \left(n + \frac{1}{2}\right)\alpha - \frac{1}{2\pi}\right).$$

(exactly in the reg. sense

$$\varphi(\mu, a) = \lim_{\varepsilon \to 0} \sum_{n=0}^{\infty} \delta\left(\mu - \left(n + \frac{1}{2}\right)\alpha e^{-n\varepsilon} - \frac{1}{2\pi}\right).$$

[In the lower left margin] 2 quantum numbers — 2 indices.
[In the lower left margin] [heavy mark] Infinitely many arithm. progressions.

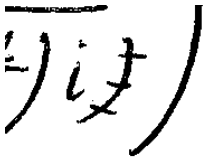


Undeciphered passage. Before the line is a heavy mark resembling the letter A.

Original source details: GELFAND-P000652

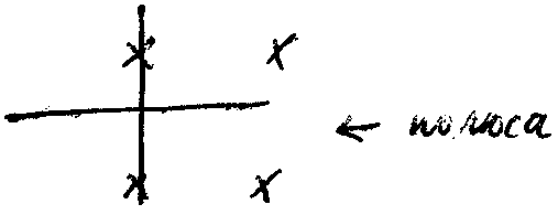
$$2) \quad \varphi(\mu, a) = c \sum_{m,n=-\infty}^{\infty} \left(\frac{1}{\mu - n \cdot 2\pi - (m + \frac{1}{2})i} + \frac{1}{n \cdot 2\pi + (m + \frac{1}{2})i \text{ [cancelled mark]}} \right)$$
$$= c \int_0^{\mu} \wp(\xi)\, d\xi,$$

$\wp$ — Weierstrass fn.



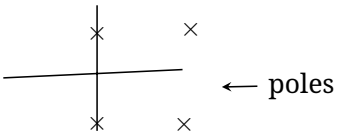
Original source details: GELFAND-P000653

Uncertain reading. A stroke with several crossbars follows i in the second denominator. The cancelled letter is undeciphered.

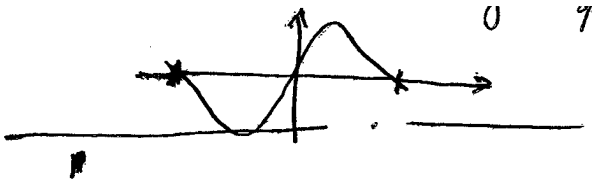


Original source details: GELFAND-P000653

Diagram and typed description.

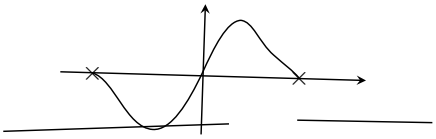


Crosses on the vertical axis and to its right.



Original source details: GELFAND-P000653

Diagram and typed description.



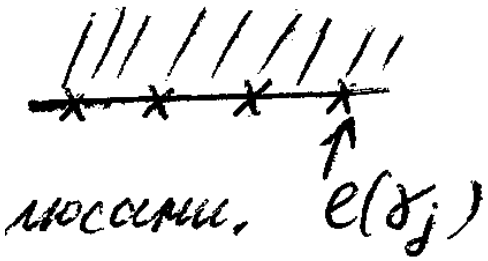
Curve with a minimum and a maximum, two crosses on the axis, and a broken lower line. A heavy illegible mark lies below the left part of the line.

Consider the fn.

$$f(z) = \sum_{k=1}^{\infty} e^{i\lambda_k z}.$$

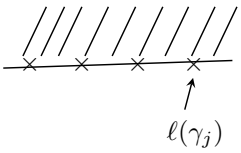
Continues analytically downwards with simple poles.

[At right]



Original source details: GELFAND-P000653

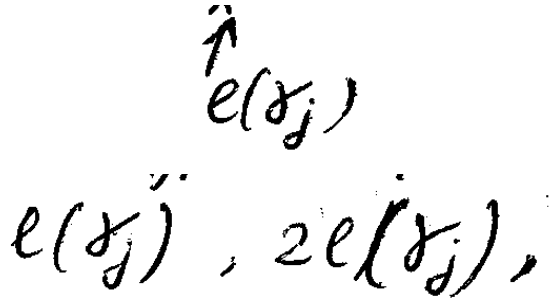
Diagram and typed description.



Four crosses on a hatched horizontal line. The arrow from below points to the rightmost cross.

Fund. sol. of the wave eq. Continuat.

Residues at $\ell(\gamma_j)$, $2\ell(\gamma_j)$, ... — Fourier coeffs., etc.



Uncertain reading. The letter with subscript j in the label and residue line reads γ, as in the geodesic notation on the preceding frame.

Original source details: GELFAND-P000653



Uncertain reading. The abbreviation before "Fourier" is heavily overwritten and reads "коэфф." (coeffs.).

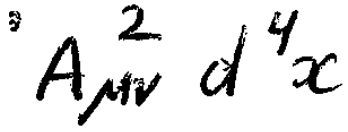
Original source details: GELFAND-P000653

GELFAND-P000654

5/IV-76. Gelfand Sem. Polyakov.

Action

$$\begin{cases} S = \dfrac{1}{4e^2}\,\mathrm{tr}\int A_{\mu\nu}^2\,d^4x \\ F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \end{cases}$$



Uncertain reading. The integrand is written $A_{\mu\nu}^2$. The next line defines $F_{\mu\nu}$.

Original source details: GELFAND-P000654

$A_\mu^+ = -A_\mu$, A_μ — a skew-Herm. matr. 2×2, which is a fn. on $\mathbb{R}^4$.

[Below A_μ] $\mu = 0, 1, 2, 3$.

A_μ — the field. S — a functional of the field.

Classical eqs. of motion — stationarity of S.

For numerical A_μ — electromagn. field.

Analogue for discrete x and t.

$$x = (t, \vec{x}).$$

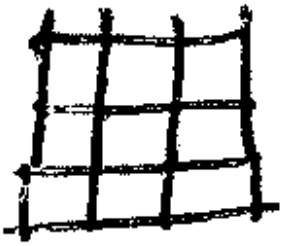


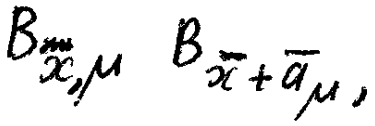
Diagram and typed description.



Four roughly horizontal lines and four vertical lines.

Original source details: GELFAND-P000654

$$S = \frac{1}{4e_0^2} \sum_{\bar{x}} \chi \left(B_{\bar{x},\mu} B_{\bar{x}+\bar{a}_\mu,\nu} B^{-1}_{\bar{x}+\bar{a}_\nu,\mu} B^{-1}_{\bar{x},\nu} \right).$$



Original source details: GELFAND-P000654

Uncertain reading. Short bars appear over x and a in the product. The bar over the first x is heavily retraced.

a_μ — frame, $\quad B_{x,\mu} \in SU(2)$.

χ — any character (crossed out)



Original source details: GELFAND-P000654

Undeciphered passage. The crossed-out word before "fn." is illegible.

(a fn. constant on classes of conj. elements — e.g. trace).

$$Z = \int \prod_{\vec{x},\mu} dB_{\vec{x},\mu}\, e^{-S(B)}.$$

$12n$ indep. variables (n — the number of lattice sites, at $\forall$ — 4 matrices).

$B_{\vec{x},\mu}$ — indep. vars. $\Longrightarrow dB_{\vec{x},\mu}$ — inv. measure ($4n$ integrals with inv. measures).

$$t \to it \quad \Longrightarrow \quad e^{iS} \to e^{-S}.$$

In the continuous case — a functional integral.

Correl. fns.

$\ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots$

GELFAND-P000655

12/IV-76. Gelfand Sem.

Fuchs. Use of computers in theor. mathematics.

Comput. of the cohom. of the alg. of formal divergence-free vect. fields

Given $\mathbb{R}^n$. consider $(\mathbb{R}^n)'$ and polynomials

$$p(\xi_1, \dots, \xi_q; \eta_1, \dots, \eta_q), \qquad \xi_i \in \mathbb{R}^n, \quad \eta_i \in (\mathbb{R}^n)'.$$

They have the property

a) if the pair ξ_i, η_i is put in the place of ξ_j, η_j, the polynomial changes sign.

$dP = P_1$ given by the formula

$$P_1(\xi_1, \dots, \xi_{q+1}, \eta_1, \dots, \eta_{q+1}) = \sum_{i<j} (-1)^{i+j-1} \big[(\xi_i, \eta_j) - (\xi_j, \eta_i)\big] \\ P(\xi_i + \xi_j, \xi_1, \dots, \widehat{\xi_i}, \dots, \widehat{\xi_j}, \dots, \xi_{q+1}, \\ \eta_i + \eta_j, \eta_1, \dots, \widehat{\eta_i}, \dots, \widehat{\eta_j}, \dots, \eta_{q+1}).$$

problem: find the cohomology of this complex (still undone).

A computer helped find nontriv. cycles.

Consider those linear in η; then this is the cohom. complex of the algebra W_n.

Undone even for $n = 1$.

All results obtained concern this case.

complex = the sum of many finite-dimensional ones.

Total degree in ξ: $m_1 + q$, in η: $m_2 + q$. [At right] ‖ subcomplex K_{m_1, m_2}.

(q varies)

polys. in $\xi_1, \dots, \xi_q, \eta_1, \dots, \eta_q$.



Uncertain reading. The first abbreviation tentatively reads “мн-ны” (polynomials).

Original source details: GELFAND-P000655

Each of them is finite-dimensional.

$m_1 \neq m_2 \Longrightarrow$ it is acyclic.

In fact, $\sum$ homol. of a countable number of subcompls.

$$H(K) = \bigoplus_m H(K_{m,m}).$$

(for now n is arbitrary)

$n = 1$.

$m = 1, 2, 3$ are acyclic.

$m = 4$ nontriv. in dim. 7 and 10 — computed on a machine (Gelfand, Fuchs, Kalinin).



Uncertain reading. The third surname reads “Калинин” (Kalinin).

Original source details: GELFAND-P000655

$m = 5$ — “work area overflow $N = 2$”.

Euler char. does not depend on the differential.

Perchik (Bott’s grad. student) computed up to $m = 35$.



Uncertain reading. The manuscript spells the surname “Першик”.

Original source details: GELFAND-P000655

$$K_{m,m} \supset K'_{m,m} \qquad H(K_{m,m}) = H(K'_{m,m}) \otimes H(S^3).$$

$$\chi^*_m \text{ — ?} \qquad (\text{for } K'_{m,m}).$$

Perchik found the generating fn. for χ_m.



Uncertain reading. After “for” the manuscript has χ_m without a star.

Original source details: GELFAND-P000655

Interesting: prove that for arbitrarily large m there are $\chi_m \neq 0$.

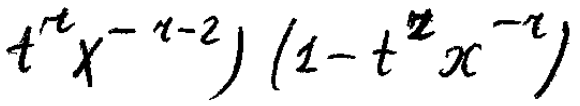


Uncertain reading. The middle letters are retraced. The word reads “Интересно” (interesting).

Original source details: GELFAND-P000655

$$\tfrac{1}{2}(1 - t^{-1}x^{-1})(1 - t^{-1}x)$$
$$(1 - x^{-2}) \qquad (1 - x^2)$$
$$(1 - tx^{-3})(1 - tx^{-1})(1 - tx)(1 - tx^3)$$
$$\dots$$

$$(1-t^{\ell}x^{-\ell-2})(1-t^{2}x^{-\ell})\dots(1-t^{\ell}x^{\ell+2}).$$



Uncertain reading. The middle exponent of t is heavily retraced and resembles 2. The outer exponents read ℓ.

Original source details: GELFAND-P000656

Take the product of all of them. Discard everything that contains x. Then one obtains

$$\sum \chi_m t^{2m} = t^{-2} + 2 - t^{8} - t^{14} - t^{22} + \dots + 10t^{70} + \dots$$

Wrote the expansion through t^{70}.

(Existence proved of $\geqslant 57$ nontriv. [inserted above: classes] of cohomology.)

Buchstaber on Adams's work on Buchstaber's work.

X — top. space. Consider $X \times X \to X$ for a group. $\Longrightarrow$

[At left, with arrow to "Consider"] If X — an actual group, then,

$$C(X \times X) \longleftarrow C(X) \quad \text{— coalgebra struct..}$$

Roughly:

$$M(f(x \cdot y)) \longleftarrow f(x)$$

(expected value of the values).

— · —

S. P. Novikov General theory of relativity.

Read: Landau and Lifshitz (together with Riem. geometry) — Theory of Fields.

$(t, \vec{x})$ — space of events. $(t, \vec{x}) \to (x^0, \vec{x}), \quad x^0 = ct.$

$$ds^2 = g_{ij}\, dx^i\, dx^j, \qquad g_{ij} = \begin{pmatrix} 1 & & & 0 \\ & -1 & & \\ & & -1 & \\ 0 & & & -1 \end{pmatrix}.$$

Proper time — length of a curve.

Let M^4; given g_{ij}. U — coord. system x^0, x^1, x^2, x^3.





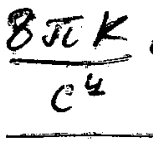
Uncertain reading. The symbol before "coord. system" tentatively reads U. Its shape resembles Cyrillic И.

Original source details: GELFAND-P000656

Einstein: the metric is called the gravit. field (1913) (+ Grossmann)

Einstein's eq. — independently Einstein and Hilbert (1916)

$$\underbrace{R_{ij} - \frac{1}{2} R g_{ij} = \frac{8\pi K}{c^4} T_{ij}}\,.$$



Uncertain reading. The numerator has uppercase K.

Original source details: GELFAND-P000656

Here $R_{j\ell} = \sum_i R^i_{ji\ell}$ — Ricci tensor.

[At right] $\| \ (R^i_j)_{k\ell}$ — 2-form (skew-symm. in k, ℓ) with values in the Lie alg. of the group $O(n)$.

T_{ij} — matter energy-momentum tens. (external mass).

[At right] T_{ij} accounts for everything except the gravit. field.

$$R = \sum_{j,\ell} g^{j\ell} R_{j\ell}.$$

In any dimension

$$\underbrace{R_{ij} - \frac{1}{2} R g_{ij} = T_{ij}}\,.$$

Let the phys. quantities be determined by the Lagrangian L.

$$S = \int L(q, g_{ij}) \underbrace{\sqrt{g}\, d^4x}_{d\sigma}\,.$$

$d\sigma$ — inv. volume elem..

Then

$$\sqrt{g}\, T_{ij} = \frac{\delta S}{\delta g_{ij}}.$$



Uncertain reading. This fraction has g_{ij}, with lower indices.

Original source details: GELFAND-P000656

Electromagn.

$$L_{\text{elm.}} = F_{\mu\nu} F^{\mu\nu}, \quad \text{where} \quad F_{\mu\nu} dx^\nu \wedge dx^\mu = d(\vec{A}).$$

[Arrow to $F^{\mu\nu}$] the metric enters.

No forces:

$$\Longrightarrow \quad \frac{\delta S}{\delta g^{ij}} = 0, \qquad S = \int R\sqrt{g}\, d^4x.$$

GELFAND-P000657

Einstein's eq.: in the grav. field there is no energy-mom. apart from that in other forms of energy.

G. Is it quantized? If so, there is no metric (geom. structure of space).

N. But this affects only 10^{-33} cm (whereas nuclear interactions at 10^{-23} cm).

As long as space makes sense, Einstein's eq. holds.

Quantum mech. is limited only by times of 10^{-43} sec.

$$T_{00} \text{ — energy density} = \varepsilon > 0.$$

[heading crossed out]



Undeciphered passage. Dense hatching obscures the heading before "Problems".

Original source details: GELFAND-P000657

Problems

1) continuum approx.: hydrodyn. form with Pascal's law

$$T_{ij} = (p + \varepsilon)u_i u_j - p g_{ij}.$$

[At right] (p — press., ε — energy density

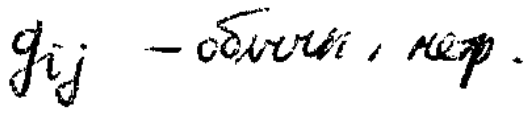
$\varepsilon, p, \vec{u}$ — fns. of the point.

$$\varepsilon = \varepsilon(p) \text{ — given} \qquad \text{or} \quad p = p(\varepsilon).$$

Problem: find (M^4, g_{ij}). Dust-like fluid $p = 0$.

or $p = \dfrac{\varepsilon}{3}$ — ultrarelativistic fluid (light

Hydrodyn. — g_{ij} — ordinary metr.



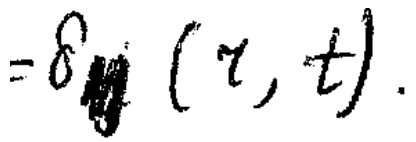
Uncertain reading. The final abbreviation tentatively reads "метр." (metric).

Original source details: GELFAND-P000657

$$\underline{\sum_k \nabla_k T_i^k = 0} \qquad \Longrightarrow \qquad \text{laws of hydrodyn.}$$

Field of a point mass

$$R_{ij} - \tfrac{1}{2} R g_{ij} = \delta_{ij}(r, t).$$



Uncertain reading. The heavily retraced signs on δ tentatively read ij.

Original source details: GELFAND-P000657

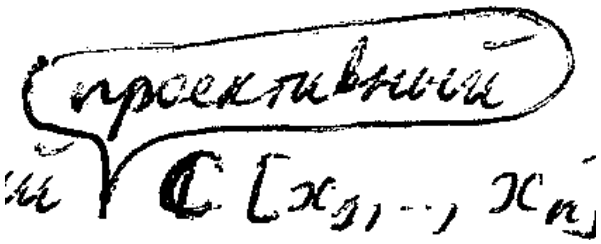
symmetry grp. $SO(3)$ — Schwarzschild soln.. First M^4 given by Kruskal in 1960.
Spherical coords. describe one chart (and not the whole manifold).
r in preferred coords. has saddles (gravit. radii).
Next time Novikov.



Original source details: GELFAND-P000657

Uncertain reading. The letters before "time" read "След." (next).

19/IV-76. Gelfand Sem.
Serre's problem. Is every [inserted in a callout: projective] $\mathbb{C}[x_1, \ldots, x_n]$-module free?



Original source details: GELFAND-P000657

Diagram and typed description.

projective

The word "projective" in an outlined callout before the ring.

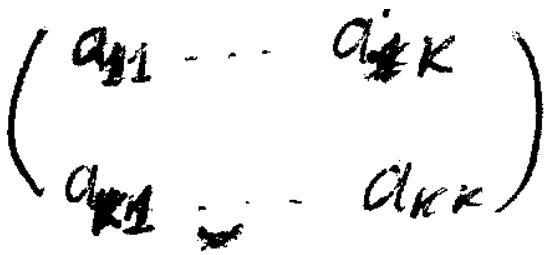
— ∘ —

Gelfand, following Vaserštein's notes

$$a_1, \ldots, a_k \in \mathbb{C}[x_1, \ldots, x_n] = A.$$

is called unimodular if $\exists p_1, \ldots, p_k : \sum p_i a_i = 1$.
Is there a matrix with entries in A, and with det $= 1$

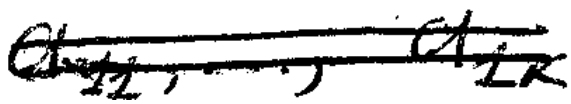
$$\begin{pmatrix} a_{11} & \ldots & a_{1k} \\ a_{k1} & \ldots & a_{kk} \end{pmatrix},$$



Original source details: GELFAND-P000657

Uncertain reading. The indices in a_{11}, a_{1k}, a_{k1} are retraced or corrected.

and with first row [crossed out: $a_{11}, \ldots, a_{1k}$] $a_1, \ldots, a_k$?



Original source details: GELFAND-P000657

Uncertain reading. The indexed row before the corrected first row is crossed out.

Drinfeld on the same topic.

B — a ring

$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} \quad \text{— a unimod. column in } B[x]$$

(x — one var.).

leading coeff. of $f_1 = 1$ Then $f(x) \sim f(0)$ (transformed by a nonsing. matrix over $B[x]$).

Main tool — resultant:

$$\prod (\xi_i - \eta_k)$$

[At right] ξ_i, η_k — roots of 2 polys. in 1 var.

GELFAND-P000658

Main lemma. Let $c = \mathrm{Res}(f_1, f_2)$. Then if f — any column, then,

$$f(Y_1 + cY_2) \sim f(Y_1) \quad \text{over } B[Y_1, Y_2].$$



Uncertain reading. The relation sign has several retraced wavy strokes. The reading $\sim$ is tentative.

Original source details: GELFAND-P000658

Proof: by an explicit construction (Suslin)

Need: $c = 1$ and set $Y_2 = -Y_1$.

Reduction to this lemma.

A change of var. reduces to the case where 1 of the leading coeffs. is 1.

Remove all vars. in turn.

S. P. Novikov.

M^n, g_{ij}, sign. $(1, -1, -1, -1)$

$$R_{ij} - \frac{1}{2} R g_{ij} = T_{ij}.$$

[At right] || Spher. symm. problem: to completion in 1960, Kruskal.

Phys. formulation of the problem — hypotheses on T_{ij}.

Continuum

$$T_{ij} = (p + \varepsilon) u_i u_j - p g_{ij}, \quad \text{where } p = p(\varepsilon);$$
$$\vec{u} = (u_0, u_1, u_2, u_3); \qquad 1 = \langle u, u \rangle.$$

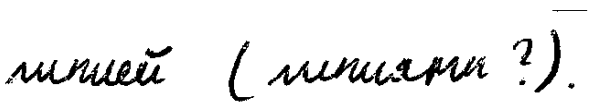
Dust: $p = 0$. Speed of sound:

$$c^2_{\text{sound}} = \frac{\partial p}{\partial \varepsilon} \geqslant 0,$$
$$c^2_{\text{sound}} \leqslant c^2 = 1.$$

There are too few eqs. But there are gauge transfs.

Hilbert wrote down the eqs. but gave them up as incorrect.

There is the metric of Gödel (that very one) with a closed timelike line (lines?).



Uncertain reading. The parenthesis reads “линиями?” (lines?) after the singular “линией”.

Original source details: GELFAND-P000658